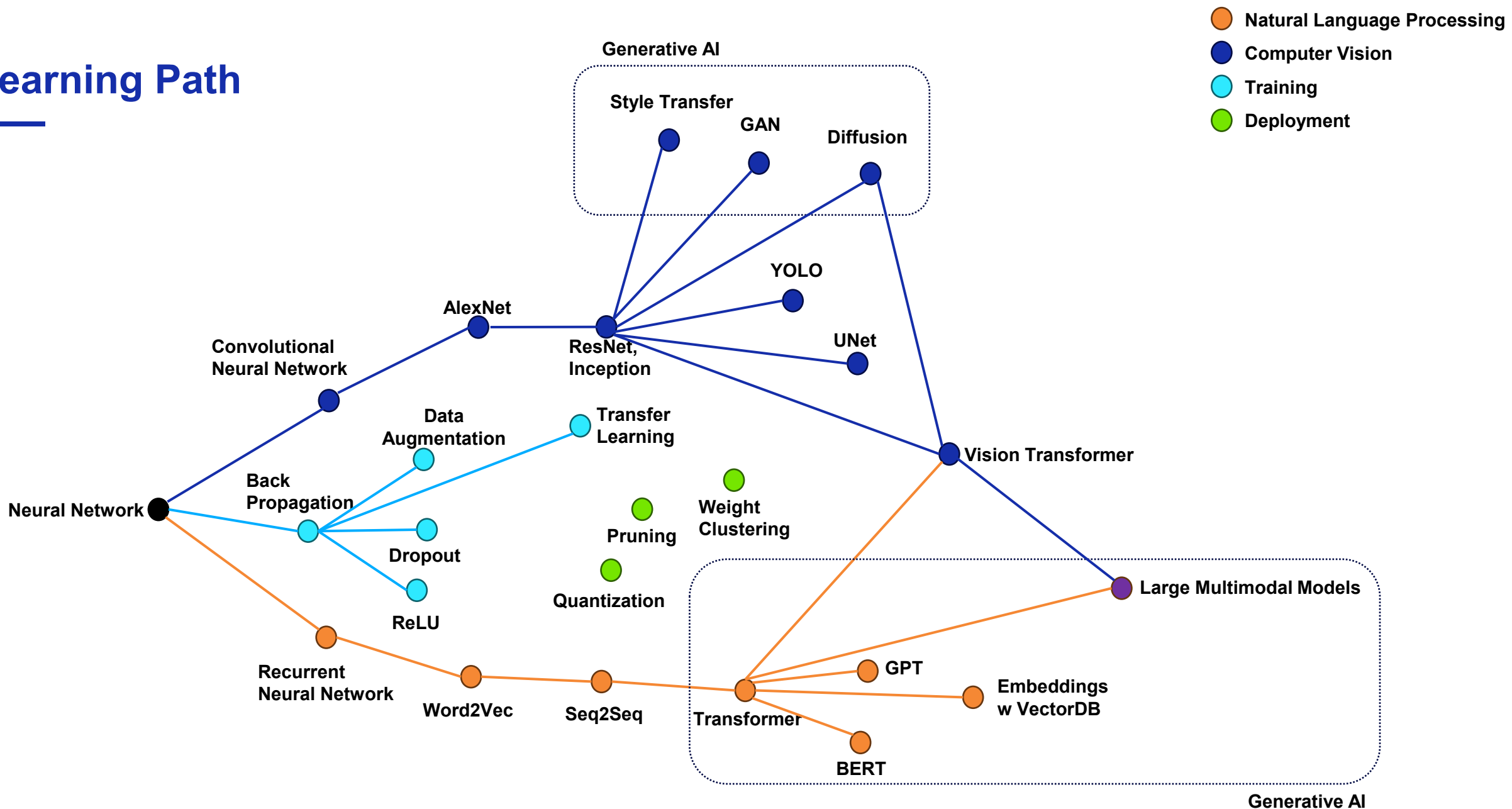


딥러닝 이해와 구현

박종현

Learning Path



Contents

Introduction

딥러닝의 탄생 이야기
vs Machine Learning
흐름, Trend

Neural network

동작 이해, 학습, 평가
(실습) Keras 뉴럴넷 만들기
(실습) 시각화 도구로 관찰하며 뉴럴넷 학습시키기

Computer Vision

CNNs
(실습) 개/고양이 이미지 분류하기
딥러닝의 시작, 발전
- AlexNet, VGGNet, GoogLeNet, ResNet & ...
Transfer Learning (Fine Tuning)
(실습) Transfer Learning
Object Detection, Segmentation
(실습) Object Detection
Generative - style transfer, VAE, GAN, Diffusion
(실습) stable diffusion
Vision Transformer

NLP

Word2Vec, RNN, LSTM, GRU, Seq2Seq
Attention
Transformer
LLM (GPT vs BERT)
AGI를 향한 도전 - Embeddings & VectorDB
(실습) GPT 응용 - Embeddings

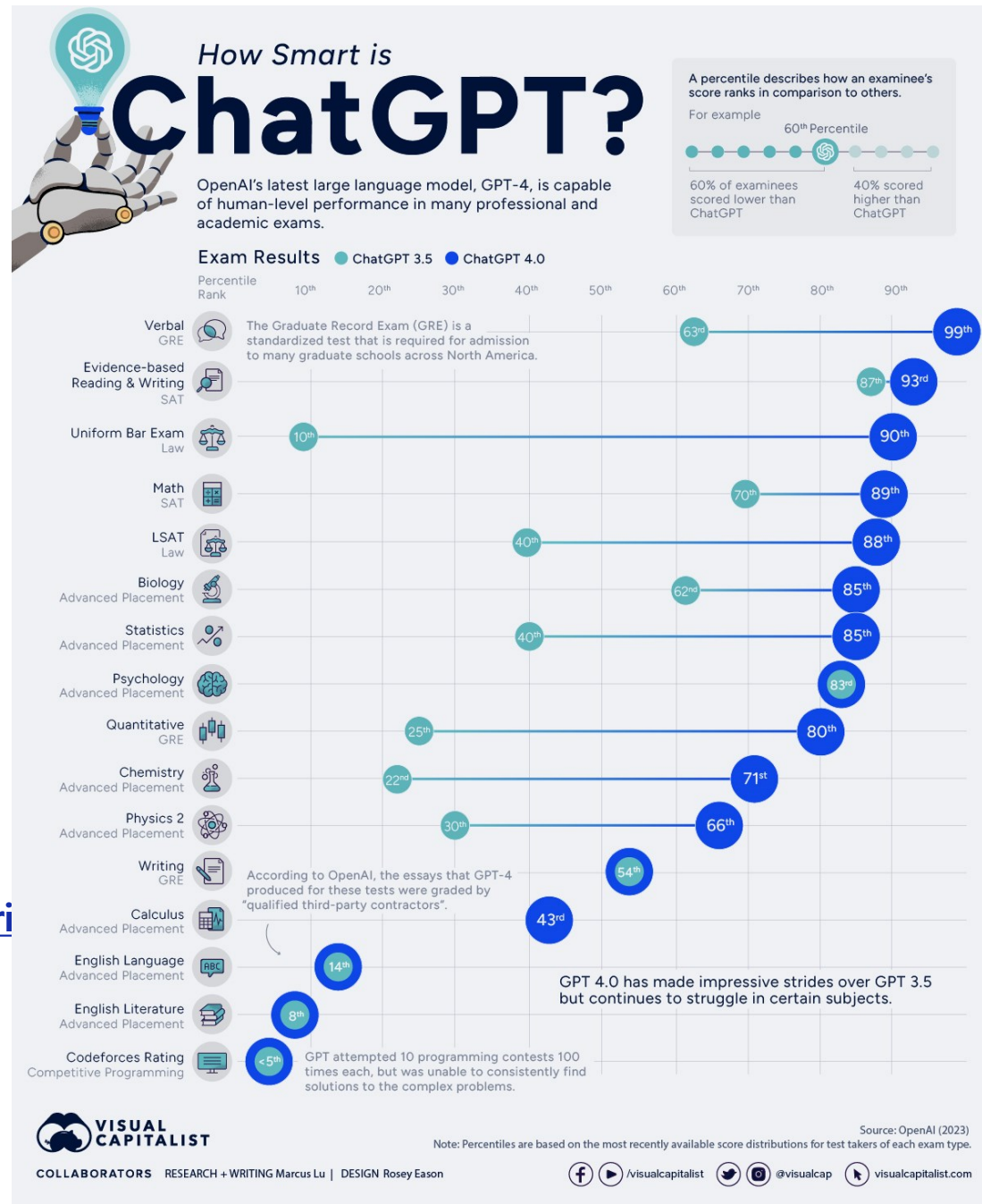
Outro

미래 - Paradigm의 변화

Appendix

Models Now

- ChatGPT (22년 11월 서비스 개시)
- Sora (24년 2월 발표)
 - <https://openai.com/sora>
 - 현재는 레드팀 테스트 중
- Google Gemini (23년~)
 - <https://deepmind.google/technologies/gemini/#gemini-1.0>
 - Phone & cloud service
 - Project Astra, Veo
- NVIDIA Robot (24년 3월)
 - <https://www.youtube.com/watch?v=51TYhPJ4zys>
- Meta Llama3 (24년)
 - <https://ai.meta.com/blog/openqa-embodied-question-answering-robotics-ar-glasses/>
- GPT-4o, OpenAI o1
 - <https://openai.com/index/hello-gpt-4o/>

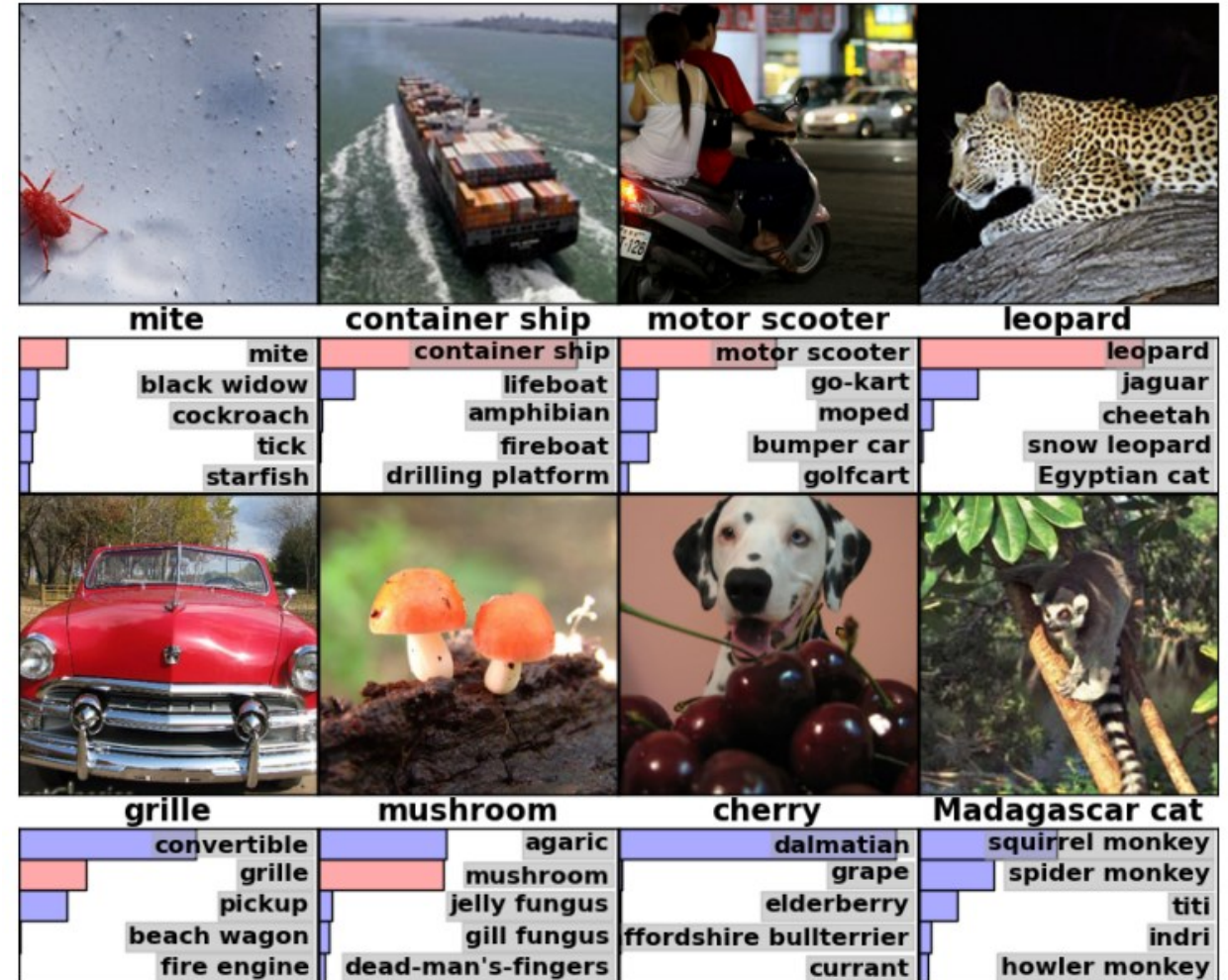


The Birth of Deep Learning



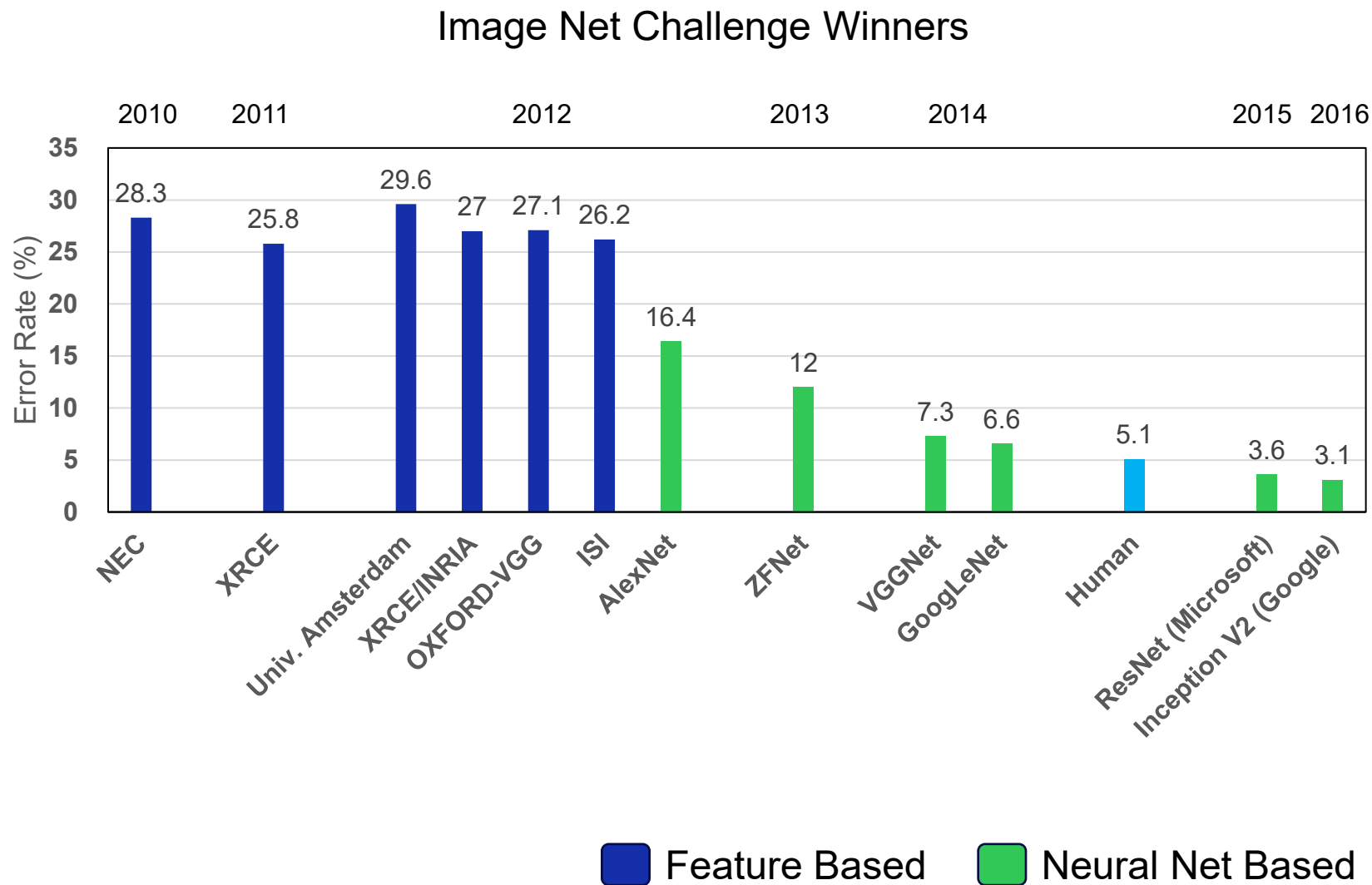
ImageNet Challenge (2010~)

- 1,200,000 개의 이미지로 학습
- 150,000 개의 이미지로 테스트
- 1000가지 항목 중 무엇인지 맞추는 대회



The Birth of Deep Learning

- 2012년 AlexNet 의 등장
- 3년 만에 사람 초월
- (참고) 2016년 알파고로 국내 관심



Geoffrey Hinton

- The Father of Deep learning
 - Yann Lecun, Ilya Sutskever ...
 - BM, AlexNet, ReLU, Dropout ...
- Toronto Univ, Google Brain
- Turing Award
- Novel Prize !!!!!



REVIEW

doi:10.1038/nature14539

Deep learning

Yann LeCun^{1,2}, Yoshua Bengio³ & Geoffrey Hinton^{4,5}

Deep learning allows computational models that are composed of multiple processing layers to learn representations of data with multiple levels of abstraction. These methods have dramatically improved the state-of-the-art in speech recognition, visual object recognition, object detection and many other domains such as drug discovery and genomics. Deep learning discovers intricate structure in large data sets by using the backpropagation algorithm to indicate how a machine should change its internal parameters that are used to compute the representation in each layer from the representation in the previous layer. Deep convolutional nets have brought about breakthroughs in processing images, video, speech and audio, whereas recurrent nets have shone light on sequential data such as text and speech.

Machine-learning technology powers many aspects of modern society: from web searches to content filtering on social networks to recommendations on e-commerce websites, and it is increasingly present in consumer products such as cameras and smartphones. Machine-learning systems are used to identify objects in images, transcribe speech into text, match news items, posts or products with users' interests, and select relevant results of search. Increasingly, these applications make use of a class of techniques called deep learning.

Conventional machine-learning techniques were limited in their ability to process natural data in their raw form. For decades, constructing a pattern-recognition or machine-learning system required careful engineering and considerable domain expertise to design a feature extractor that transformed the raw data (such as the pixel values of an image) into a suitable internal representation or feature vector from which the learning subsystem, often a classifier, could detect or classify patterns in the input.

Representation learning is a set of methods that allows a machine to be fed with raw data and to automatically discover the representations needed for detection or classification. Deep-learning methods are

intricate structures in high-dimensional data and is therefore applicable to many domains of science, business and government. In addition to beating records in image recognition¹⁻⁴ and speech recognition⁵⁻⁷, it has beaten other machine-learning techniques at predicting the activity of potential drug molecules⁸, analysing particle accelerator data^{9,10}, reconstructing brain circuits¹¹, and predicting the effects of mutations in non-coding DNA on gene expression and disease^{12,13}. Perhaps more surprisingly, deep learning has produced extremely promising results for various tasks in natural language understanding¹⁴, particularly topic classification, sentiment analysis, question answering¹⁵ and language translation^{16,17}.

We think that deep learning will have many more successes in the near future because it requires very little engineering by hand, so it can easily take advantage of increases in the amount of available computation and data. New learning algorithms and architectures that are currently being developed for deep neural networks will only accelerate this progress.

Supervised learning

The most common form of machine learning, deep or not, is supervised learning.

갑자기? 어떻게?

Machine Learning 의 한 분야인 Neural Network 는 이미 존재했으나
뉴런을 많이 사용하지 못했음

1. 학습 데이터가 부족함

Big Data Era

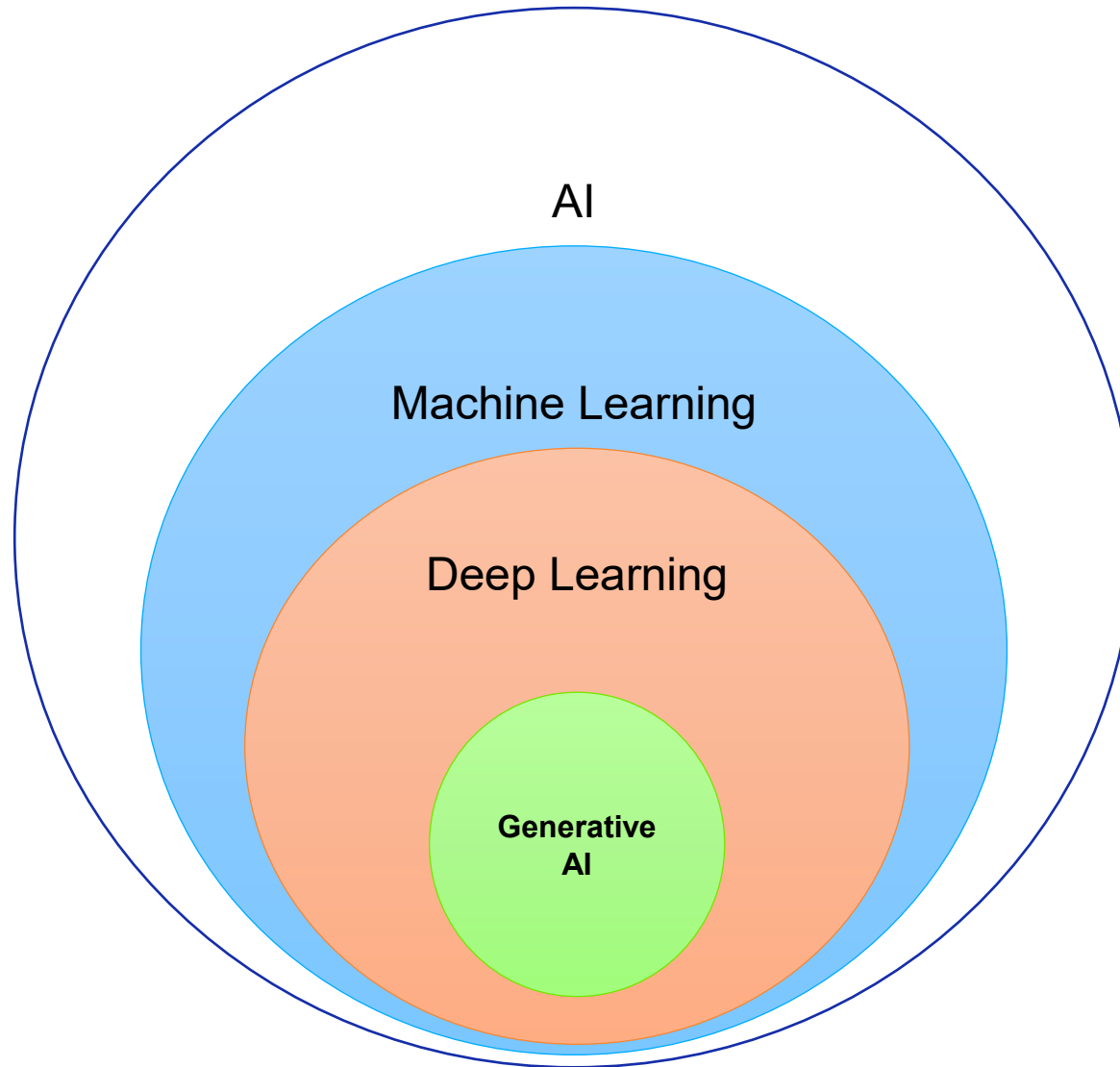
2. 아무리 가르쳐도
학습이 잘 안됨

Algorithms for NN
- Relu, Dropout, ...

3. 너무 오래 걸림

Hardware
- GPU, GPU, GPU

AI, Machine Learning, Deep Learning, Generative AI



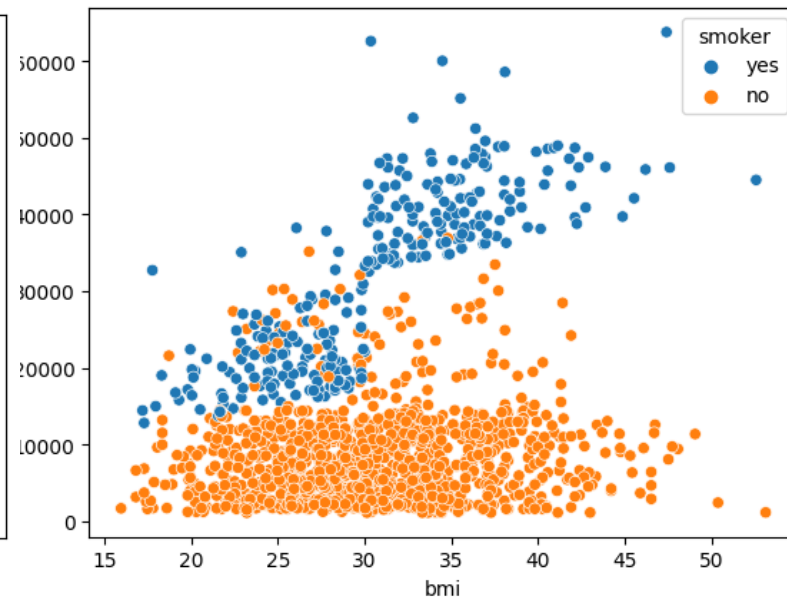
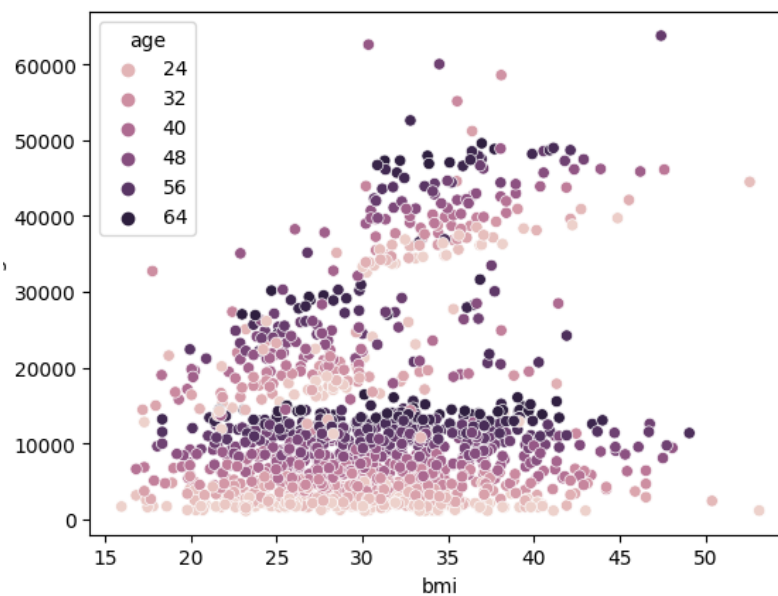
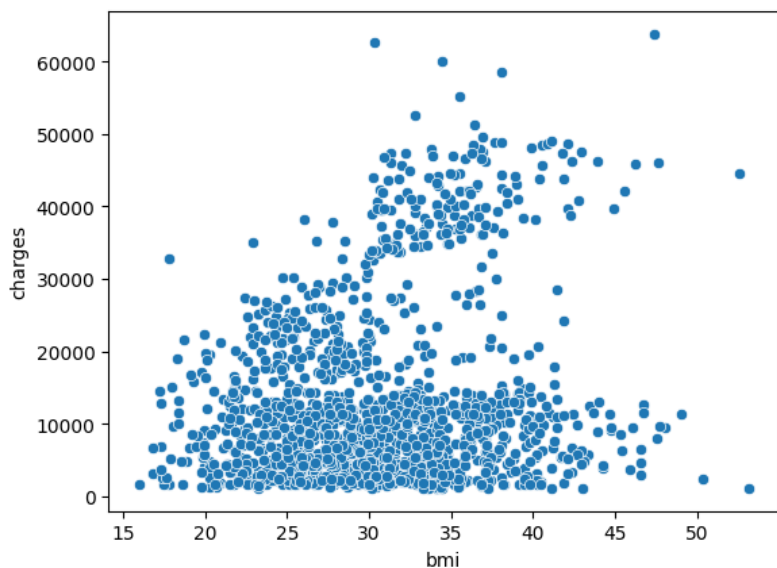
Machine Learning Example – Decision Tree

- 보험 계리 문제를 풀어봅시다!

	age	sex	bmi	children	smoker	region	charges
0	19	female	27.900	0	yes	southwest	16884.92400
1	18	male	33.770	1	no	southeast	1725.55230
2	28	male	33.000	3	no	southeast	4449.46200
3	33	male	22.705	0	no	northwest	21984.47061
4	32	male	28.880	0	no	northwest	????

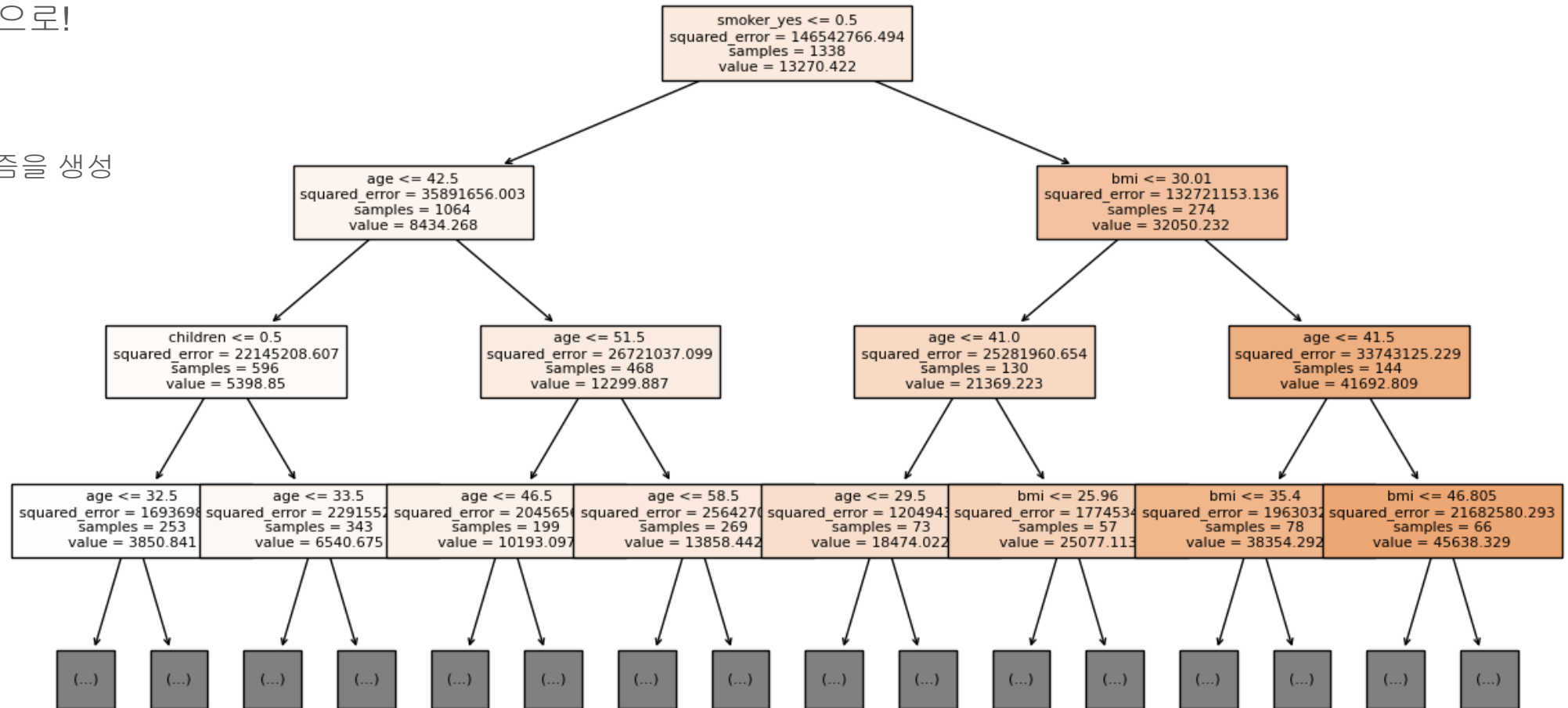
Machine Learning Example – Decision Tree

- 보험 계리 문제를 풀어봅시다!
- Human Analysis
 - BMI 랑 보험료랑 어떤 관계가 있을까?
 - 나이랑은 어떤 관계가 있을까?
 - 흡연여부와는 어떤 관계가 있을까?

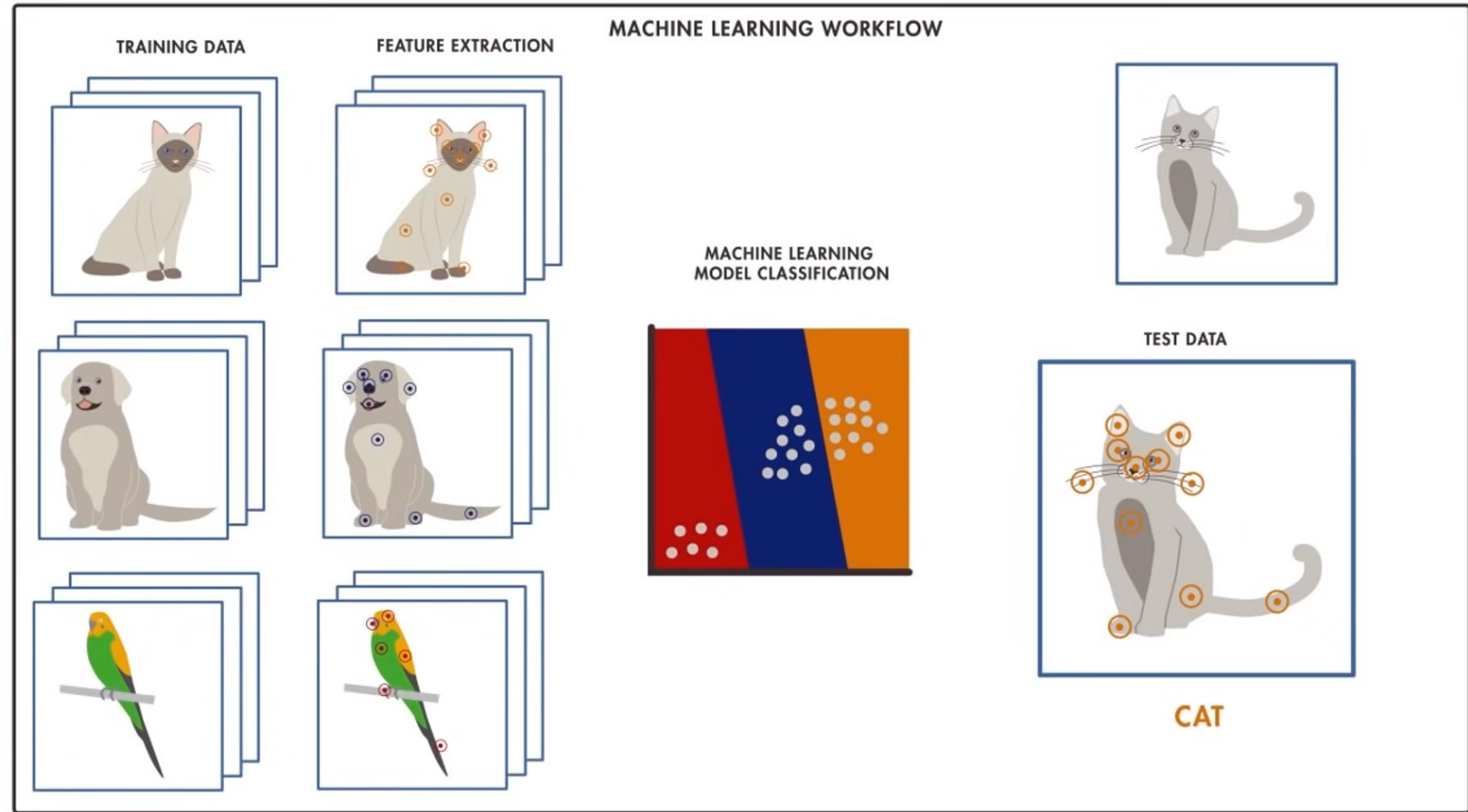
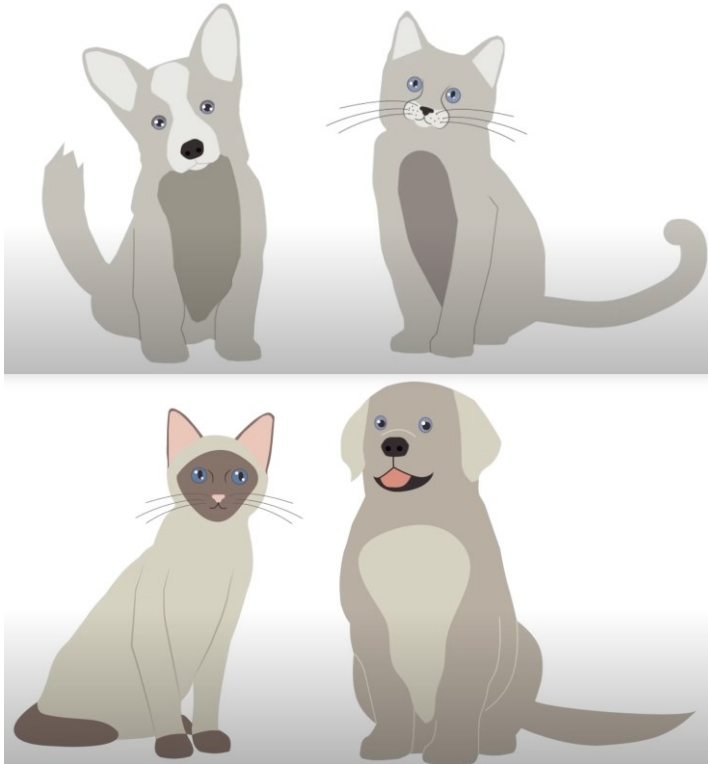


Machine Learning Example – Decision Tree

- 보험 계리 문제
- Machine Learning 으로!
 - Decision Tree
 - Data 로 알고리즘을 생성

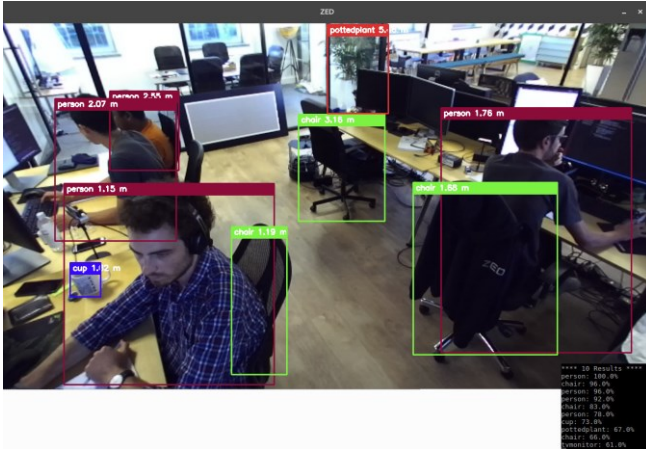


Machine Learning – Vision



쓰이는 곳?

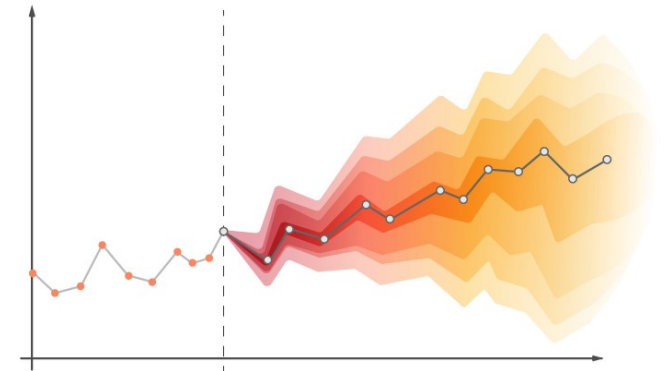
Computer Vision



Natural Language Processing



Time-Series, Recommendation, Speech Recognition ...



1. 문제 종류에 따라 필요한 지식이 많이 다르다!

Computer Vision

- Convolutional Neural Network
- Classification
 - ResNet, Inception ...
- Object Detection
 - YOLO, ...
- Segmentation
 - Unet, ...
- Vision Transformer, Diffusion...

Natural Language Processing

- Transformer
- GPT vs BERT
- Embeddings
- Vector Database

Time-Series, Recommendation, Speech Recognition ...

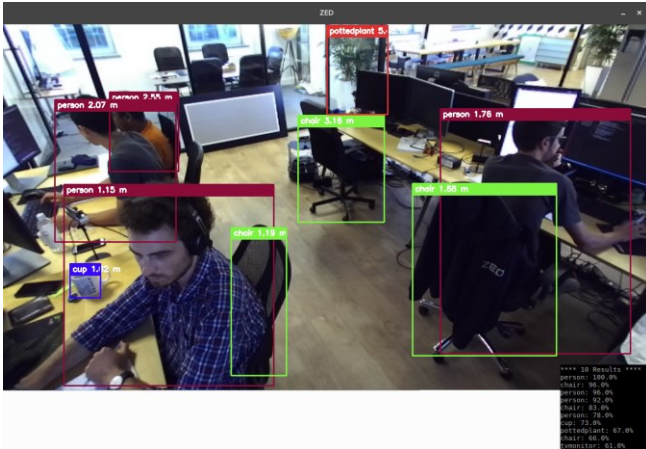
- RNN, Conv1D, ...
- DLRM

2. 문제 난이도에 따라서도 필요한 지식이 많이 다르다!

Inference

(ex) 물체인식 CCTV 을 만들자!

- YOLO! 를 쓰면 된다



Tuning

(ex) 사원인식 CCTV 을 만들자!

- Fine-Tuning YOLO!
(사원 데이터로 튜닝하기)

Research

(ex) 사원들 감정을 파악하는 CCTV 을 만들자!

- 걸음걸이나 표정으로 인식을 해야 할까?
- 새로운 모델을 정의해야 할까?
- 어떤 데이터로 학습을 시킬까?

3. 배포 범위에 따라서도 필요한 지식이 많이 다르다!

Local

(ex) 부서내에서 사용하기

- 코드 (또는 프로그램)를 배포하거나
- 접속을 열어주자

Cloud

(ex) B2B 또는 B2C 배포

- Scalable 클라우드 서버위에 배포
- 가능하다면 경량화 ; Pruning

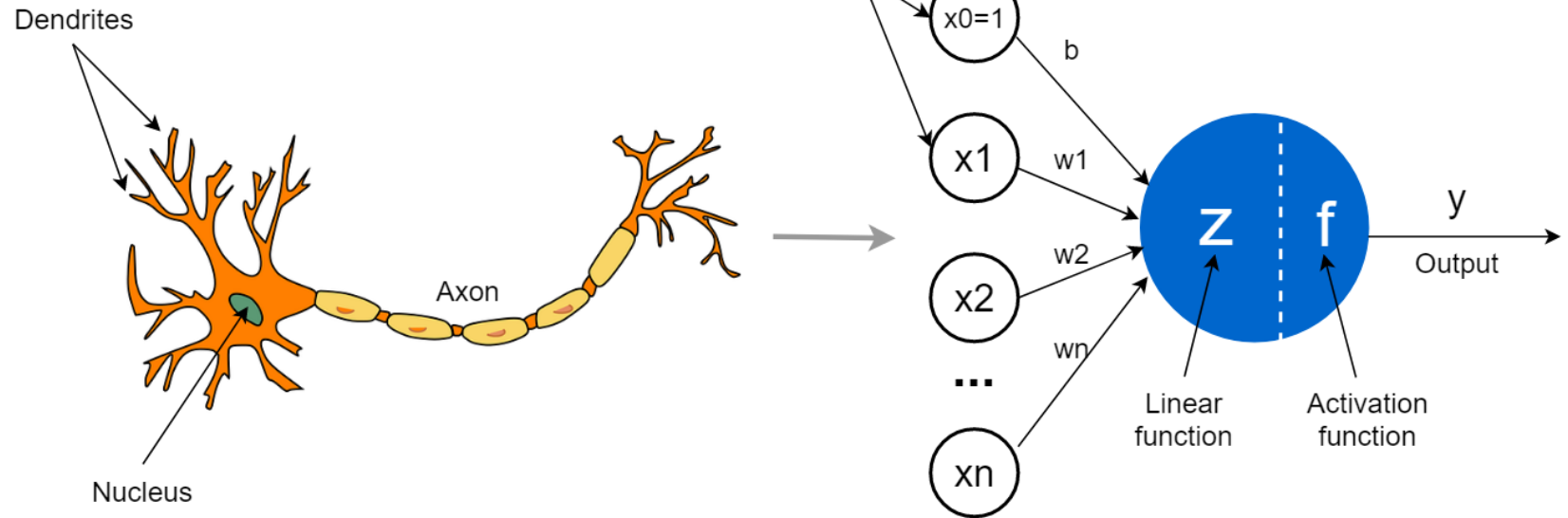
Embedded

(ex) 개인의 핸드폰에서 돌아가도록 배포

- 경량화 필수
- 모델 재정의
- Quantization aware Learning

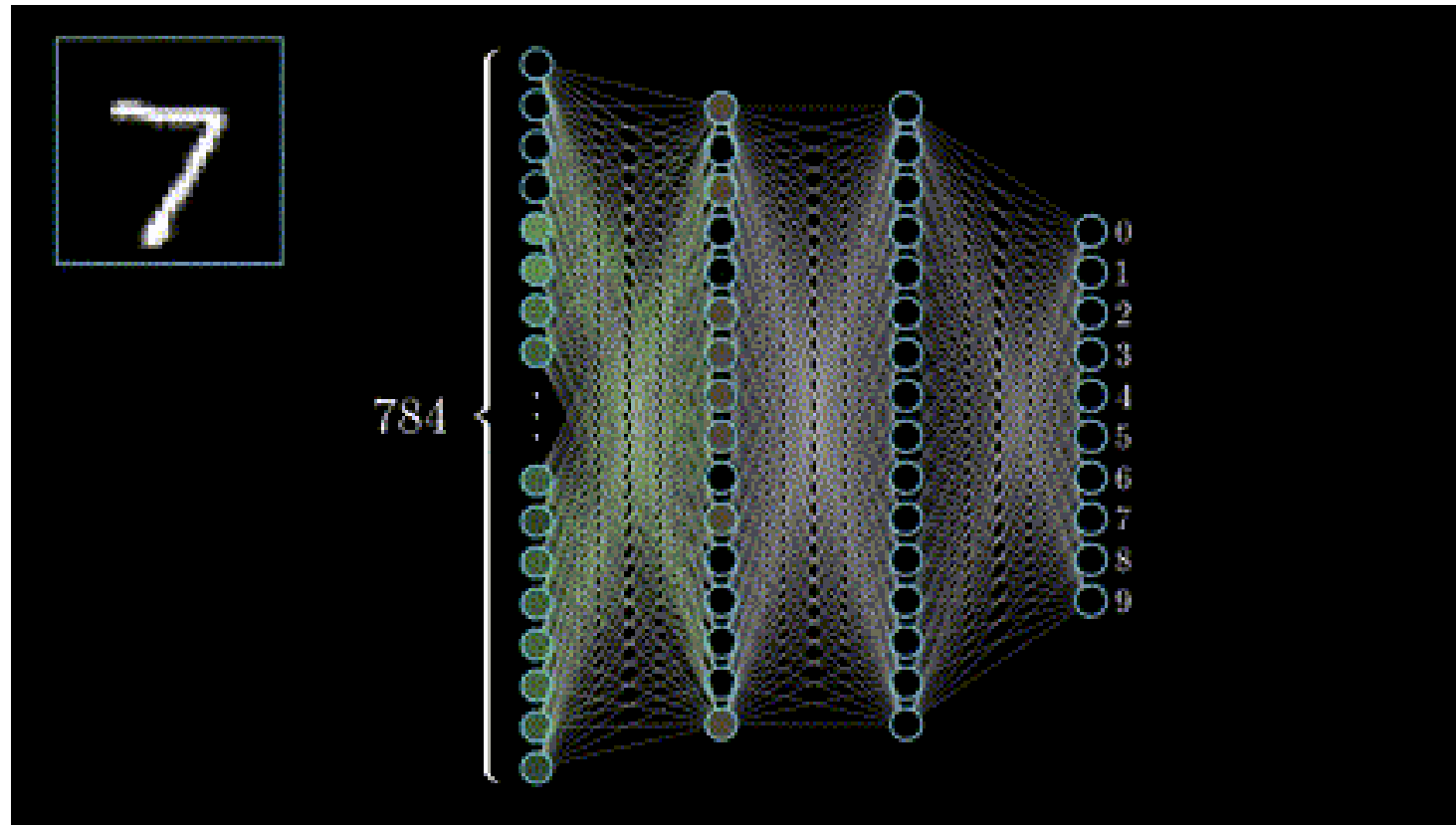
Artificial Neural Network

- Neurons to Artificial Neural Networks
- Weights = Parameter
- Activation Functions
 - Sigmoid
 - ReLU !



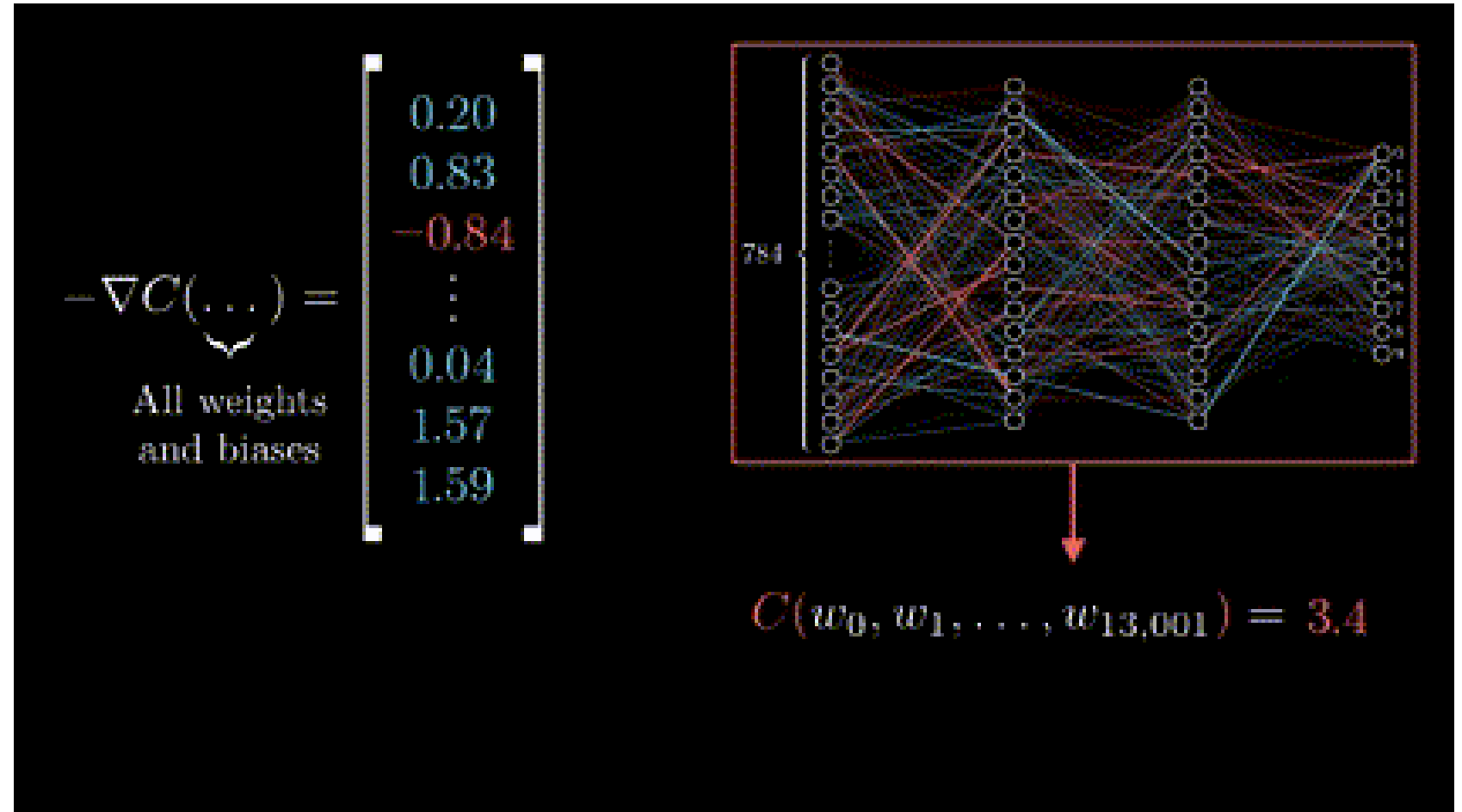
How the NN works

Videos!



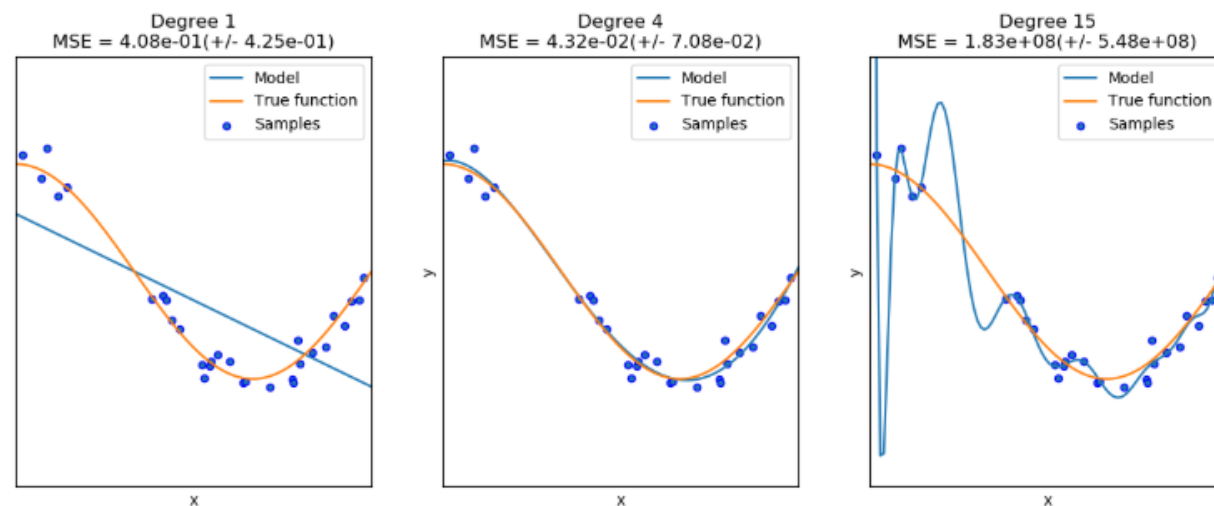
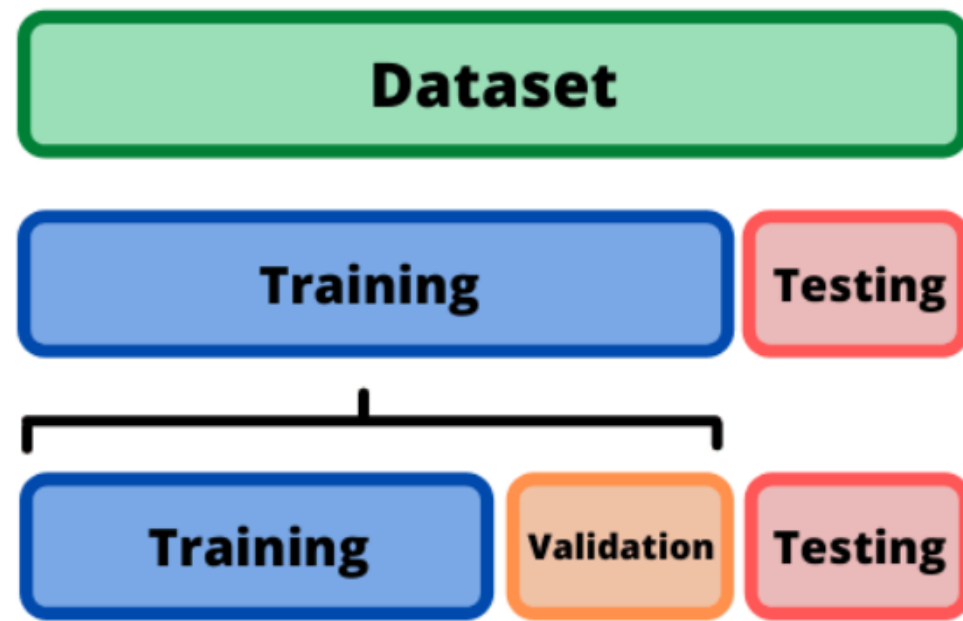
Training : Supervised Learning

- Loss function
- Gradient Descent
- Back Propagation
- Batch
- Local Minimum



Evaluate

- Epoch
- Validation
 - Overfitting 판단 (TBD)
 - unseen data 평가
 - Hyper Parameter 를 정의하기 위함!
- Testing



Practice – TensorFlow vs PyTorch

Beginner



▪ Keras

- 사용/학습 쉬움!



▪ Tensorflow

- 양질의 교육 자료



▪ PyTorch

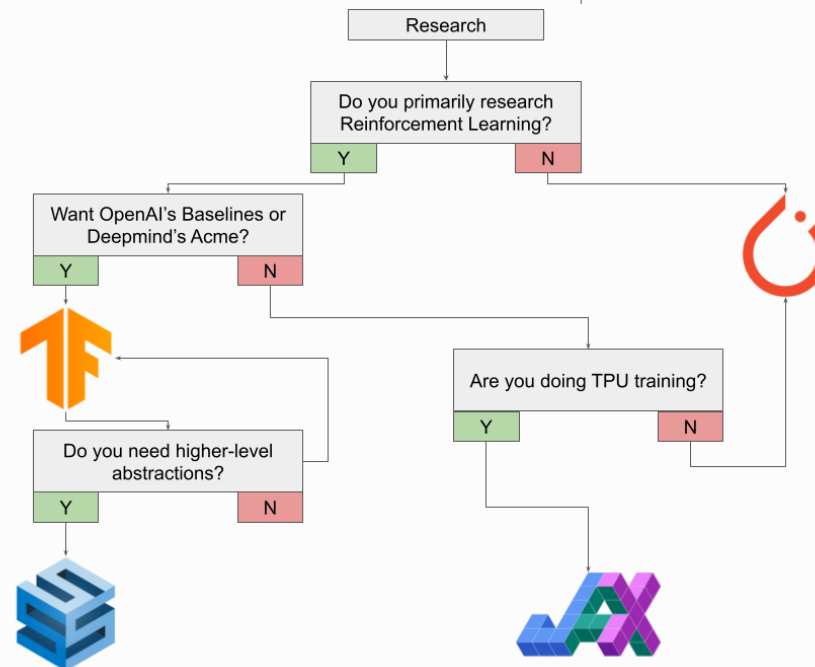
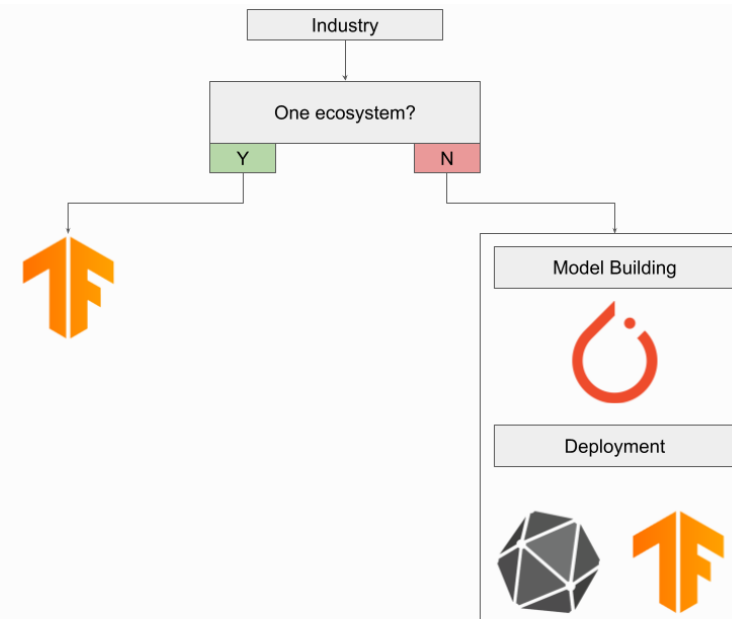
- 가장 활발한 연구

▪ 2022년 TF vs PyTorch

(<https://news.hada.io/topic?id=5578>)

▪ PyTorch는 죽었다

(<https://news.hada.io/topic?id=16369>)

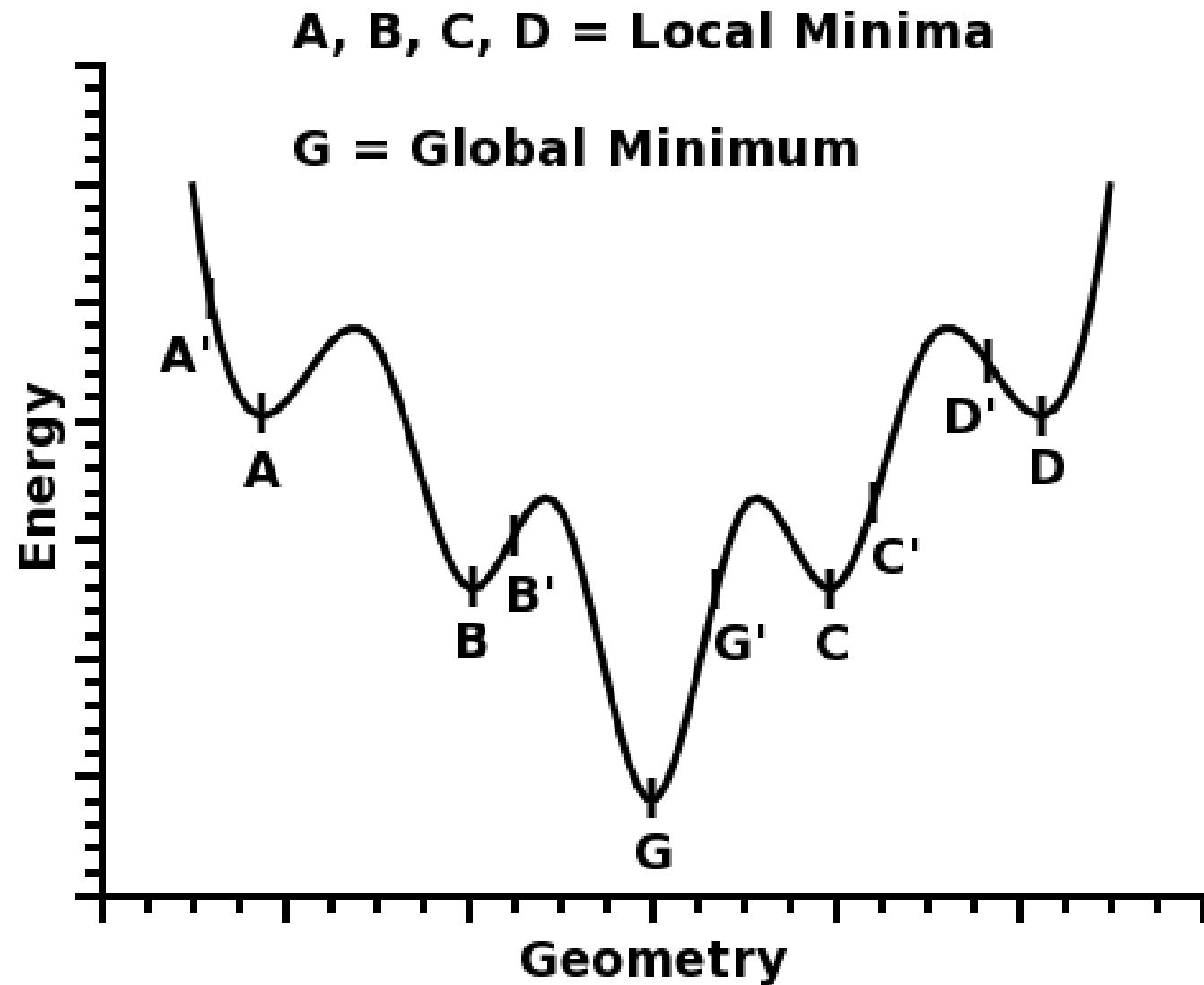


Practice 1 – 뉴런 만들기

- Build A Brain!
- 간단한 뉴런으로 Fashion MNIST 문제 풀기

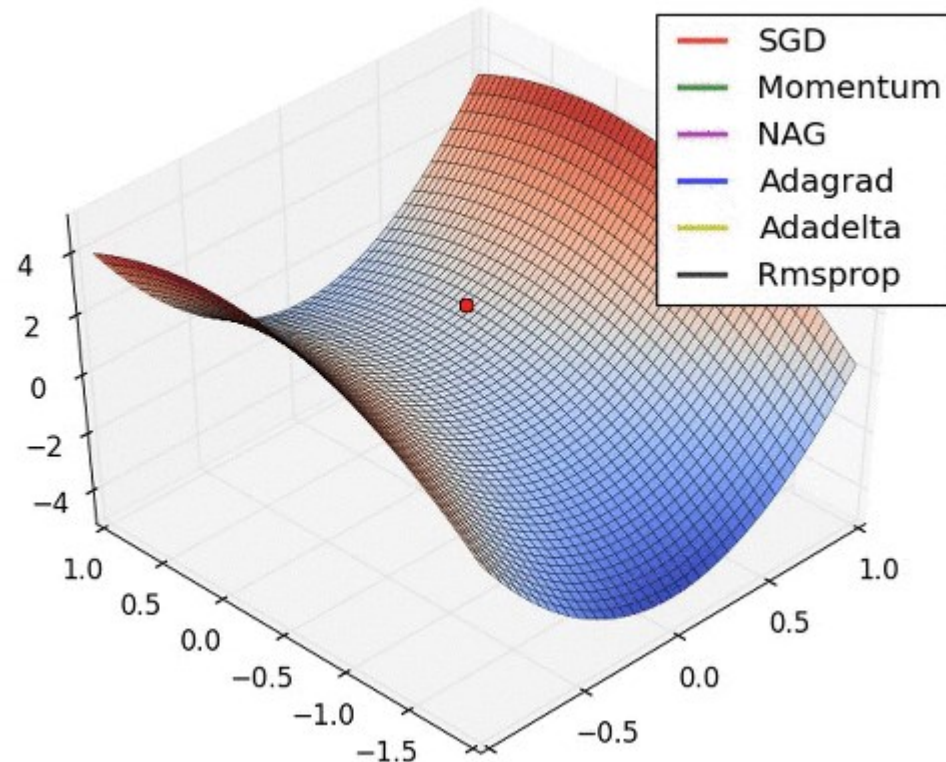
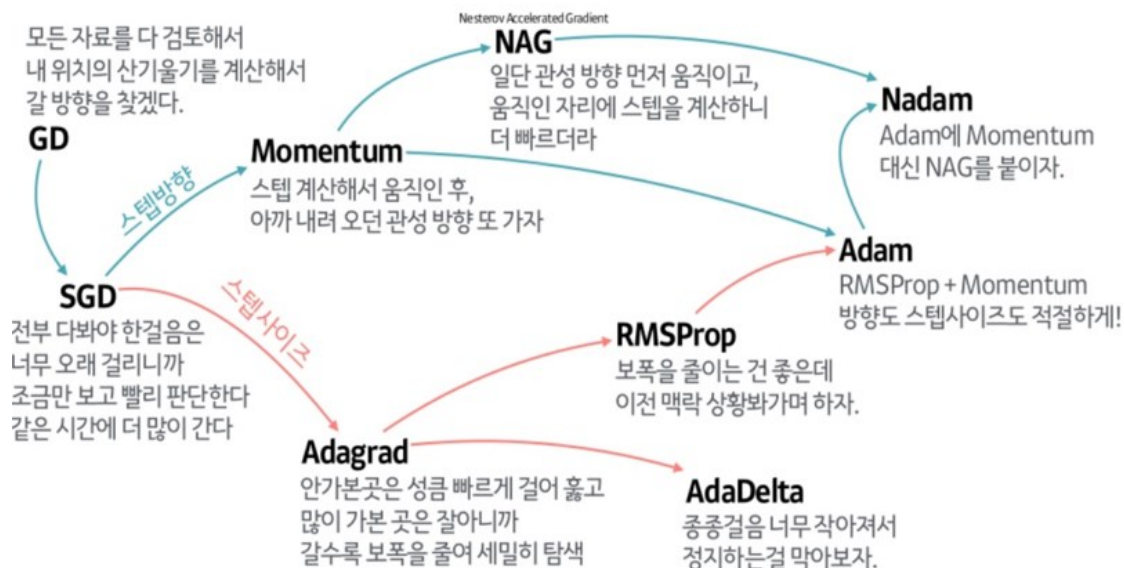
Optimizer

- GD, SGD, RMSProp, ADAM
- Learning Rate
- Momentum



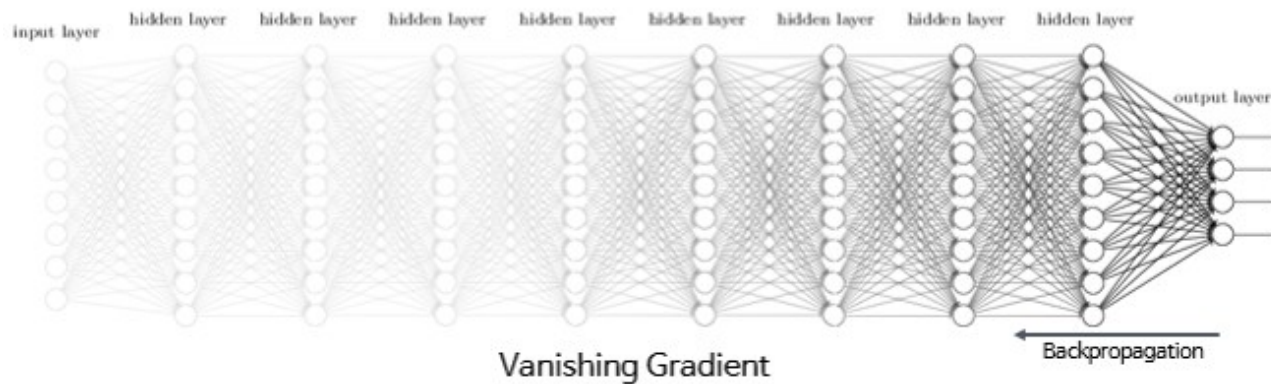
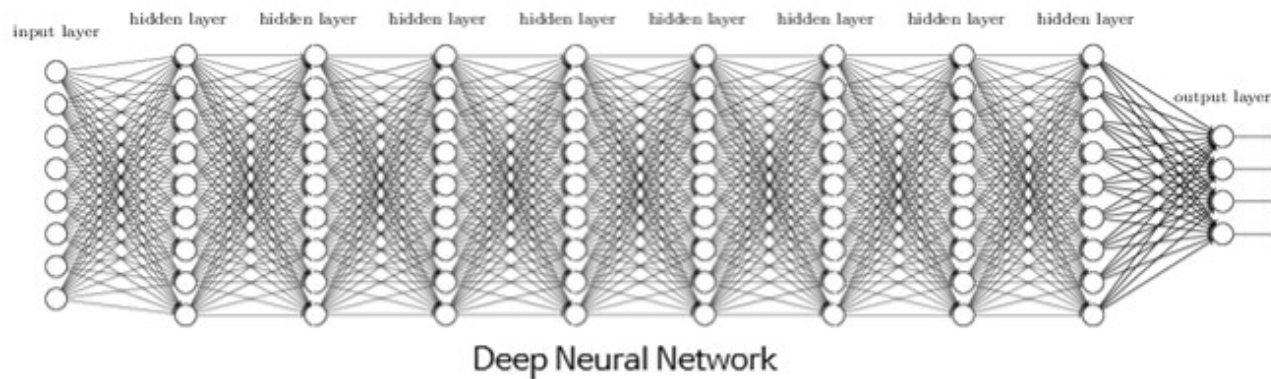
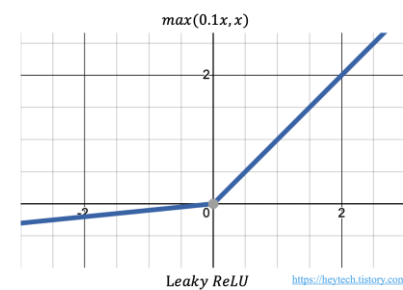
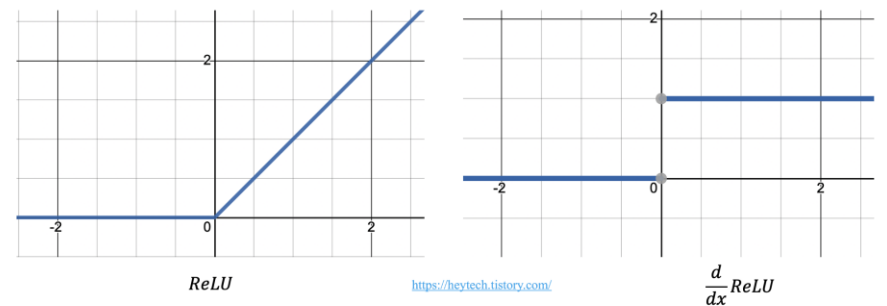
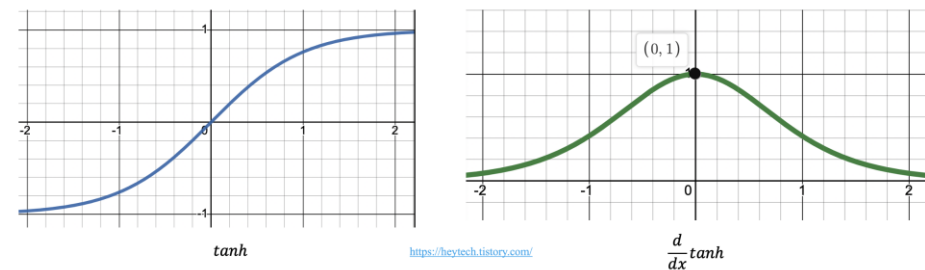
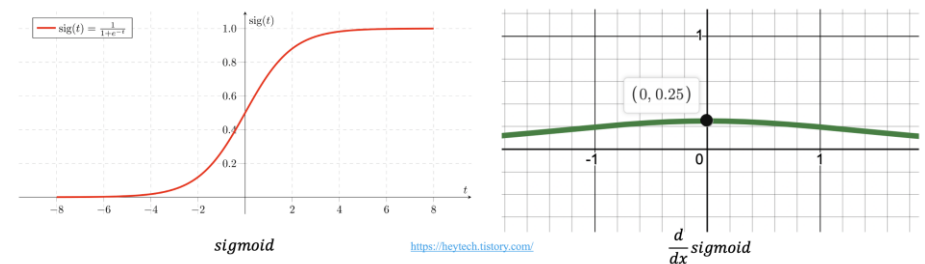
Optimizer

- GD, SGD, RMSProp, ADAM
- Learning Rate, Momentum
- Batch, Epoch



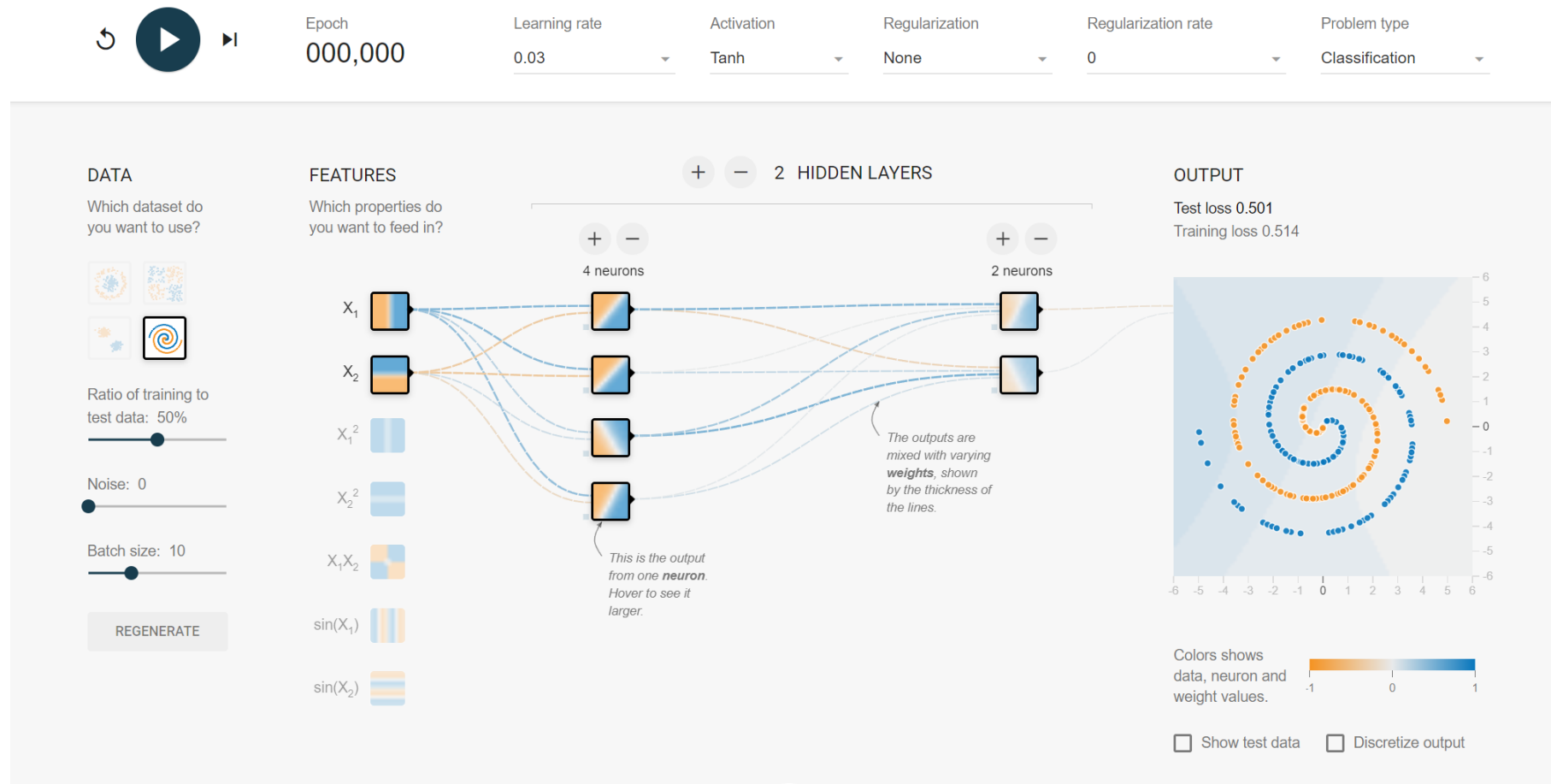
Vanishing Gradients

- Sigmoid / Tanh 때문
- ReLU, Leaky ReLU 로 어느정도 해결
- 그 외 다른 기법들 도입



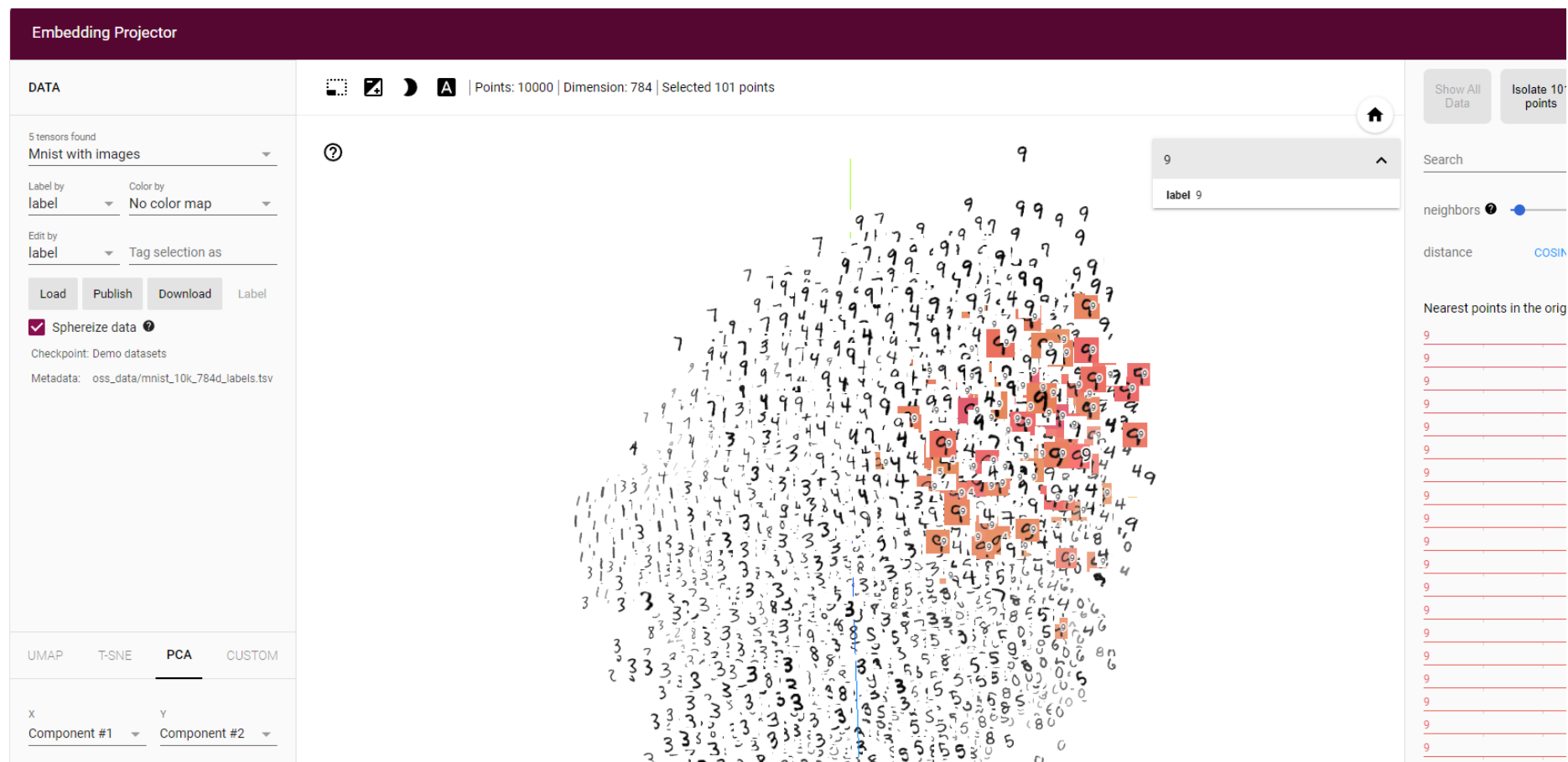
Practice – Visualization, Playground

- [A Neural Network Playground \(tensorflow.org\)](https://tfplayground.tensorflow.org/)

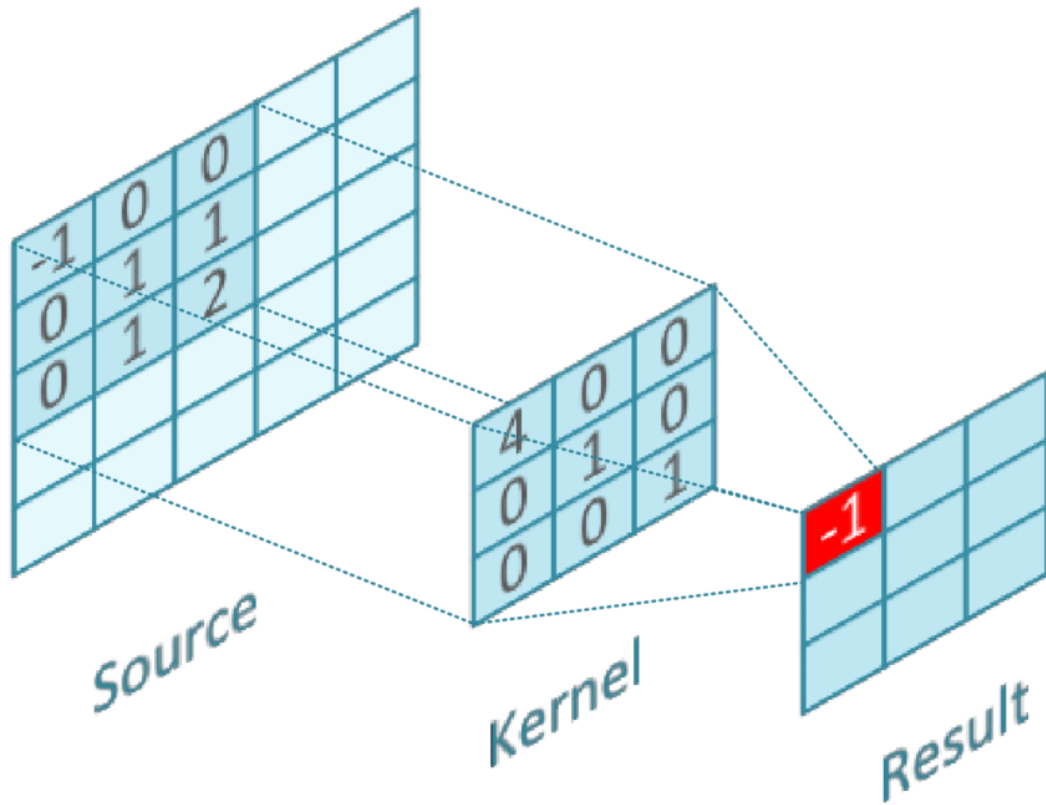


MNIST Data - Visualization

- [Embedding projector - visualization of high-dimensional data \(tensorflow.org\)](https://tfhub.dev/tensorflow/mnist-embedding-projector/1)



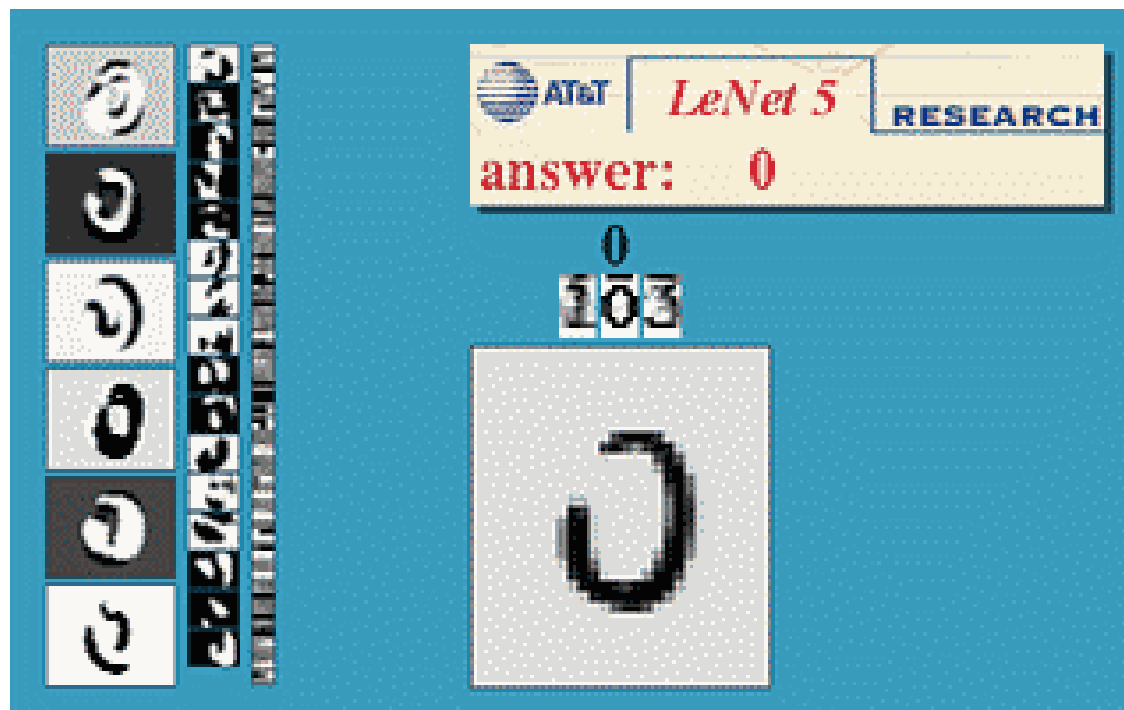
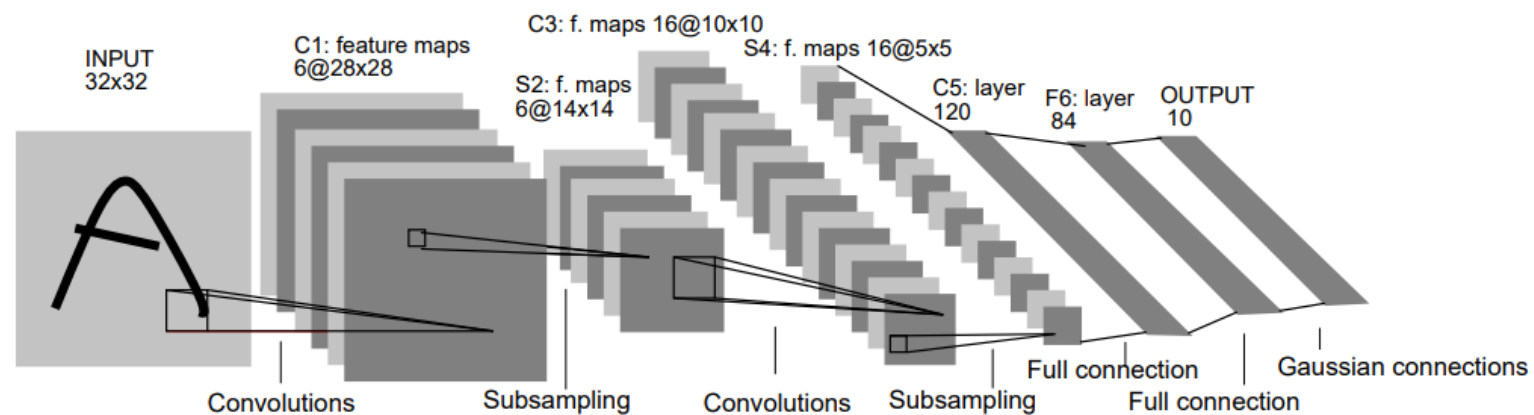
Background - Convolution



Sharpen	$\begin{bmatrix} 0 & -1 & 0 \\ -1 & 5 & -1 \\ 0 & -1 & 0 \end{bmatrix}$	
Box blur (normalized)	$\frac{1}{9} \begin{bmatrix} 1 & 1 & 1 \\ 1 & 1 & 1 \\ 1 & 1 & 1 \end{bmatrix}$	

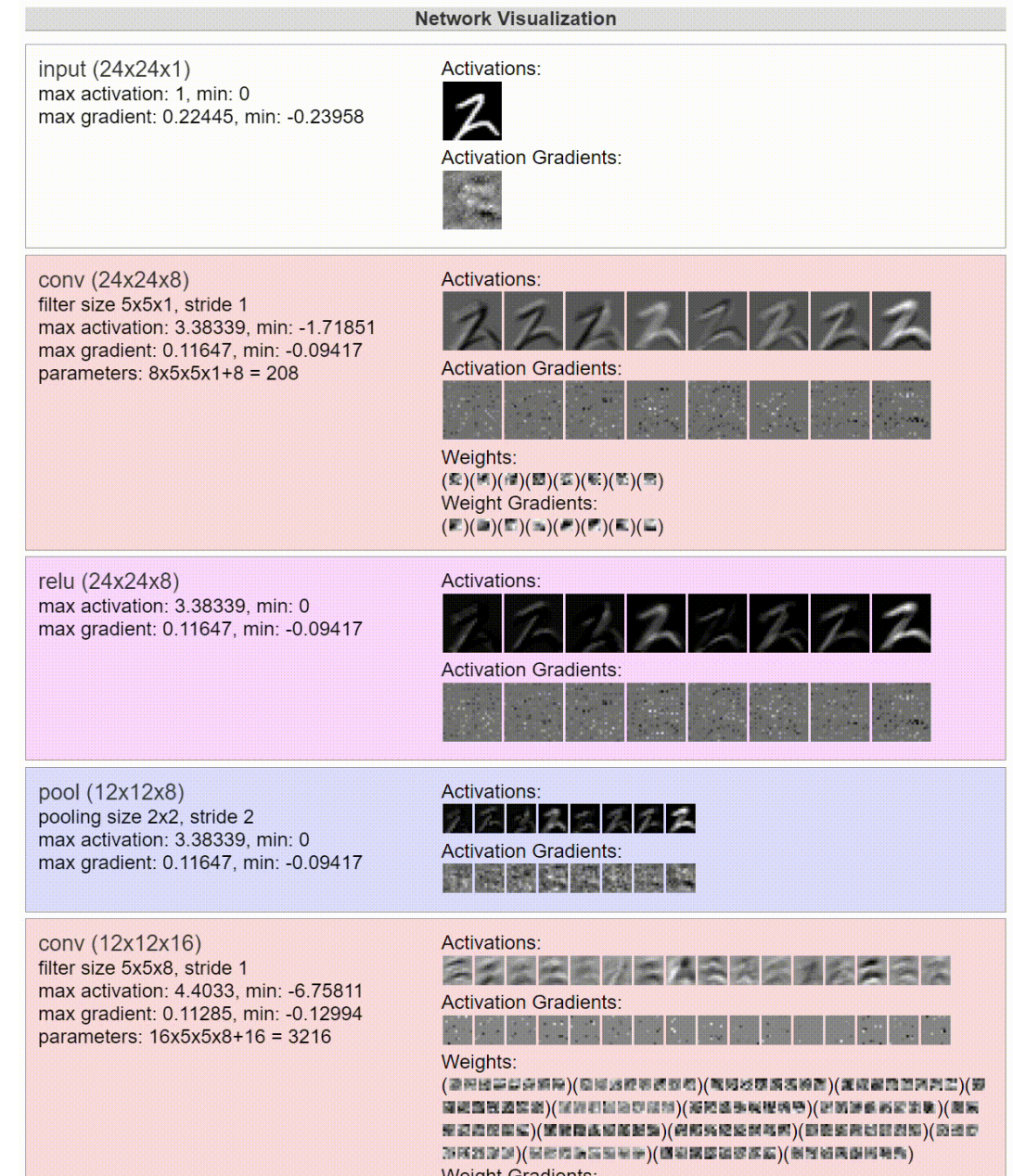
Convolutional Neural Network

- LeNet



Visualization - ConvNetJS

- <https://cs.stanford.edu/people/karpathy/convnetjs/>

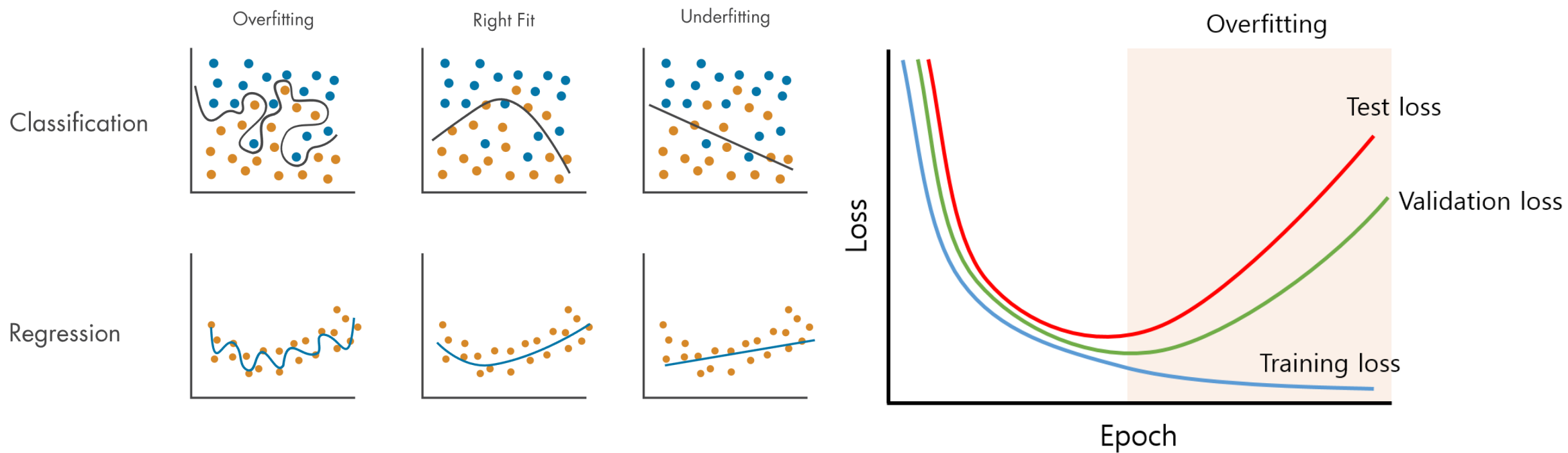


Practice 2 – Cat vs Dog

- 직접 바닥부터 학습해서 풀어보기! (from the scratch)
- 다양한 옵션으로 성능 올려보기
- (TBD) Transfer Learning 으로 성능 올려보기
- (Optional) Practice 1, CNN을 도입해서 성능 올려보기

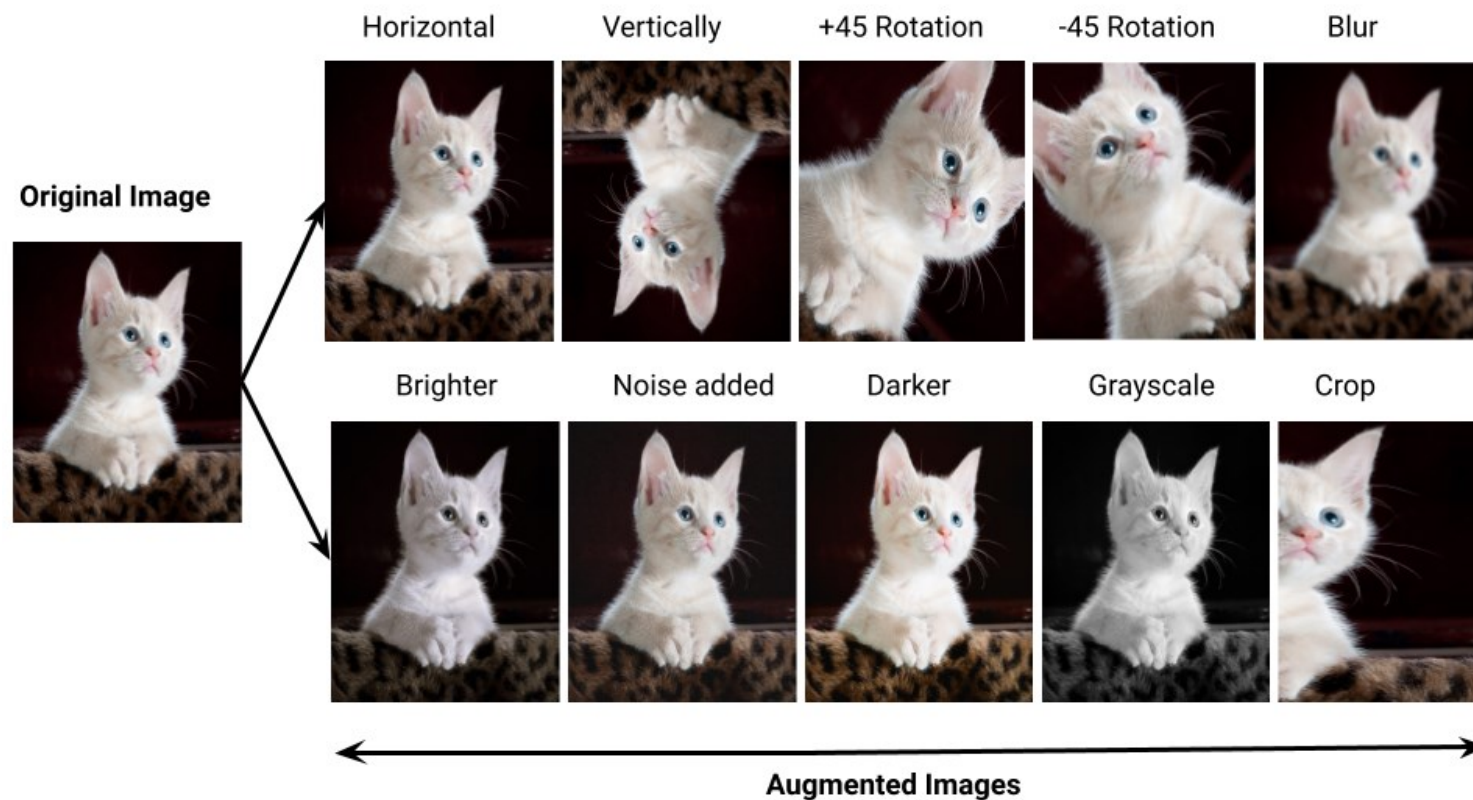
Overfitting

- 존재하는 데이터에 대해 너무 맞춤
- Training data 가 아닌 Testing data 에 대해서는 안 좋은 성능을 보임



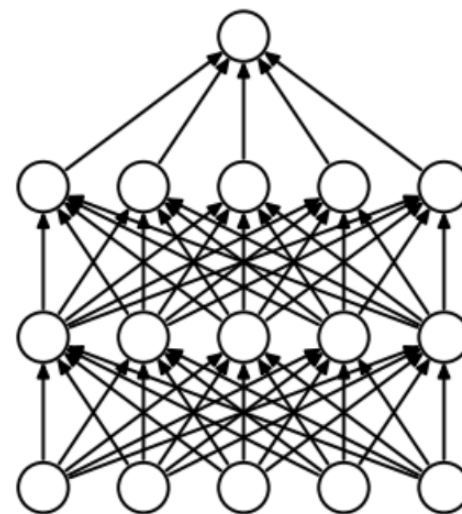
Data Augmentation

- 회전 축소, 조정 등 다양한 처리를 통해 데이터의 양을 늘림
- Overfitting 문제 해결

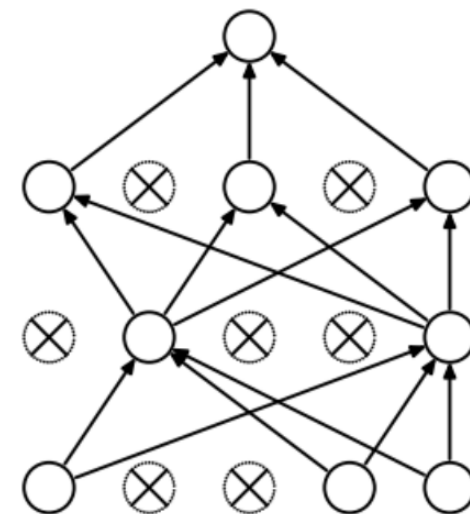


Dropout, Normalization, Regularization

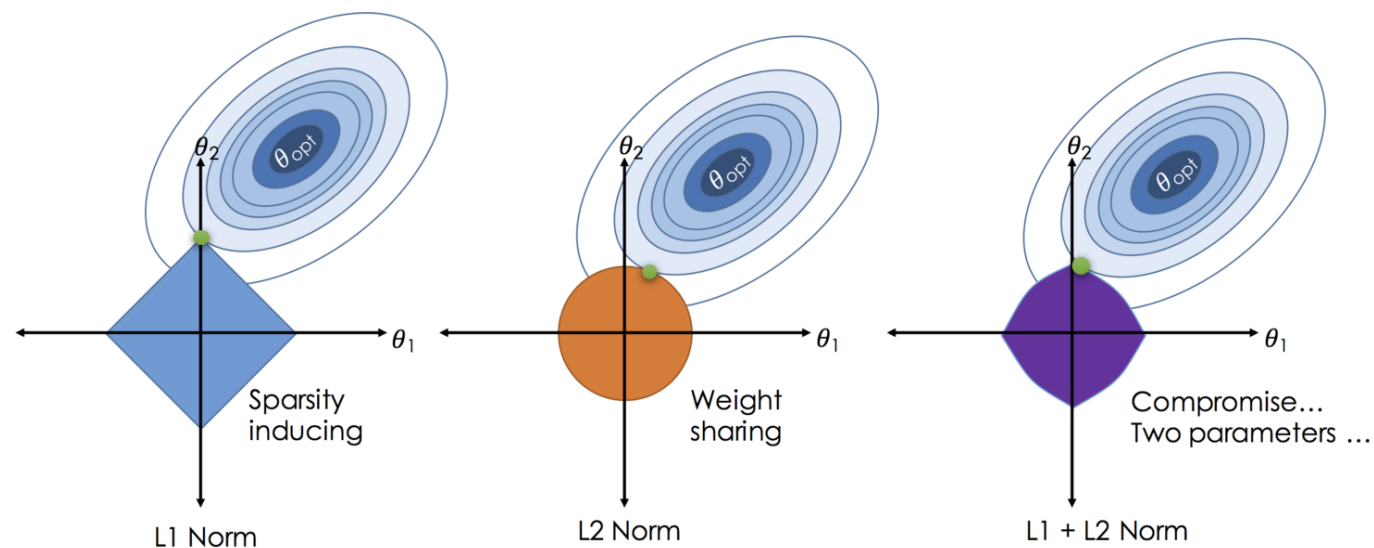
- Dropout
 - 일 못하는 뉴런도 학습 시키기
- Normalization
 - Batch, Layer, Weight, etc
- L1, L2 Regularization
 - Weight 복잡성에 제한 두기



(a) Standard Neural Net



(b) After applying dropout.

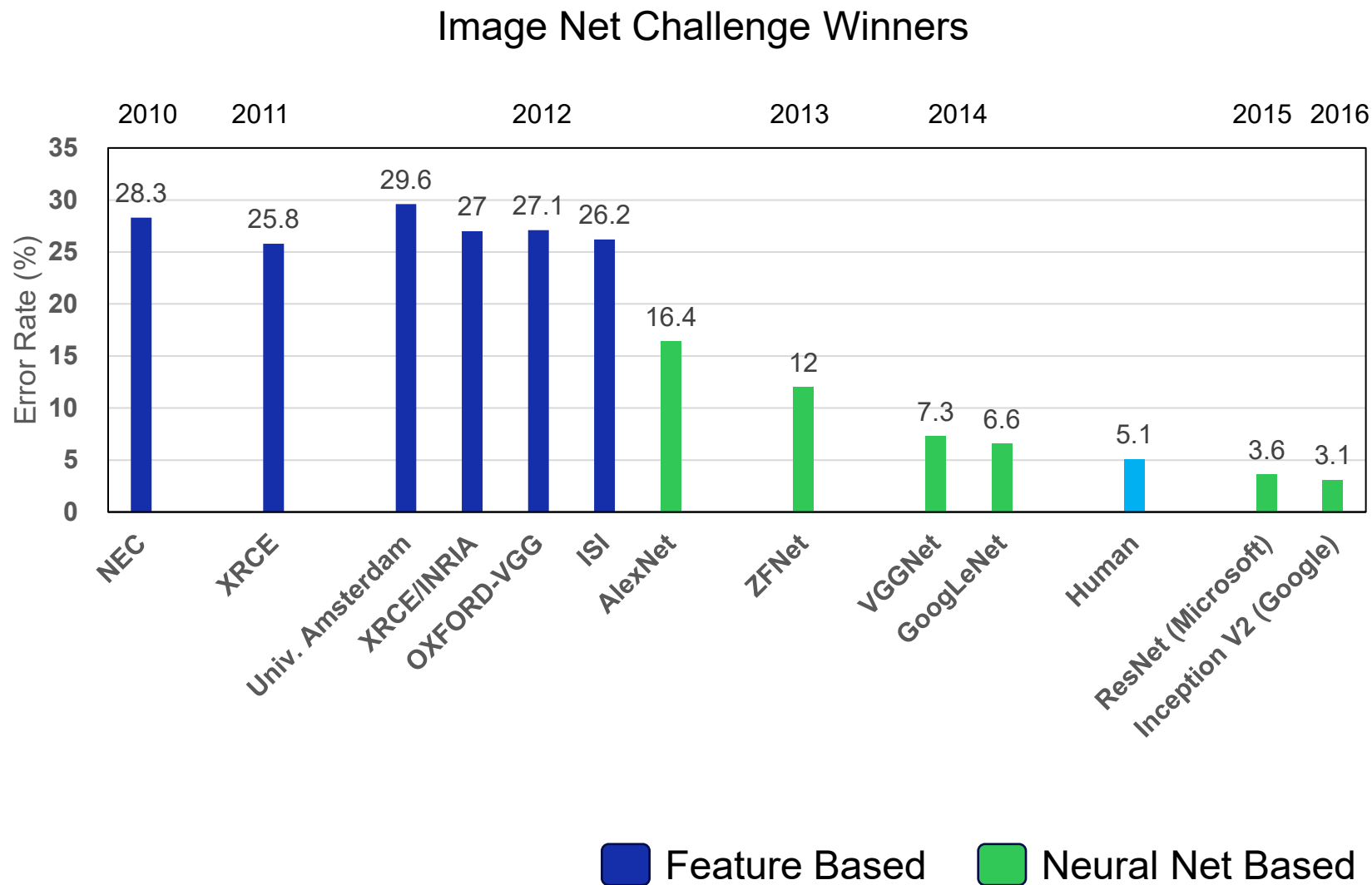


Practice 2 – Cat vs Dog (cont)

- Solving Overfitting
- Self Practice – Build Model from the scratch

The Birth of Deep Learning

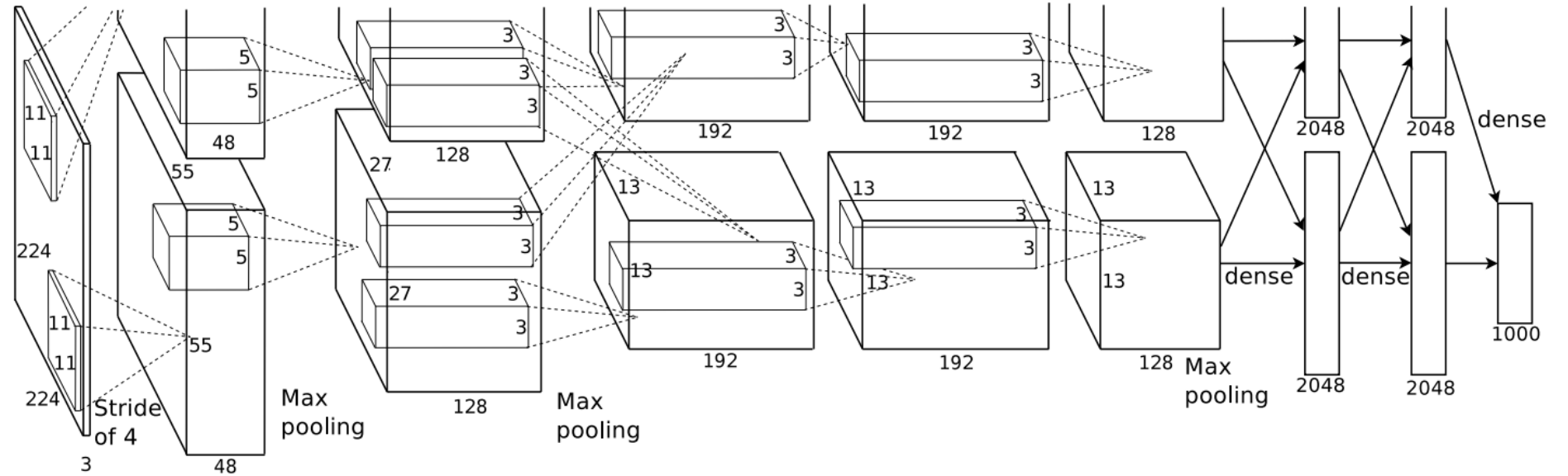
- 2012년 **AlexNet** 의 등장
- 3년 만에 사람 초월
- (참고) 2016년 알파고로 국내 관심



AlexNet

- Highlights

- 8 layers
- ReLU
- 2-GPU

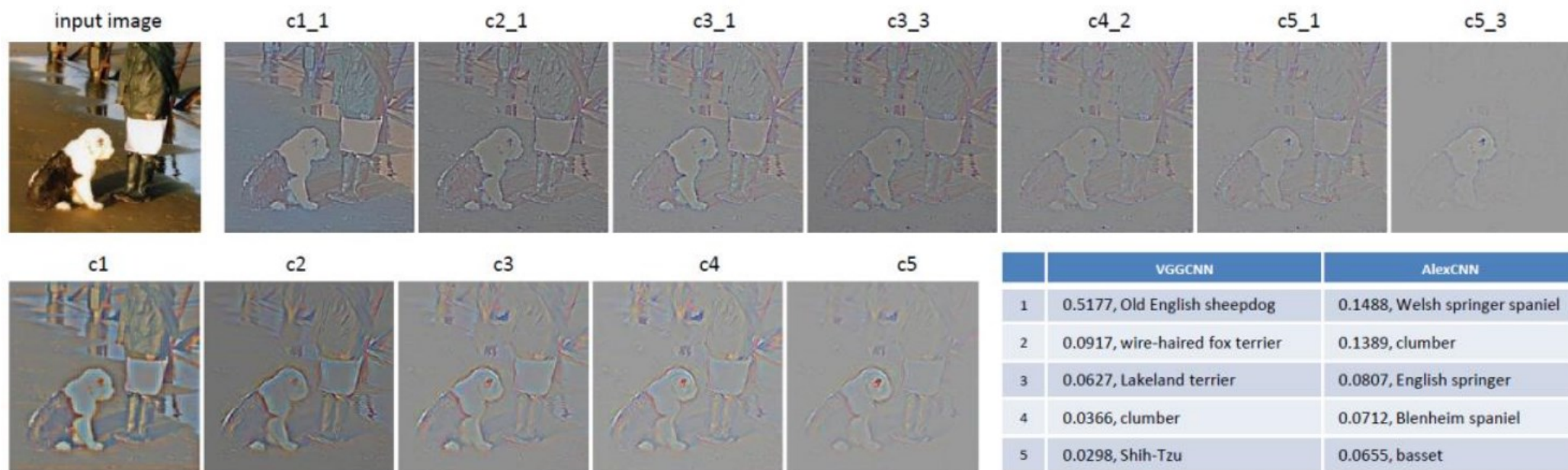
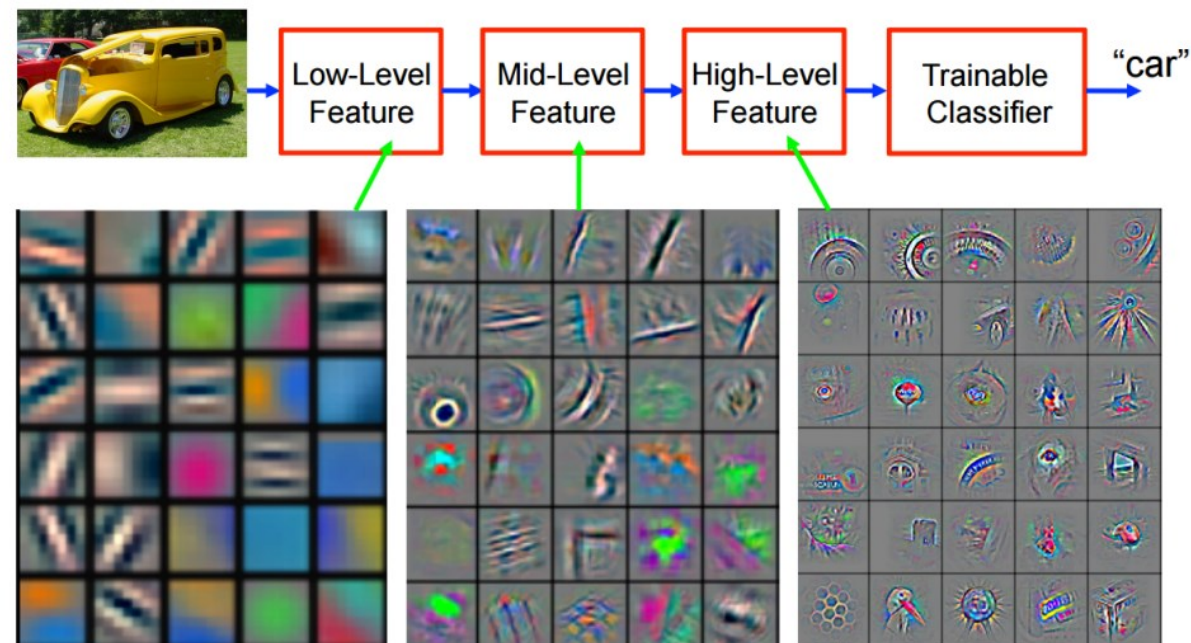


- Others

- Local Response Normalization
- Overlapping pooling
- Data Augmentation

Understanding CNN

- Low level features to High level features



	VGGCNN	AlexCNN
1	0.5177, Old English sheepdog	0.1488, Welsh springer spaniel
2	0.0917, wire-haired fox terrier	0.1389, clumber
3	0.0627, Lakeland terrier	0.0807, English springer
4	0.0366, clumber	0.0712, Blenheim spaniel
5	0.0298, Shih-Tzu	0.0655, basset

[8] Wei et al. (<https://icmlviz.github.io/icmlviz2016/assets/papers/4.pdf>)

VGGNet

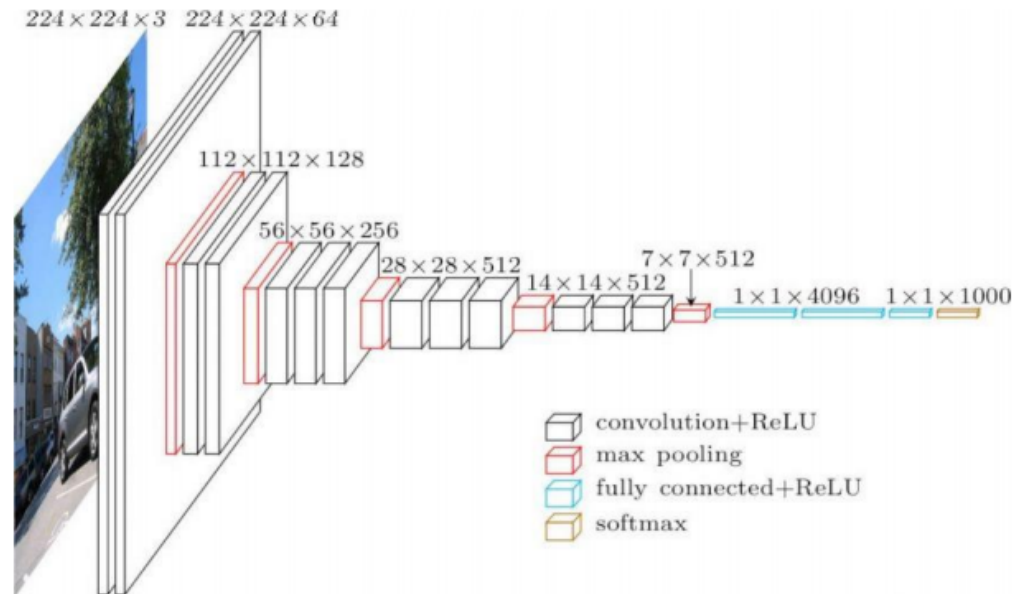
- Highlights
 - 19 layers
 - 3x3 Smaller Filters with Padding
 - Simple Architecture

VGG-16



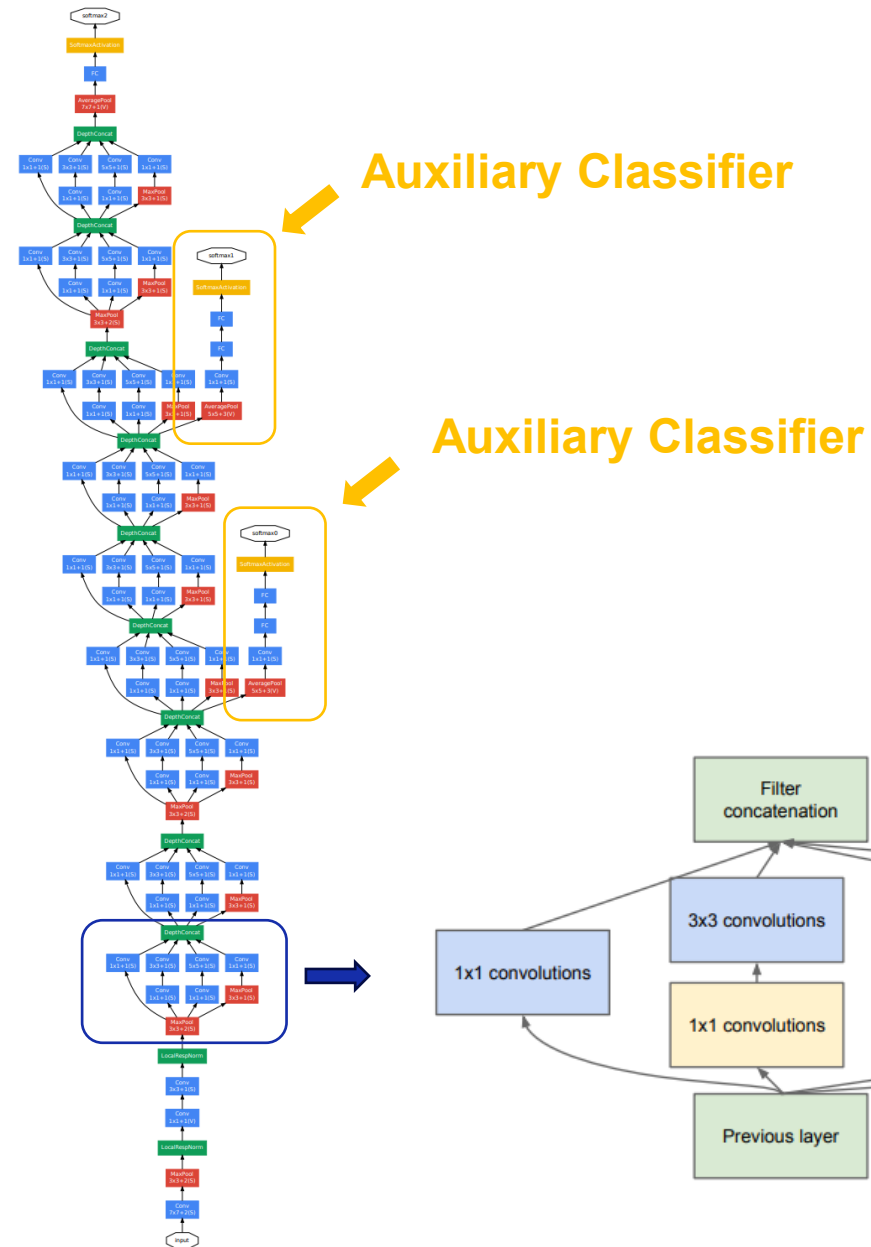
The Architecture

The architecture depicted below is VGG16.



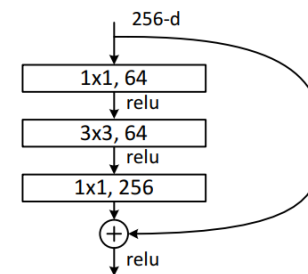
GoogLeNet – Inception V1

- Highlights
 - 22 layers
 - Inception Module
 - Auxiliary Classifier



11/11/2019

- 42



[11] Kaiming et al. (<https://arxiv.org/pdf/1512.03385v1.pdf>)

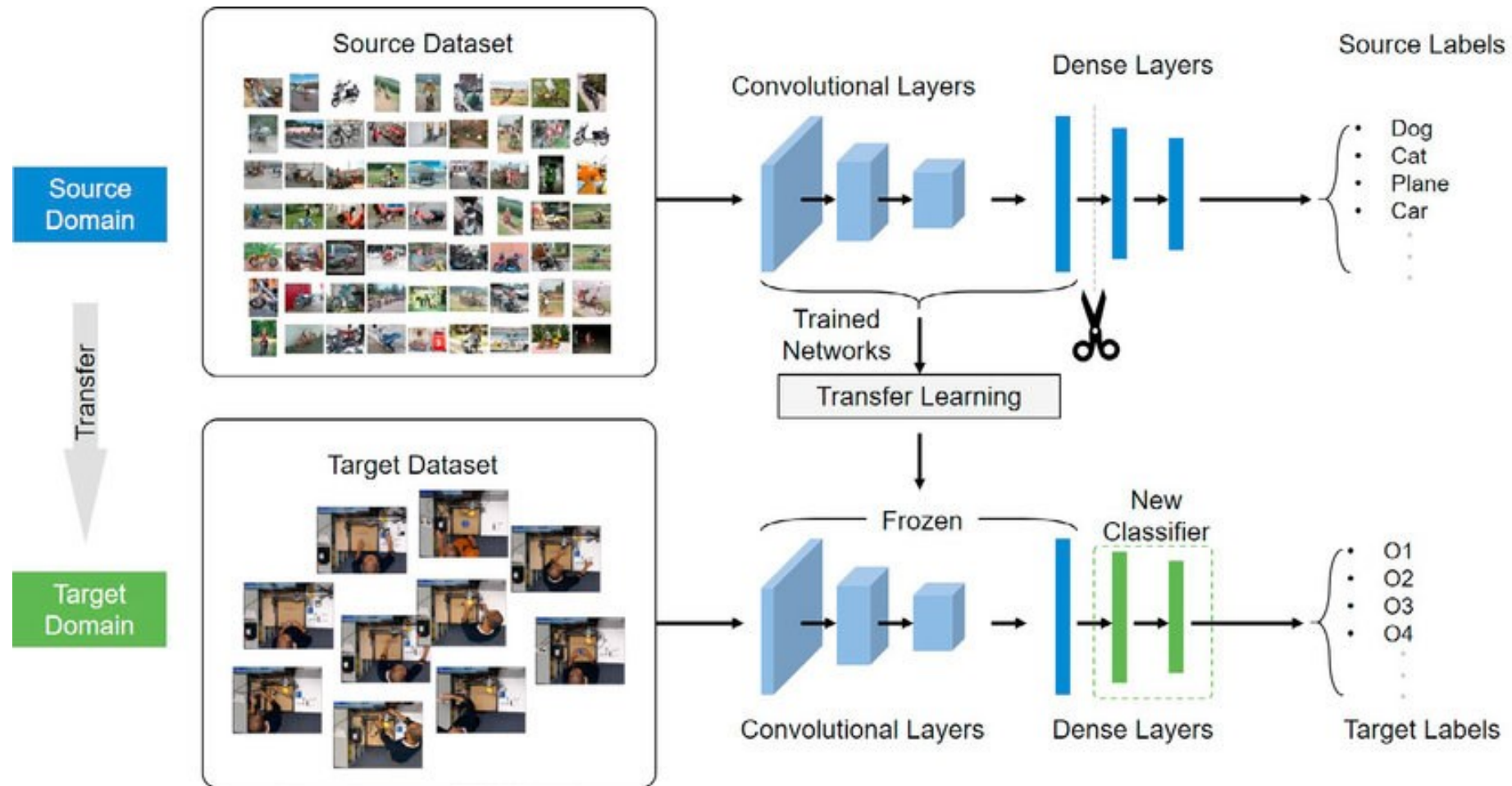
Inceptions, ResNets, EfficientNets, and so on...

- Inception (GoogLeNet) v2, v3, v4 ...
- ResNet + Inceptions, ResNeXt, ...
- SqueezeNet, MobileNet, EfficientNets

Using CNNs from HUB or Huggingface

- TensorHUB (<https://tfhub.dev/> → <https://www.kaggle.com/models>)
- Huggingface (<https://huggingface.co/>)

Transfer Learning



Practice 3 - Transfer Learning

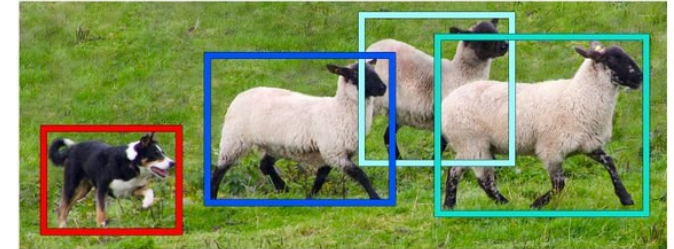
- 꽃 종류 맞추기
- Practice 2 – Cat vs Dog, Transfer Learning 으로 풀어보기
- Self Practice – Transfer learning

Computer Vision Tasks

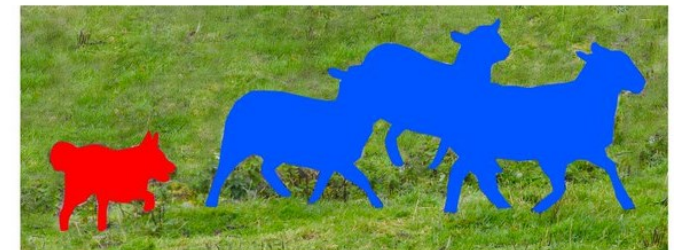
- Object Detection
 - YOLO, R-CNN
- Image Segmentation
 - U-NET
- Generative
 - VAE, GAN, Diffusion
- Super-resolution, videos, pose detection, etc.



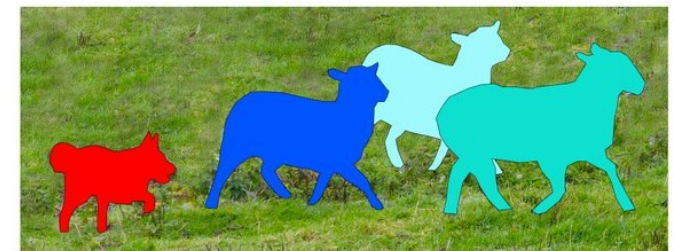
Image Recognition



Object Detection



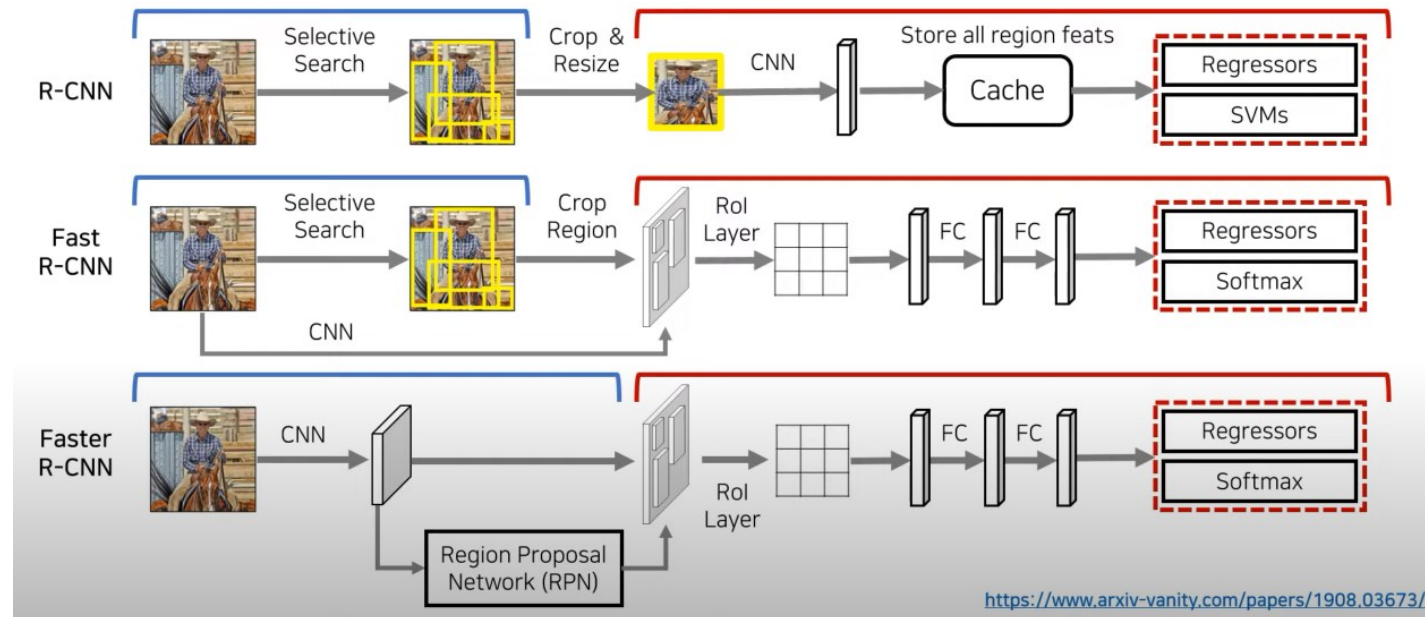
Semantic Segmentation



Instance Segmentation

Object Detection

- Two stage Object detection
 - R-CNN, Fast R-CNN, Faster R-CNN
- One stage Object detection
 - YOLO, SSD



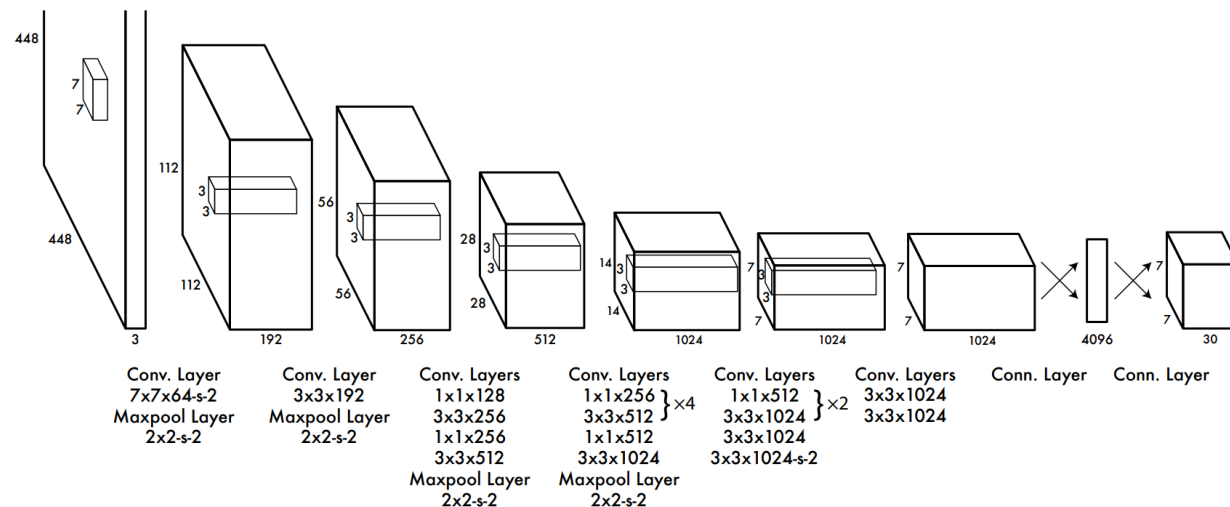
Speed

vs

Accuracy

Object Detection

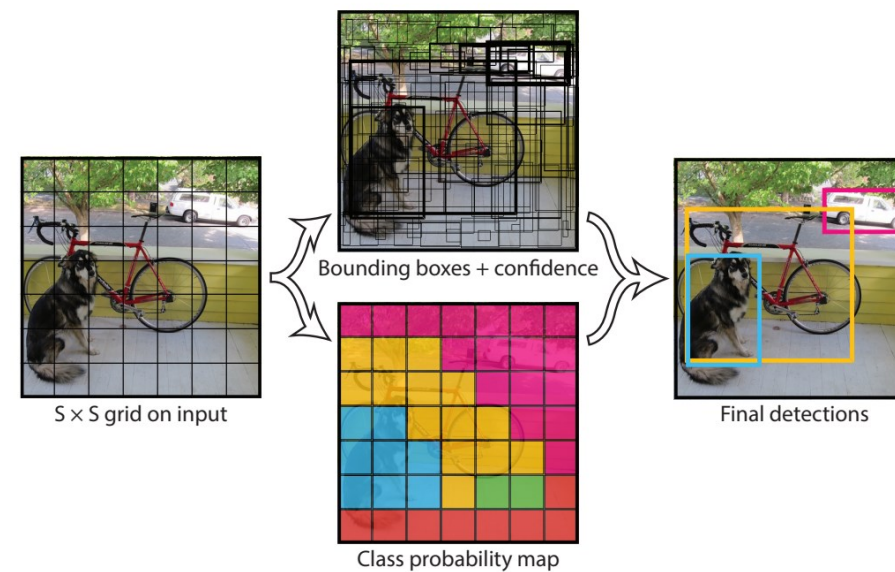
- Two stage Object detection
 - R-CNN, Fast R-CNN, Faster R-CNN
- One stage Object detection
 - YOLO, SSD



Speed

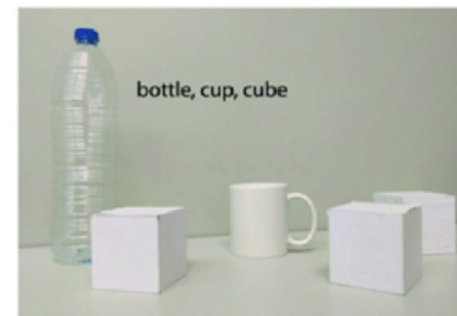
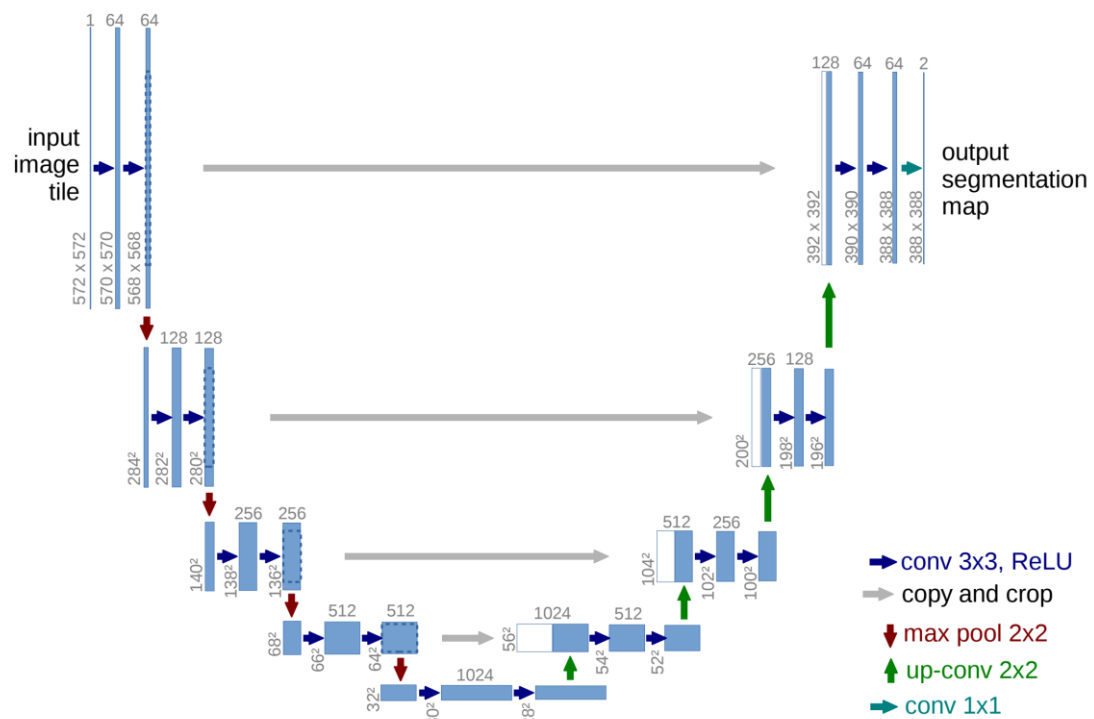
vs

Accuracy

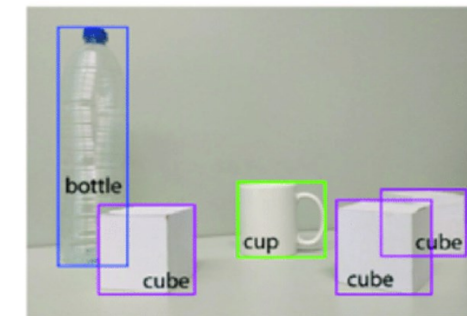


Segmentation

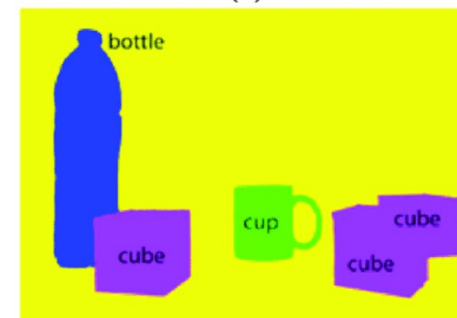
- U-Net



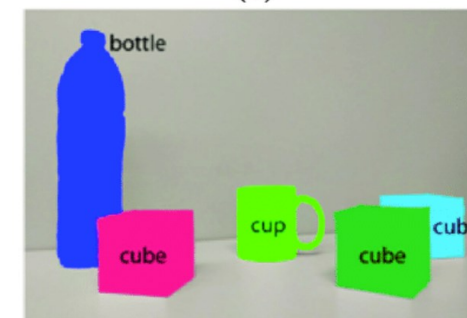
(a)



(b)

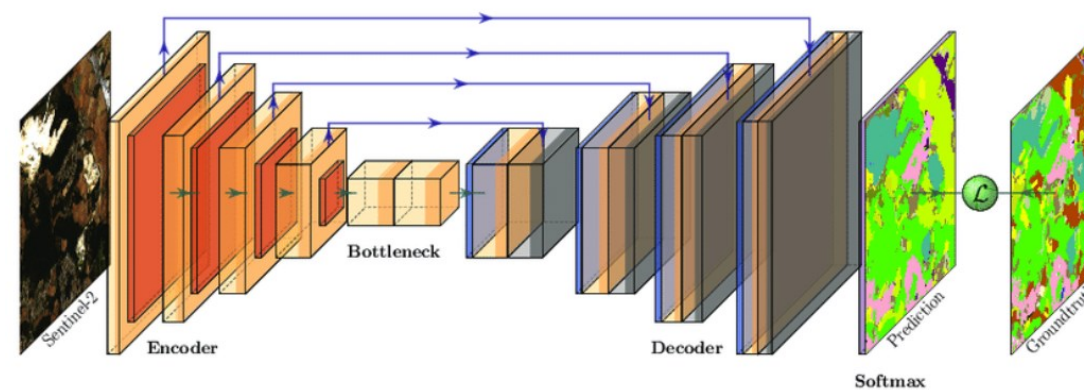


(c)



(d)

An example of different visual perception problems [75]: (a) image classification, (b) object detection, (c) semantic segmentation, and (d) instance segmentation.

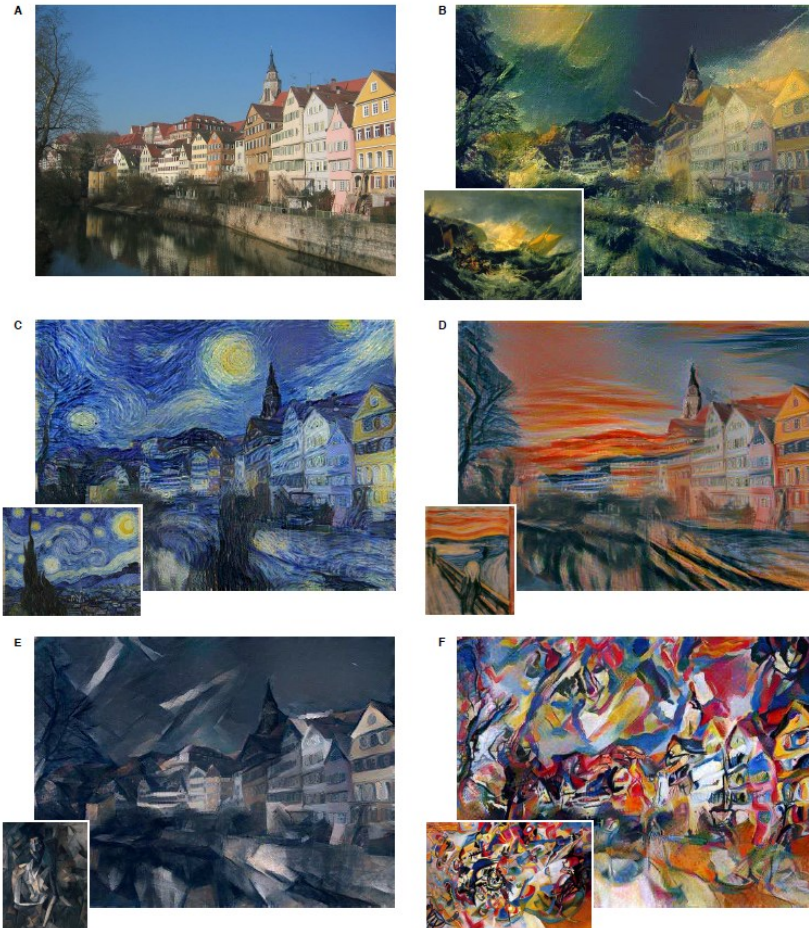


Practice - Object Detection

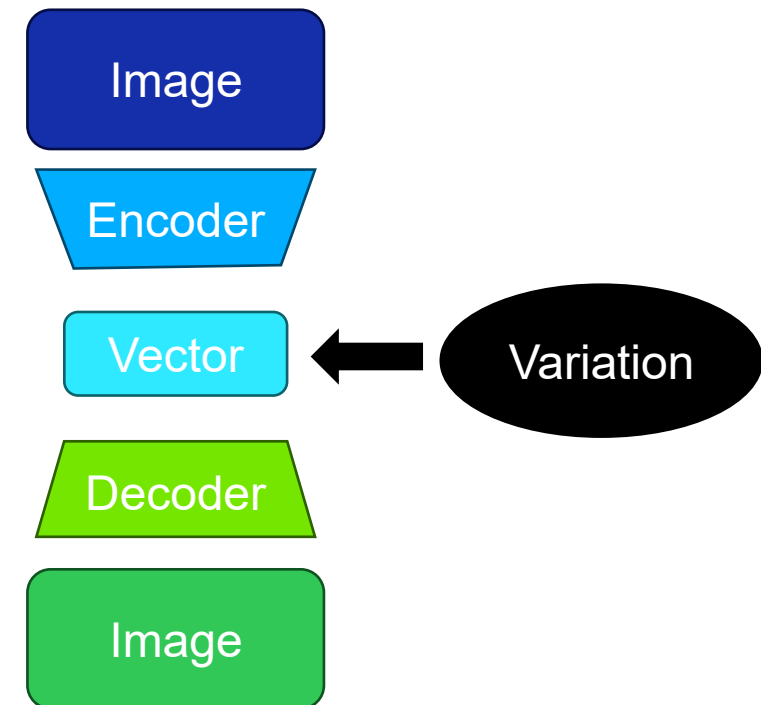
- Colab!
 - windows visualization tool setup is too hard
- R-CNN 계열 모델로 Object Detection, Image Segmentation 사용해보기

Generative – Deprecated?

- Style Transfer

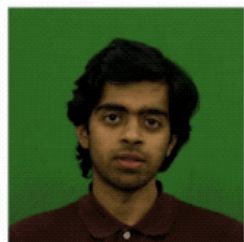
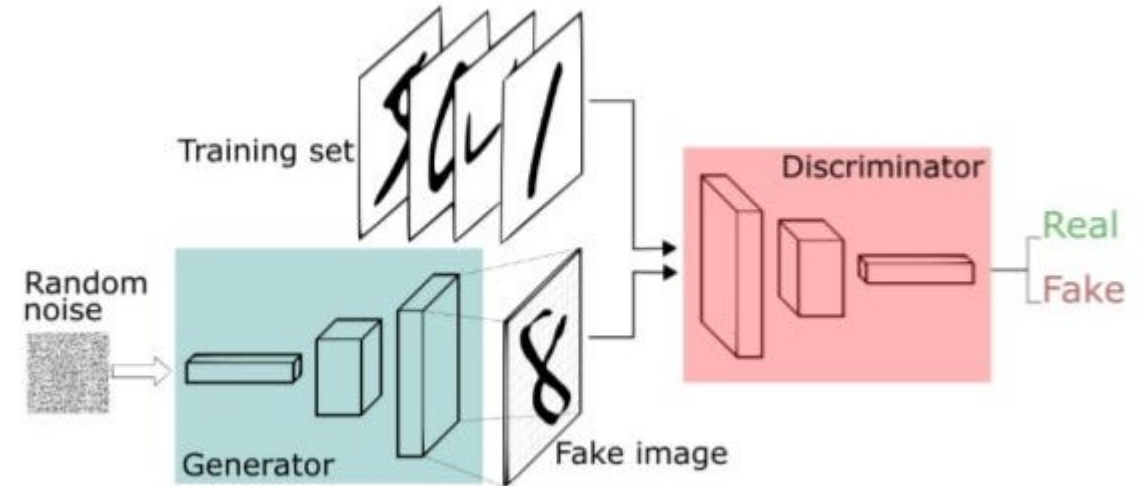


- VAE (Variational Auto-Encoder)

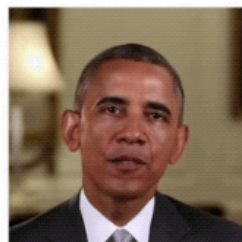


Generative – Depreciated?

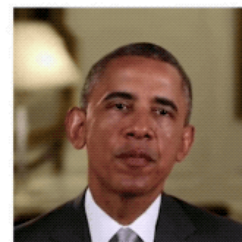
- GAN (Generative Adversarial Networks)
 - (ex) Deepfakes
 - Limitation: Model Collapse



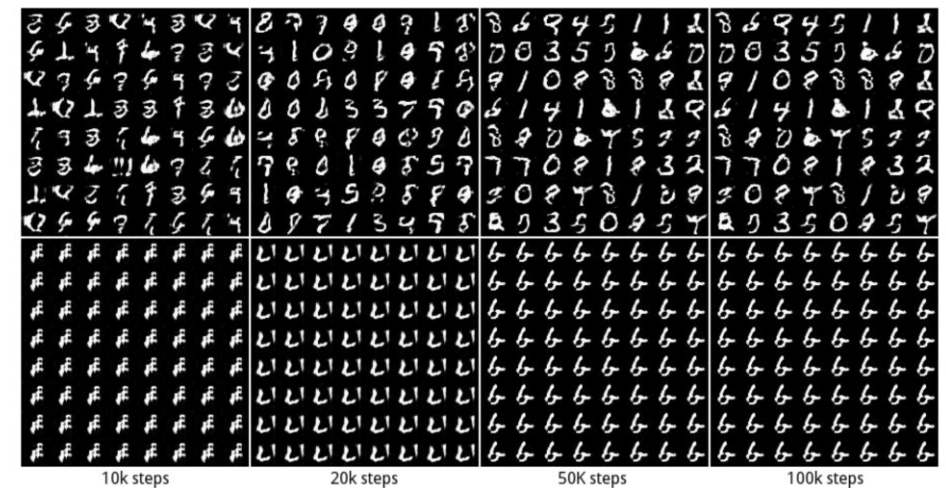
Source Sequence



Our Reenactment
(Full Head)



Averbuch-Elor et al. 2017



Diffusion

- OpenAI DALL-E - <https://labs.openai.com/>
- Stable Diffusion - <https://clipdrop.co/stable-diffusion>
- Deepmind - Imagen2 → GalaxyAI

Denoising diffusion models

● Forward / noising process

- Sample data $p(\mathbf{x}_0) \rightarrow$ turn to noise



● Reverse / denoising process

- Sample noise $p_T(\mathbf{x}_T) \rightarrow$ turn into data

Diffusion – 2024 수능

- 노이즈 생성기 & 노이즈 예측기
- 잠재 표현
- 확산 이미지

[10~13] 다음 글을 읽고 물음에 답하시오.

문장이나 영상, 음성을 만들어 내는 인공 지능 생성 모델 중 확산 모델은 영상의 복원, 생성 및 변환에 뛰어난 성능을 보인다. 확산 모델의 기본 발상은, 원본 이미지에 노이즈를 점진적으로 추가하였다가 그 노이즈를 다시 제거해 나가면 원본 이미지를 복원할 수 있다는 것이다. 노이즈는 불필요하거나 원하지 않는 값을 의미한다. 원하는 값만 들어 있는 원본 이미지에 노이즈를 단계별로 더하면 노이즈가 포함된 확산 이미지가 되고, 여러 단계를 거치면 결국 원본 이미지가 어떤 이미지였는지 전혀 알아볼 수 없는 노이즈 이미지가 된다. 역으로, 단계별로 더해진 노이즈를 알 수 있다면 노이즈 이미지에서 원본 이미지를 복원할 수 있다. 확산 모델은 노이즈 생성기, 이미지 연산기, 노이즈 예측기로 구성되며, 순확산 과정과 역확산 과정 순으로 작동한다. 순확산 과정은 이미지에 노이즈를 추가하면서 노이즈 예측기를 학습시키는 과정이다. 첫 단계에서는, 노이즈 생성기에서 노이즈를 만든 후 이미지 연산기가 이 노이즈를 원본 이미지에 더해서 노이즈가 포함된 확산 이미지를 출력한다. 다음 단계부터는 노이즈 생성기에서 만든 노이즈를 이전 단계에서 출력된 확산 이미지에 더한다. 이러한 단계를 충분히 반복하면 최종적으로 노이즈 이미지가 출력된다. 이때 더해지는 노이즈는 크거나 분포 양상 등 그 특성이 단계별로 다르다. 따라서 노이즈 예측기는 단계별로 확산 이미지를 입력받아 이미지에 포함된 노이즈의 특성을 추출하여 수치들로 표현하고, 이 수치들을 바탕으로 노이즈를 예측한다. 노이즈 예측기 내부의 이러한 수치들을 **잠재 표현**이라고 한다. 노이즈 예측기는 잠재 표현을 구하고 노이즈를 예측하는 방식을 학습한다.

노이즈 예측기의 학습 방법은 기계 학습 중에서 지도 학습에 해당한다. 지도 학습은 학습 데이터에 정답이 주어져 출력과 정답의 차이가 작아지도록 모델을 학습시키는 방법이다. 노이즈 예측기를 학습시킬 때는 노이즈 생성기에서 만들어 낸 이 준 노이즈가 정답에 해당하며 이 노이즈와 예측된 노이즈 사이의 차이가 작아지도록 학습시킨다.

역확산 과정은 노이즈 이미지에서 노이즈를 제거하여 원본 이미지를 복원하는 과정이다. 노이즈를 제거하려면 이미지에 단계별로 어떤 특성의 노이즈가 더해졌는지 알아야 하는데 노이즈 예측기가 이 역할을 한다. 노이즈 이미지 또는 중간 단계에서의 확산 이미지를 노이즈 예측기에 입력하면 이미지에 포함된 노이즈의 특성을 추출하여 잠재 표현을 구하고 이를 바탕으로 노이즈를 예측한다. 이미지 연산기는 입력된 확산 이미지로부터 이 노이즈를 빼서 현 단계의 노이즈를 제거한 확산 이미지를 출력한다. 확산 이미지에 이런 단계를 반복하면 결국 노이즈가 대부분 제거되어 원본 이미지에 가까운 이미지만 남게 된다.

한편, 많은 종류의 이미지를 학습시킨 후 학습된 이미지의 잠재 표현에 고유 번호를 붙이면 역확산 과정에서 이미지를 선택하여 생성할 수 있다. 또한 잠재 표현의 수치들을 조정하면 다른 특성의 노이즈가 생성되어 여러 이미지를 혼합하거나 실재하지 않는 이미지를 만들어 낼 수도 있다.

10. 학생이 첫글을 읽은 방법으로 적절하지 않은 것은?

- ① 확산 모델이 지도 학습을 사용한다는 점에 주목하고, 지도 학습 방법이 확산 모델에 어떻게 적용되는지 확인하며 읽었다.
- ② 확산 모델이 두 가지 과정으로 이루어진다는 점에 주목하고, 두 과정 중 어느 과정이 선행되어야 하는지 살펴며 읽었다.
- ③ 확산 모델에서 노이즈의 중요성을 파악하고, 사용되는 노이즈의 종류가 모델의 성능에 미치는 영향을 이해하며 읽었다.
- ④ 잠재 표현의 개념을 파악하고, 그 개념을 바탕으로 확산 모델이 노이즈를 예측하고 제거하는 원리를 이해하며 읽었다.
- ⑤ 확산 모델의 구성 요소를 파악하고, 그 구성 요소가 노이즈 처리 과정에서 어떤 기능을 하는지 확인하며 읽었다.

11. 첫글을 이해한 내용으로 가장 적절한 것은?

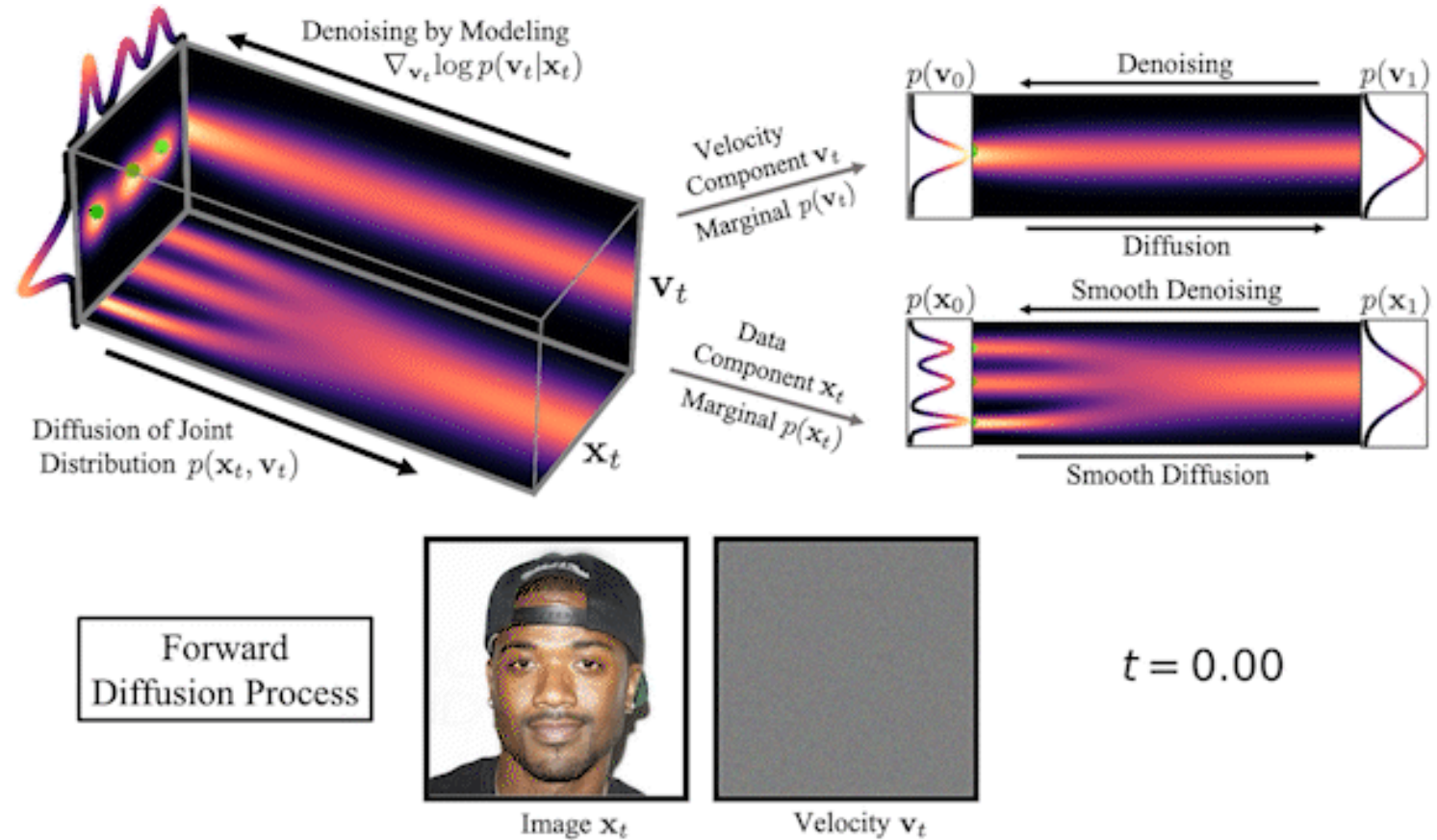
- ① 노이즈 생성기는 순확산 과정에서만 작동한다.
- ② 확산 모델에서의 학습은 역확산 과정에서 이루어진다.
- ③ 이미지 연산기와 노이즈 예측기는 모두 확산 이미지를 출력한다.
- ④ 노이즈 예측기를 학습시킬 때는 예측된 노이즈가 정답으로 사용된다.
- ⑤ 역확산 과정에서 단계를 반복될수록 출력되는 확산 이미지는 원본 이미지와의 유사성이 줄어든다.

12. **잠재 표현**에 대한 설명으로 적절하지 않은 것은?

- ① 잠재 표현의 수치들을 조정하면 여러 이미지를 혼합할 수 있다.
- ② 역확산 과정에서 잠재 표현이 다르면 예측되는 노이즈가 다르다.
- ③ 확산 모델의 학습에는 잠재 표현을 구하는 방식이 포함되어 있다.
- ④ 잠재 표현은 이미지에 더해진 노이즈의 크거나 분포 양상에 따라 다른 값들이 얻어진다.
- ⑤ 잠재 표현은 노이즈 예측기가 원본 이미지를 입력받아 노이즈의 특성을 추출한 결과이다.

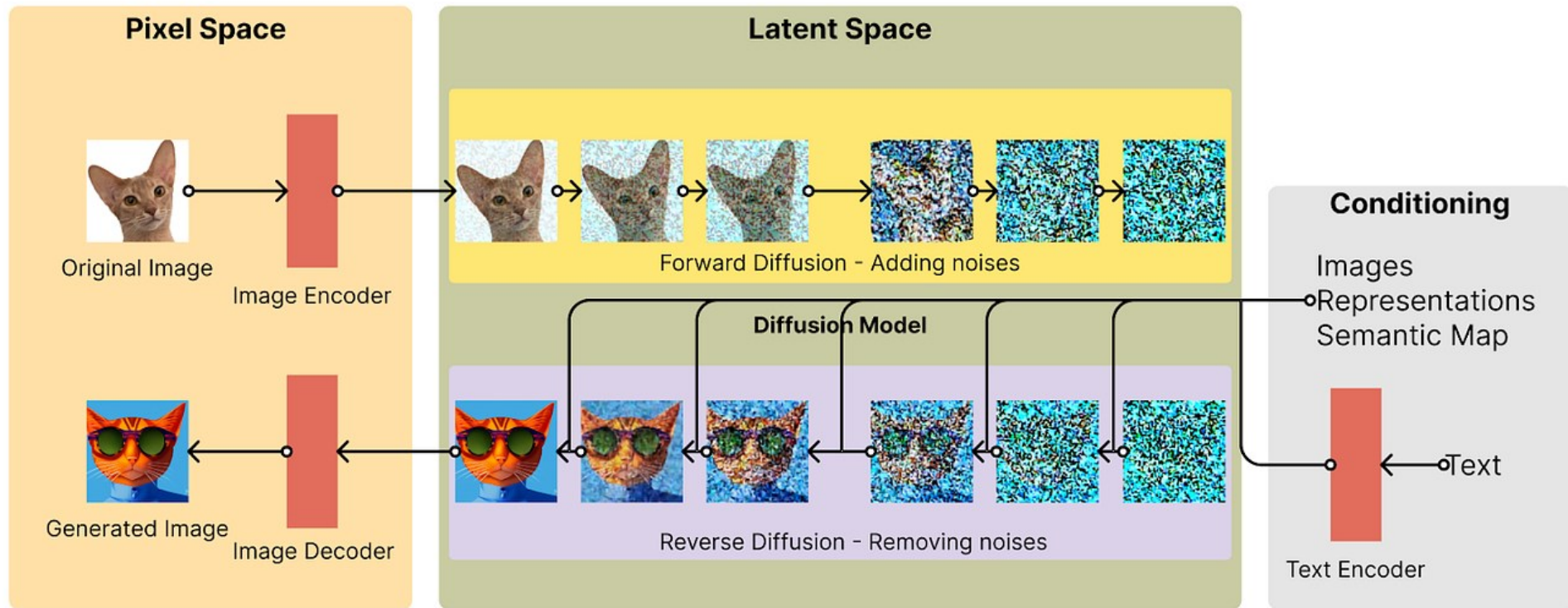
Diffusion - Visualization

- Denoising
- Train Reverse Denoising
- U-Net like Architecture



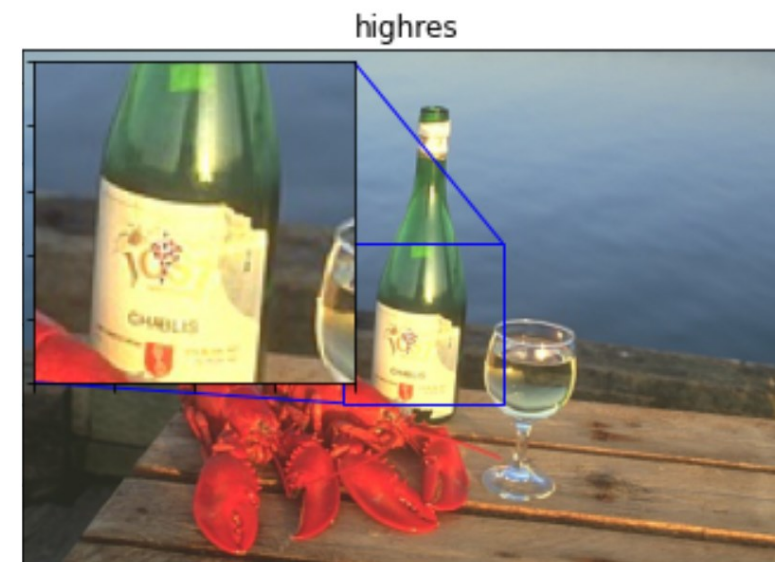
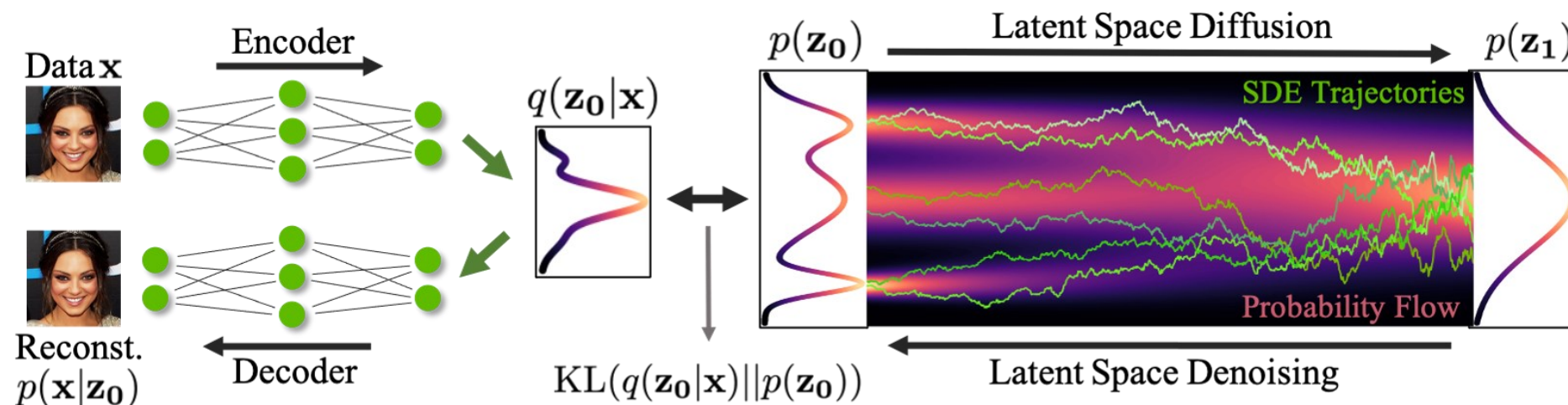
Stable Diffusion by stability.ai

- open-source - <https://github.com/CompVis/stable-diffusion>
- Light Model that could be used in PC
- In October 2022, raised US\$101 million



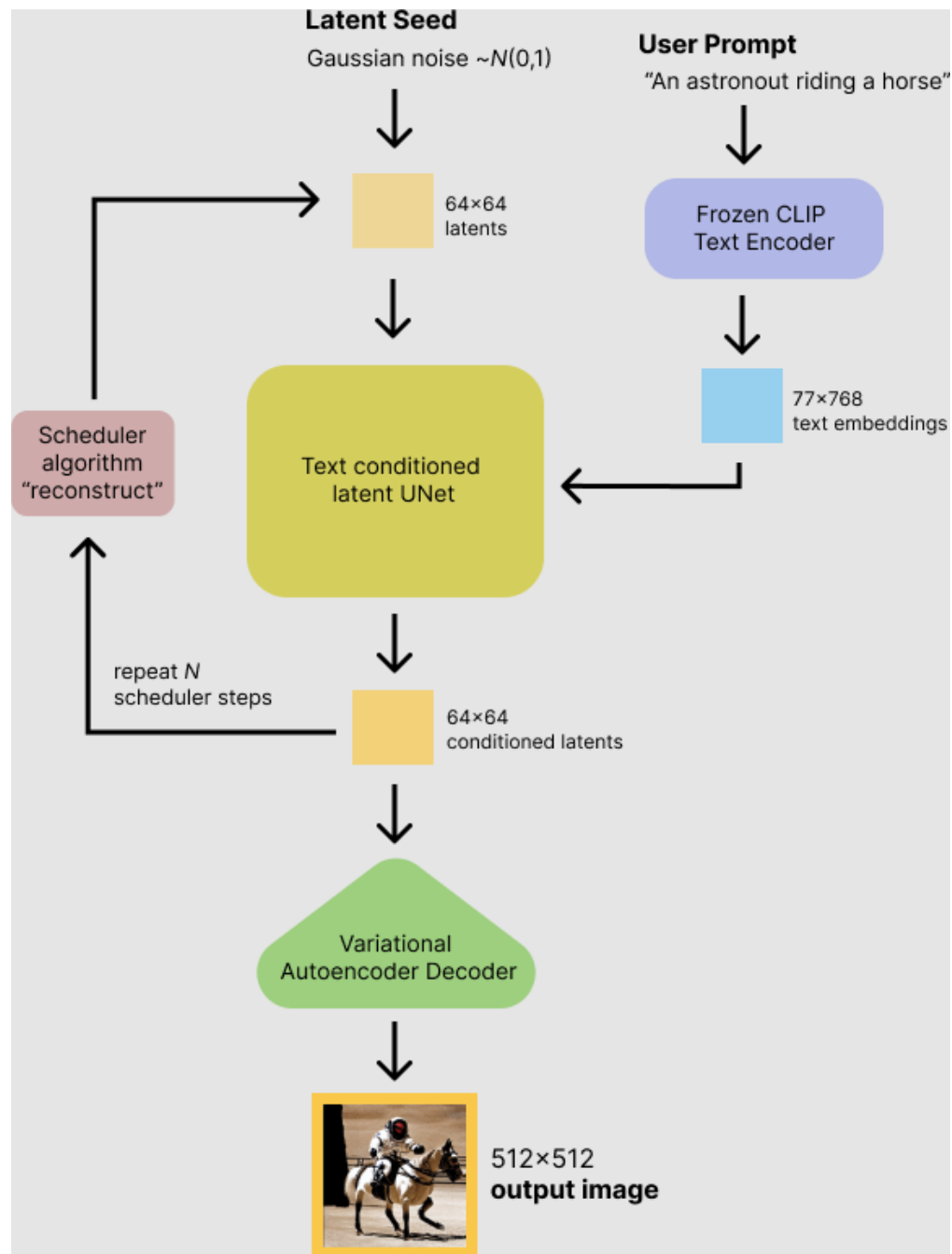
Stable Diffusion

- Latent Diffusion
- + Super Resolution



Stable Diffusion

- Latent Diffusion
- + Super Resolution



Practice – Stable Diffusion

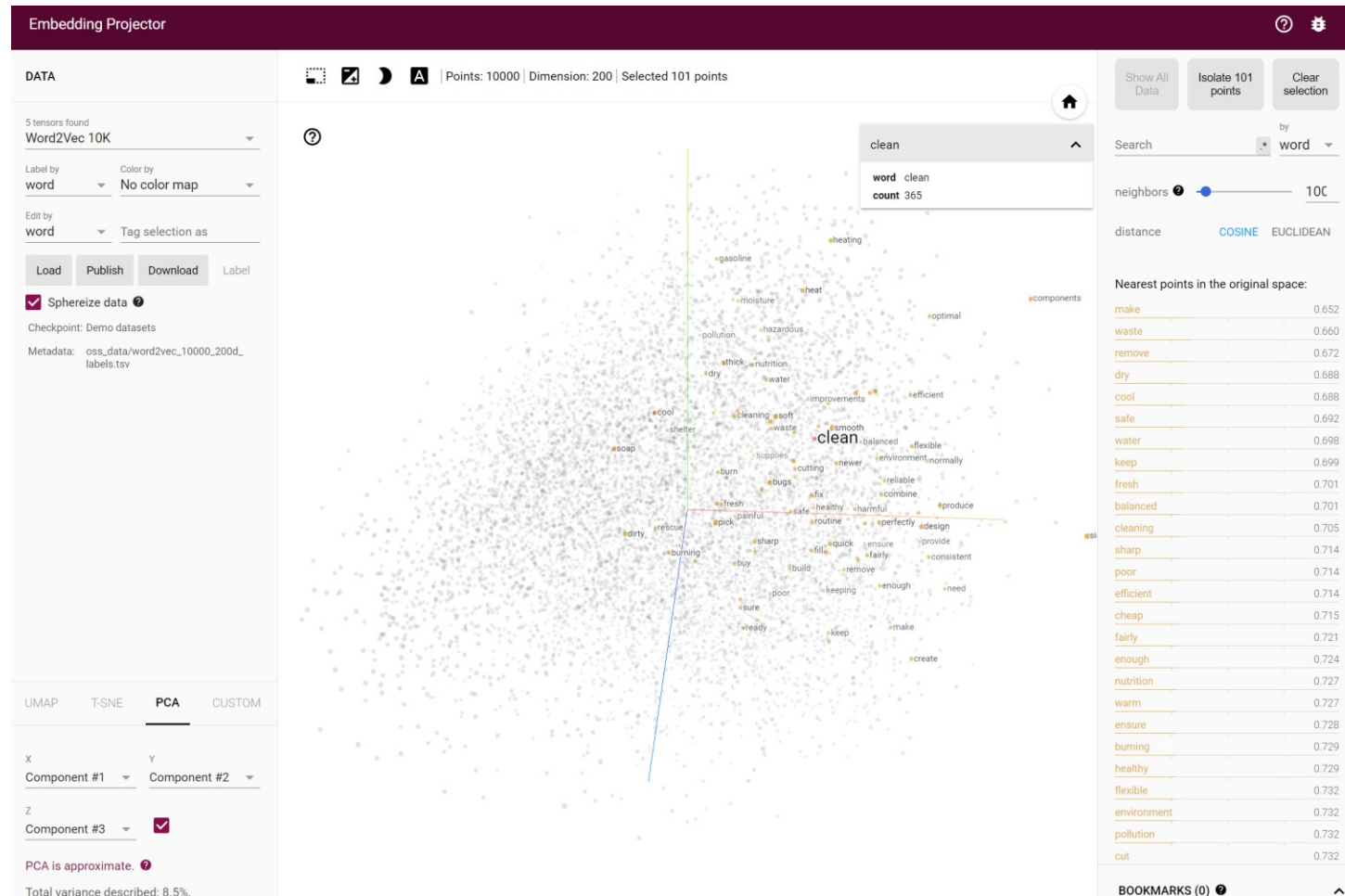
- Prompt to Image by my own
 - Ipynb – COLAB
- Other ways
 - [Automatic1111 \(aka A1111\)](#)
 - [Civitai: The Home of Open-Source Generative AI](#) - LoRA

Vision Transformer

- TBD after transformer
- 이미지도 언어처럼!
 - 언어나 이미지나 임베딩 (인코딩) 시키면 같은 데이터 벡터일 뿐이다!

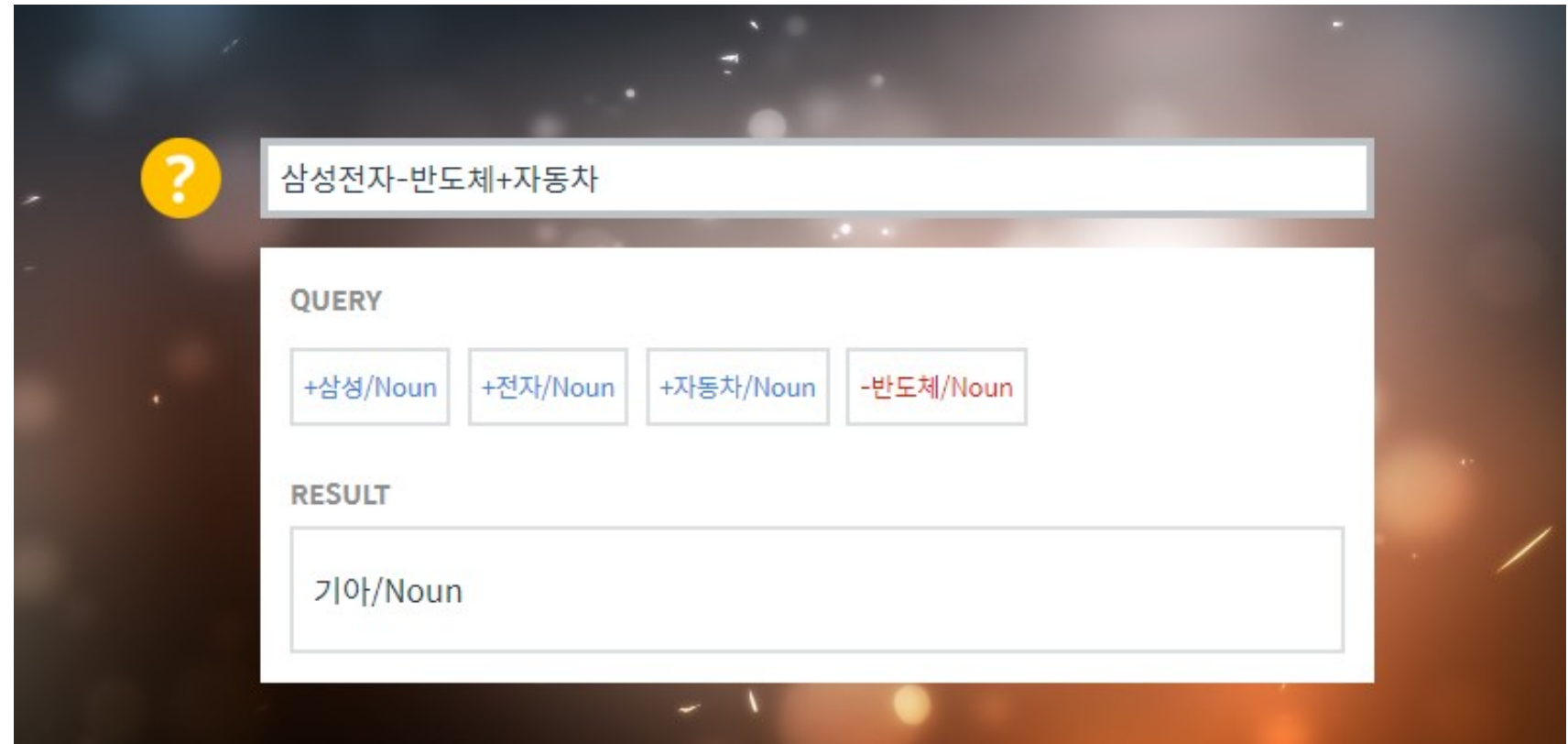
Word2Vec - Visualization

- [Embedding projector - visualization of high-dimensional data \(tensorflow.org\)](https://tfhub.dev/tensorflow/word2vec-visualizer/1)



Word2Vec

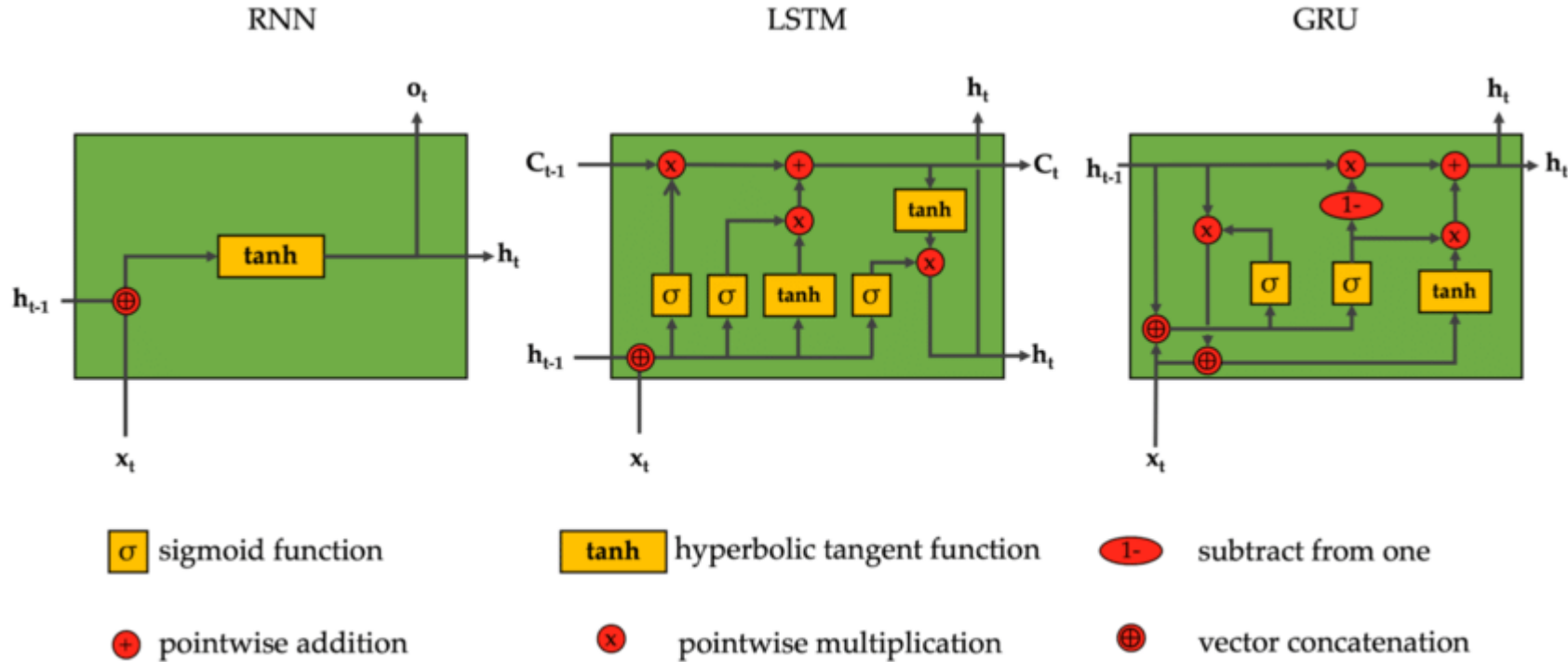
- Make Word Vector based on corpus
- Cosine similarity, Recommendation, Search engine
- <https://word2vec.kr/search/>



The screenshot shows a web interface for Word2Vec search. At the top, there is a search bar with a yellow question mark icon on the left and the text "삼성전자-반도체+자동차" (Samsung Electronics-Semiconductor+Automobile) inside. Below the search bar, the word "QUERY" is displayed. Underneath, there are four buttons: "+삼성/Noun" (blue text), "+전자/Noun" (blue text), "+자동차/Noun" (blue text), and "-반도체/Noun" (red text). Below these buttons, the word "RESULT" is displayed. Underneath, there is a single button with the text "기아/Noun" (blue text).

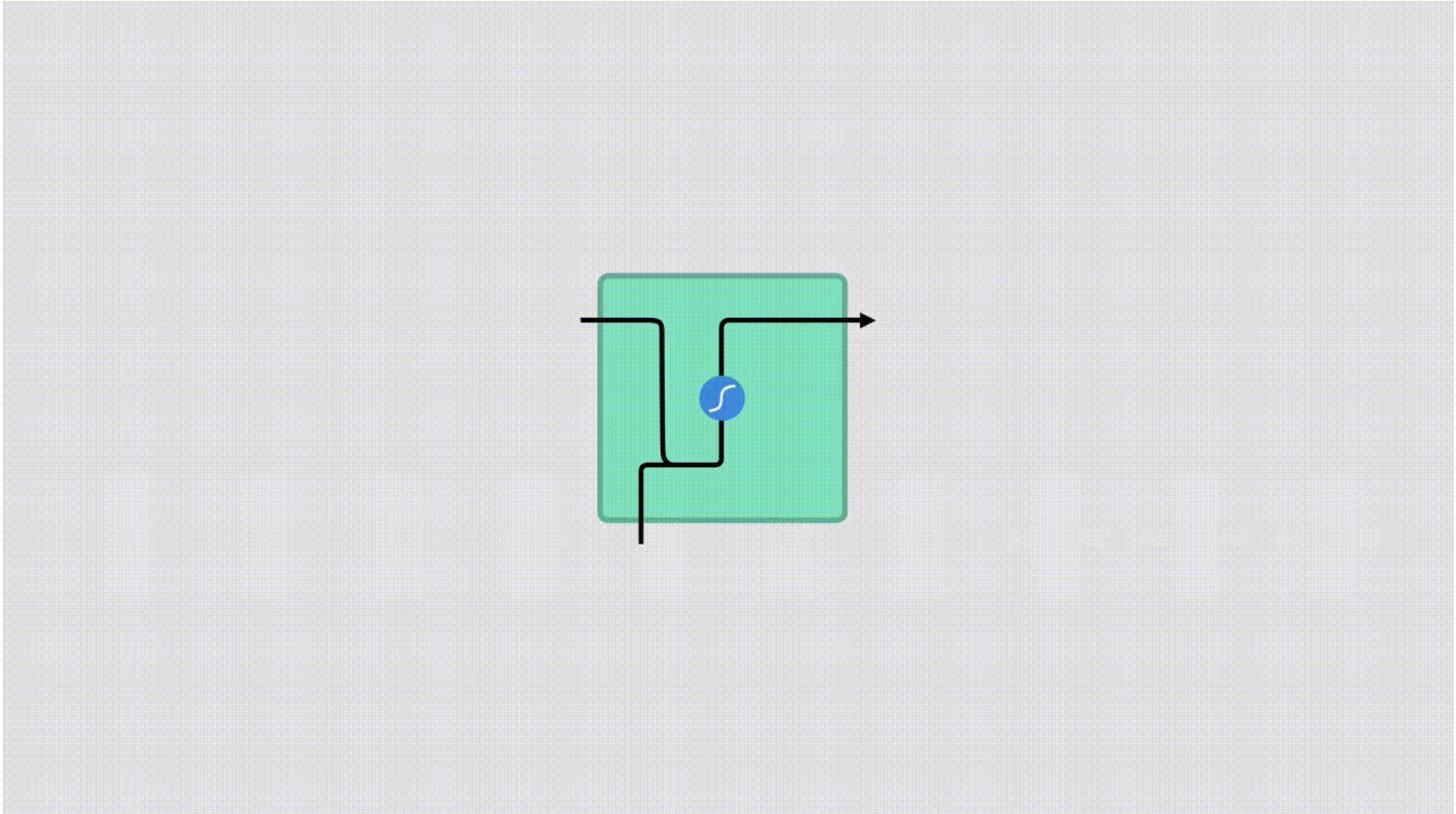
RNN, LSTM, GRU ...

- Neural Networks for Natural Language Processing
- 양옆 단어를 input 으로 받아야 함
- Text Classification, ...



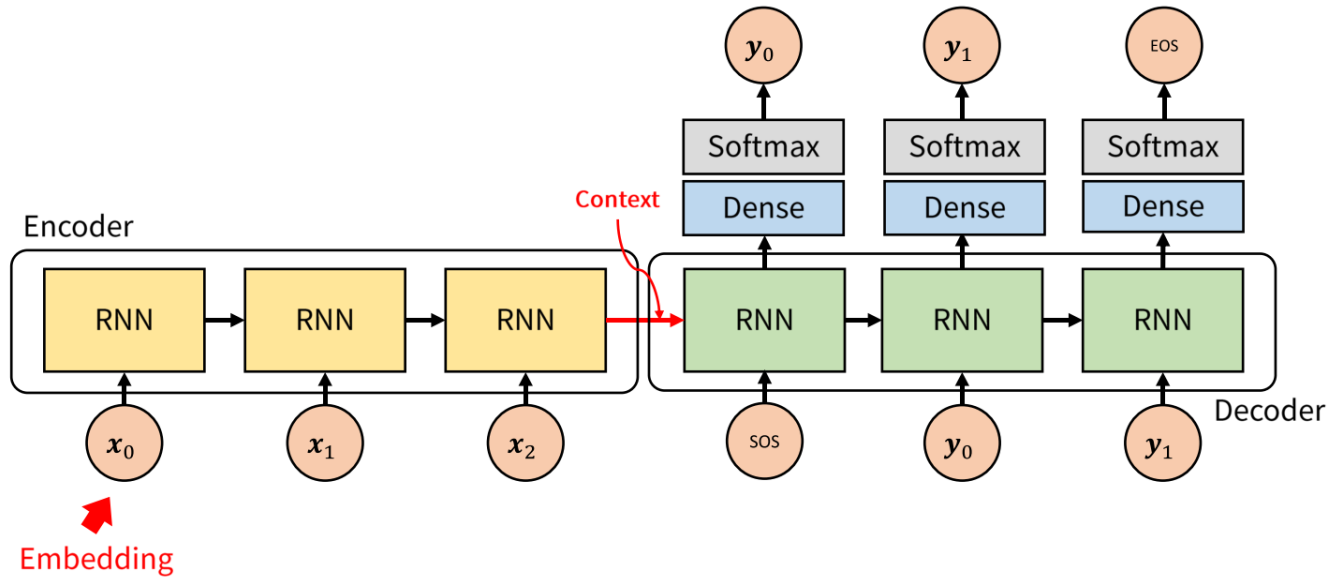
Visualization

- Videos

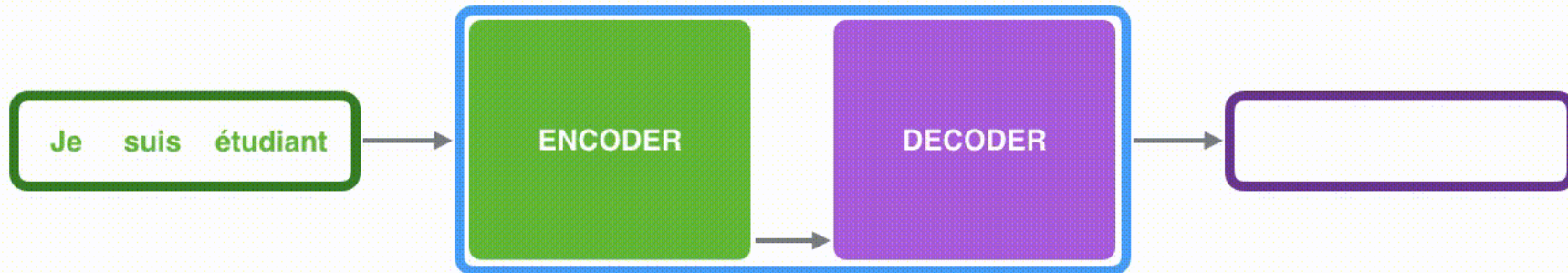


Seq2Seq

- For Machine Translation
- Fixed-size Context Vector
 - 정보 손실
 - 긴 글에 대한 context 유지 어려움

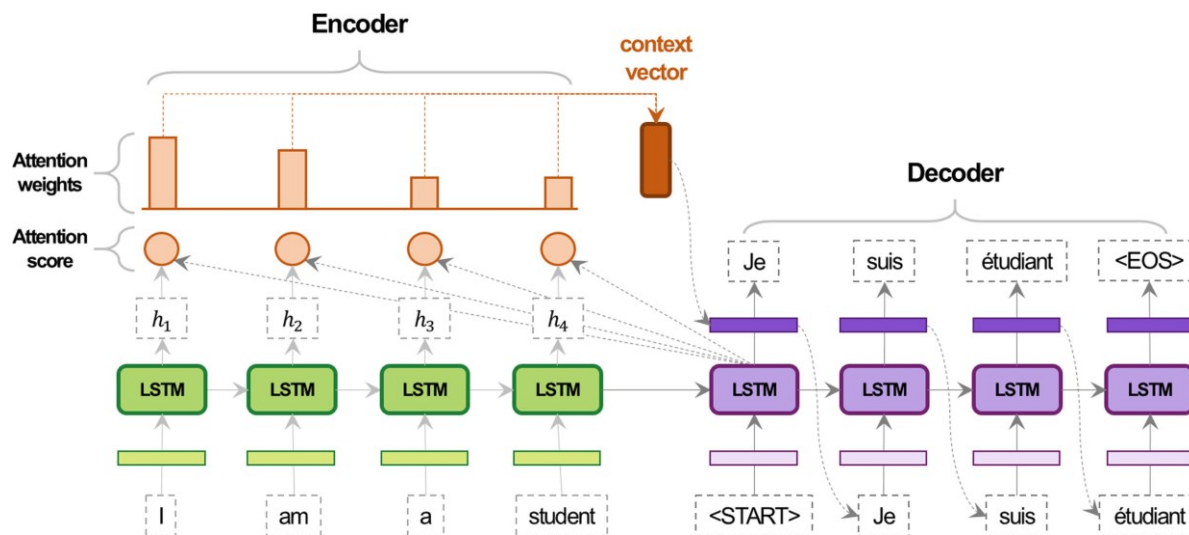
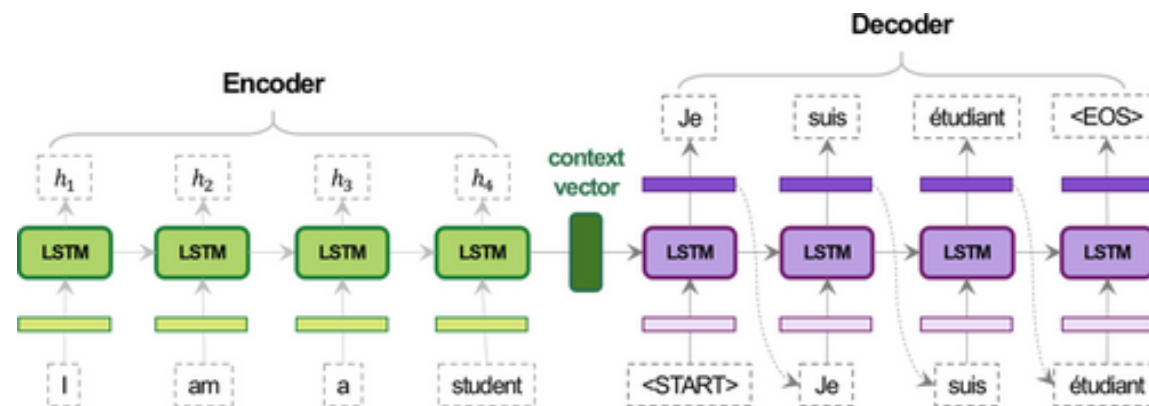


Neural Machine Translation SEQUENCE TO SEQUENCE MODEL



Attention

- 어떤 주변 단어에 주목할지를 추가하는 메커니즘
- Convolution in Vision ?
 - 주변 픽셀만을 관찰하면 된다!
- Attention
 - 어딘가 만 관찰하면 된다!
 - 어디?? 주목할 위치도 학습하면 된다!



Transformer !!!

- “Attention is all you need”
- RNN 기반 구조는 버리고, attention 만으로 가자는 것이 핵심
- 현재 모든 State of the Art 는 transformer 기반
- Encoder & Decoder
- Positional Encoding
- Self-attention

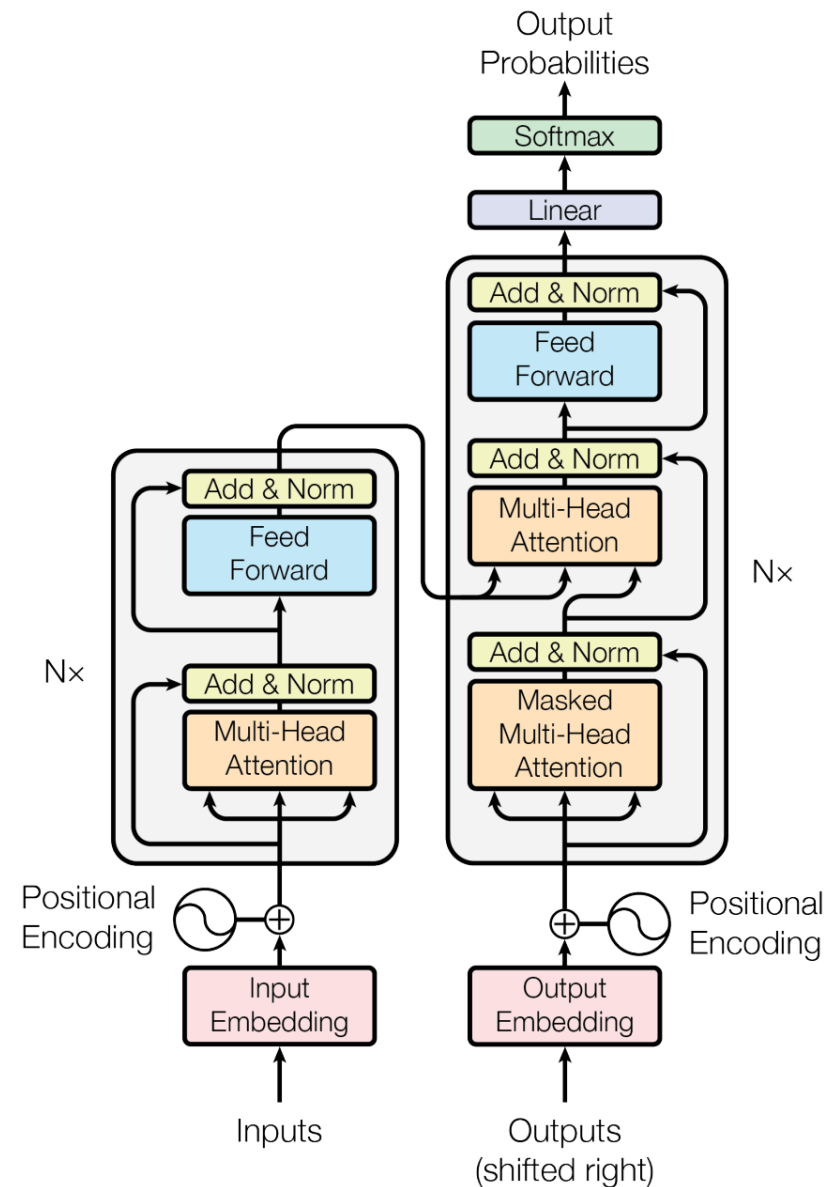
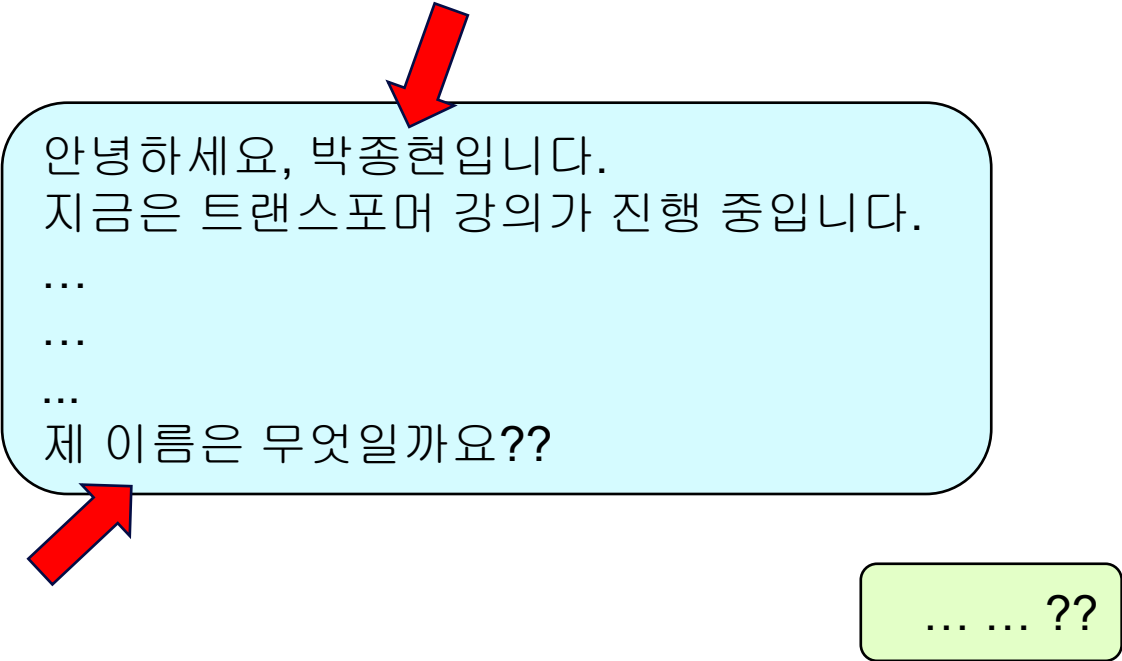


Figure 1: The Transformer - model architecture.

Transformer – How it works & Why it works

- Covers Long Token
- Could (Should) learn a looooooooooooooot of data
 - Inductive bias
- Works in Parallel

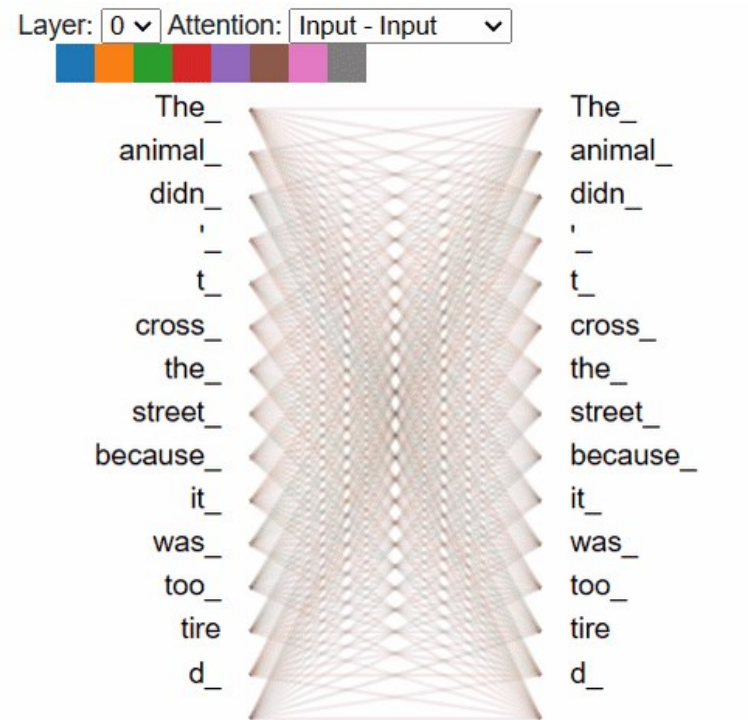


안녕하세요, 박종현입니다.
지금은 트랜스포머 강의가 진행 중입니다.
...
...
...
제 이름은 무엇일까요??

... ??

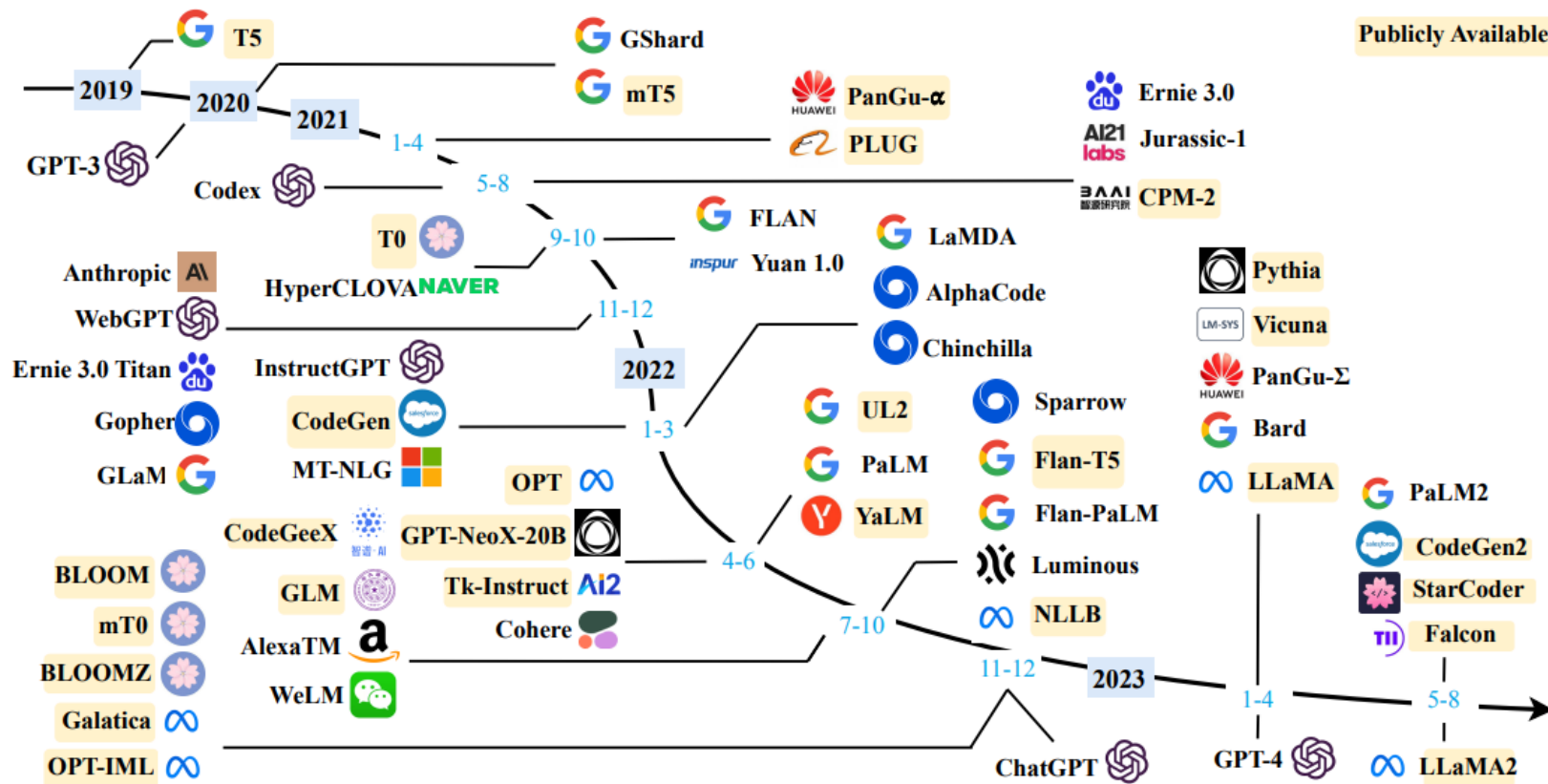
Attention & LLM - Visualize

- [GPT Visualization](#)
- [Attention Visualization Demo](#)

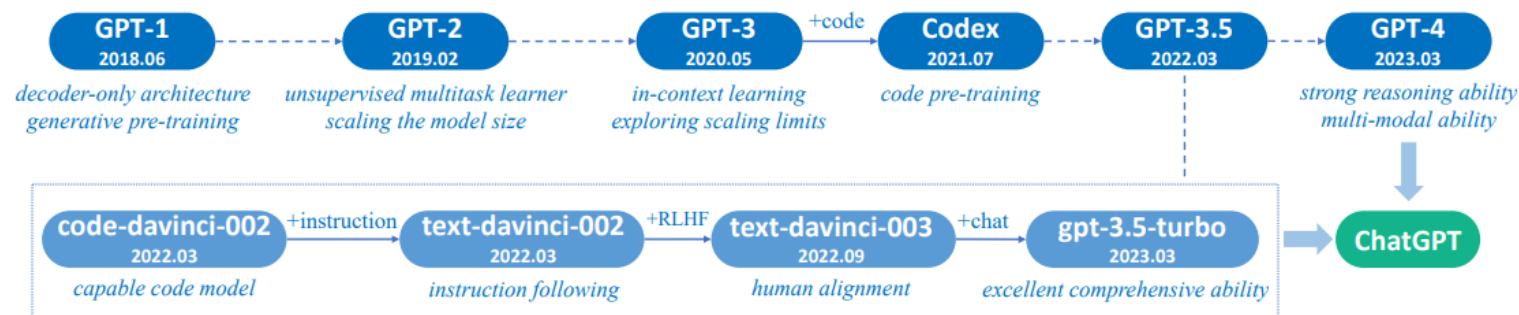


LLMs

- Large Language Models



OpenAI



LLM - BERT vs GPT

- Google BERT
 - Encoders Only
 - google search
 - Gmail, Docs, Assistants
- OpenAI GPT
 - Decoders Only
 - Writing articles, github, Bing

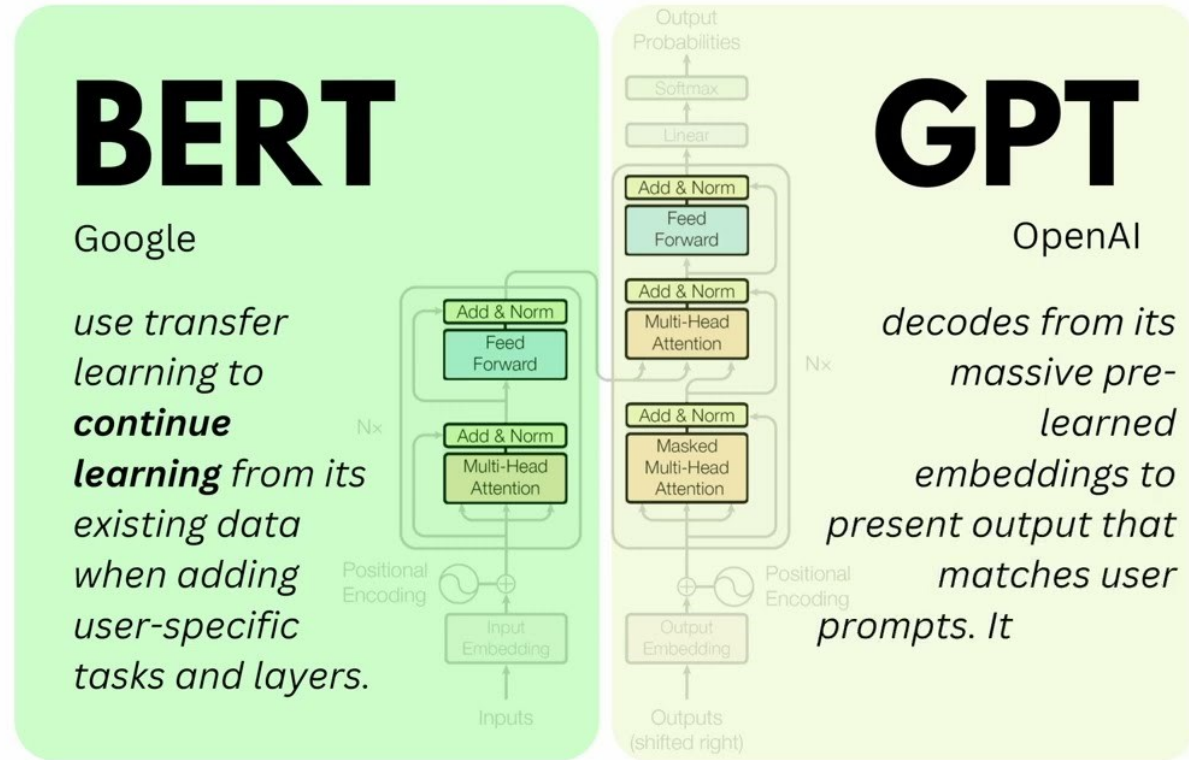


Figure 1: The Transformer - model architecture.

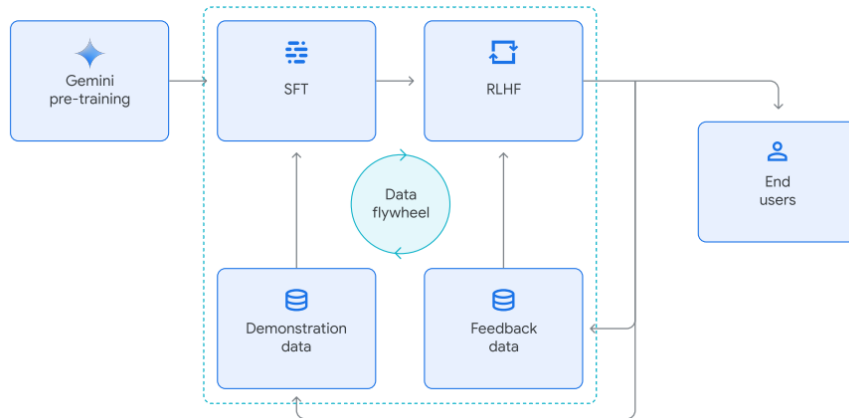
LLM – Models Now

- OpenAI GPT
- Google Gemini / Gemma
- Meta llama3
 - 마크 주커버그 인터뷰 ([ref](#))
- Microsoft Phi3
- Mistral / Mixtral
- Claude

RLHF (Reinforcement Learning by Human Feedback)

- Reward Model
- RLHF

(4) **RLHF**: Applying reinforcement learning from human feedback (RLHF) to our models provides further gains over SFT alone. Our approach creates an iterative process in which RL continually pushes the boundaries of the RM, while the RM is continuously improved through evaluation and data collection, leading to progressive improvements in both.



ChatGPT 4

You

```
import seaborn as sns

sns.set(style="darkgrid")

plot1 = sns.catplot(
    data=df_filtered3, x='계약년월', y='거래금액(만원)', hue='전용면적(m²)',
    native_scale=True, zorder=1
)

sns.regplot(
    data=df_filtered3, x='계약년월', y='거래금액(만원)',
    scatter=False, truncate=False, order=2, color="2",
)

i want to set a name of x, y, hue axis and title
```

Which response do you prefer?
Your choice will help make ChatGPT better.

Response 1

To set the names of the x-axis, y-axis, hue, and the title of your plot in Seaborn, you can utilize the `'set_axis_labels'` method for your `'catplot'` and `'set_title'` method for setting the title. Additionally, you might want to adjust the legend to reflect the hue meaning more accurately. Since `'catplot'` returns a `'FacetGrid'` object, you can use these methods directly on the object returned by `'catplot'`. For `'regplot'`, since it doesn't directly support setting axis titles or a title, you'll incorporate it into the figure that `'catplot'` creates.

Here's how you could do it:

```
python
import seaborn as sns
import matplotlib.pyplot as plt

# Assuming your DataFrame and other setup code is correct
sns.set(style="darkgrid")

# Create the catplot
plot1 = sns.catplot(
```

Response 2

To customize your plot with Seaborn and set names for the x-axis, y-axis, and hue, as well as adding a title, you can follow these steps. Based on the code snippet you provided, it looks like you're combining a categorical plot (`'sns.catplot'`) with a regression line (`'sns.regplot'`). Given that `'sns.catplot'` generates a `'FacetGrid'`, you'll have to set axis labels and titles a bit differently compared to `'sns.regplot'`.

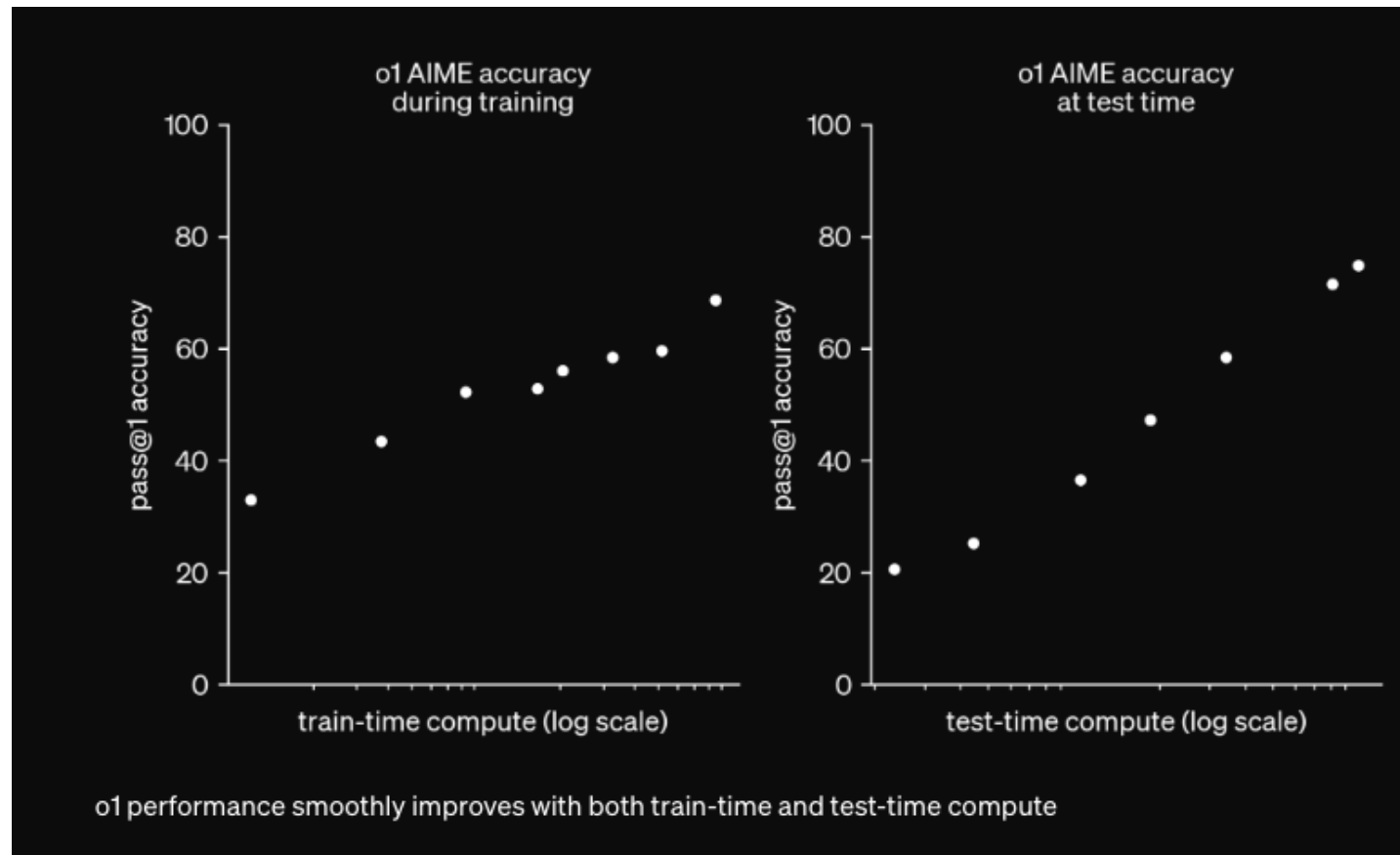
Let's assume you want to set the x-axis label as "Contract Month", the y-axis label as "Transaction Amount (10,000 KRW)", the hue label as "Area (m²)", and the title of the plot as "Real Estate Transactions Analysis". Here's how you can do it:

```
python
import seaborn as sns
import matplotlib.pyplot as plt

# Assuming df_filtered3 is already defined and sns has been set
# Create the catplot
plot1 = sns.catplot(
```

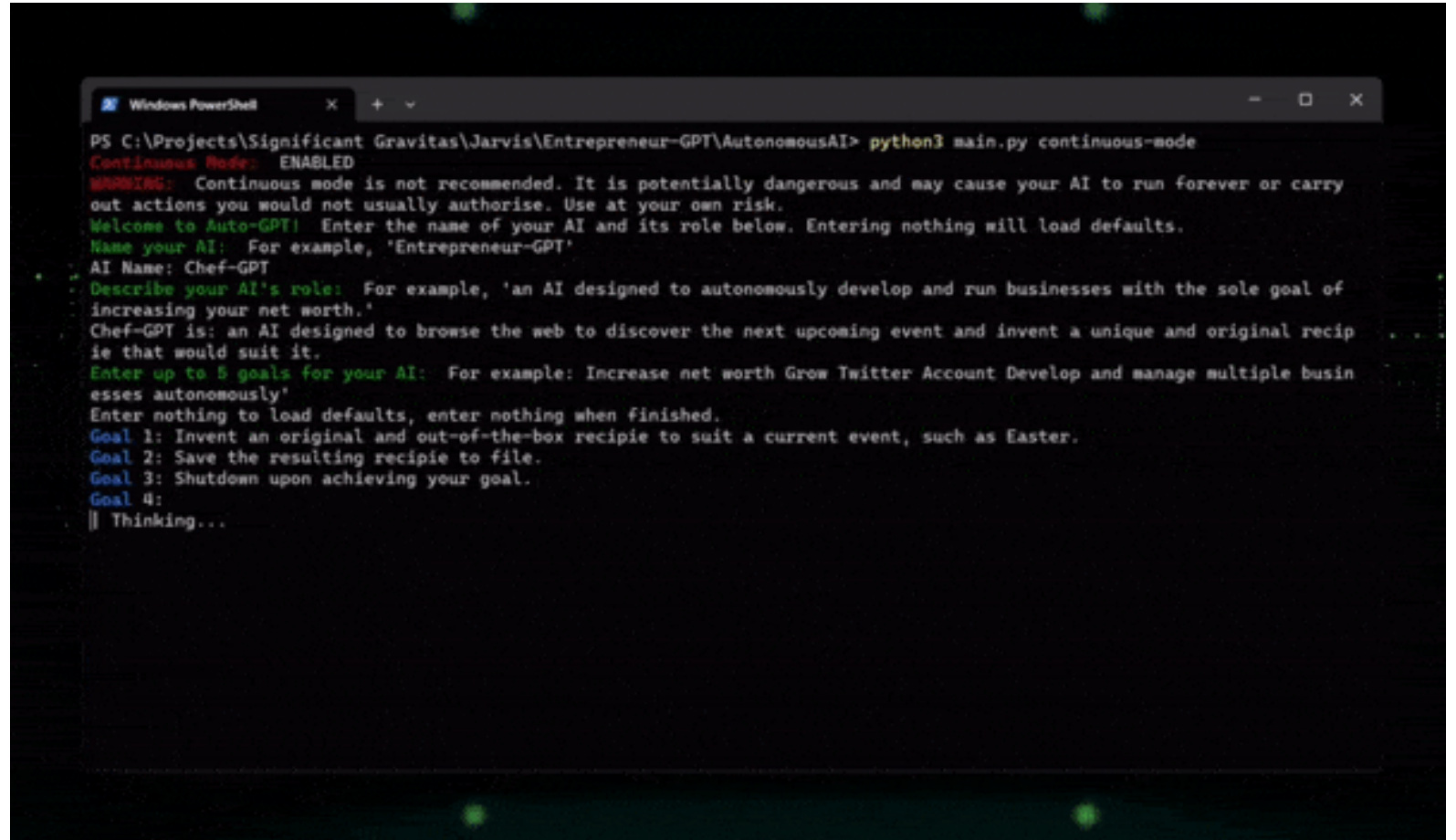
Train Time & Compute Time

- OpenAI o1
- Learning To Reason (Reinforcement Learning)
 - <https://openai.com/index/learning-to-reason-with-llms/>



Next Wave ? to AGI !

- Auto-GPT
- Baby AGI
- LangChain



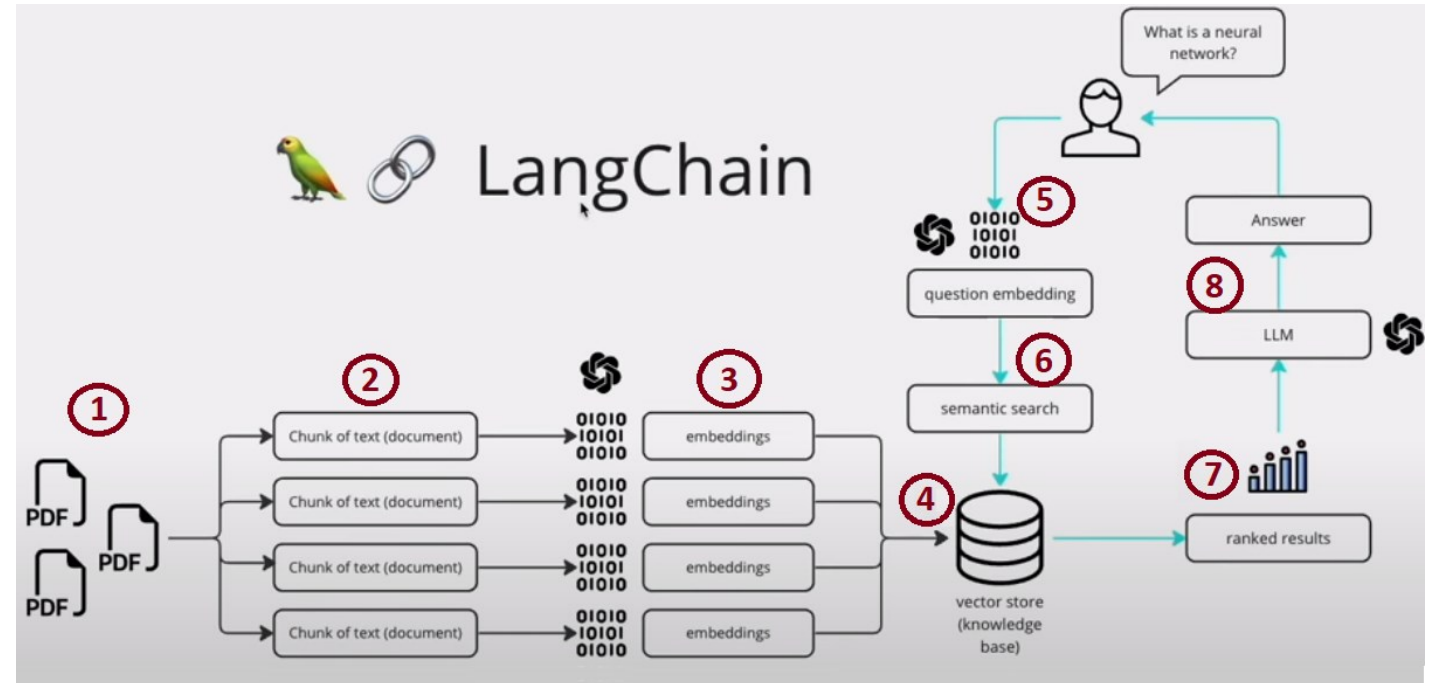
```
PS C:\Projects\Significant Gravitas\Jarvis\Entrepreneur-GPT\AutonomousAI> python3 main.py continuous-mode
Continuous Mode: ENABLED
WARNING: Continuous mode is not recommended. It is potentially dangerous and may cause your AI to run forever or carry out actions you would not usually authorise. Use at your own risk.
Welcome to Auto-GPT! Enter the name of your AI and its role below. Entering nothing will load defaults.
Name your AI: For example, 'Entrepreneur-GPT'
AI Name: Chef-GPT
Describe your AI's role: For example, 'an AI designed to autonomously develop and run businesses with the sole goal of increasing your net worth.'
Chef-GPT is: an AI designed to browse the web to discover the next upcoming event and invent a unique and original recipe that would suit it.
Enter up to 5 goals for your AI: For example: Increase net worth Grow Twitter Account Develop and manage multiple businesses autonomously
Enter nothing to load defaults, enter nothing when finished.
Goal 1: Invent an original and out-of-the-box recipe to suit a current event, such as Easter.
Goal 2: Save the resulting recipe to file.
Goal 3: Shutdown upon achieving your goal.
Goal 4:
|| Thinking...
```

Practice - Using GPT !

- Doc to Question Answering bot using GPT

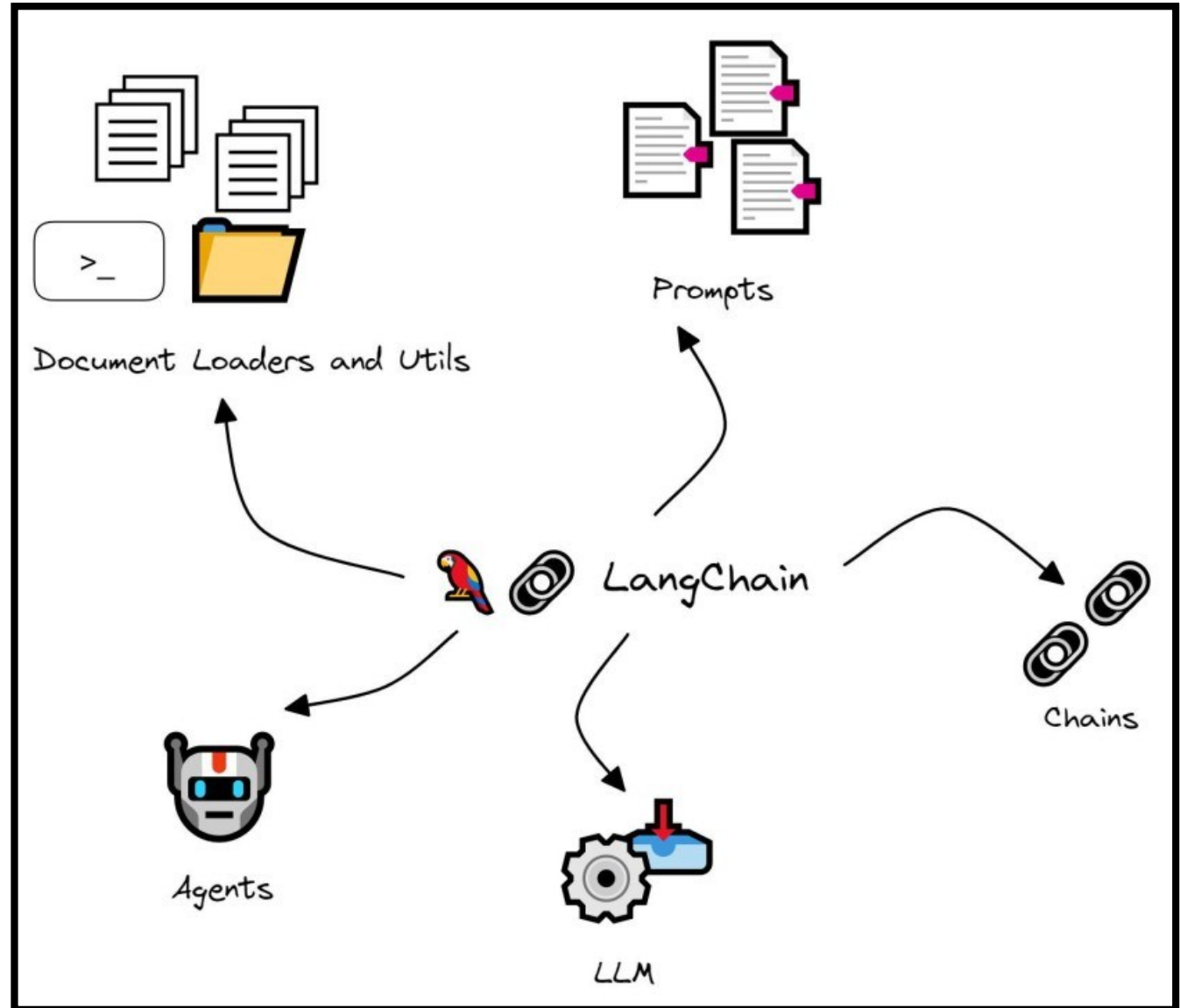
Embeddings & VectorDB

- Knowledge after trained
- Token length is not infinite
- Ideas
 - Let's make a knowledge to Database using embeddings!
 - Word2Vec → Articles2Vec → Vector Database
- Example)
 - Document Answering GPT
 - Semantic Searching



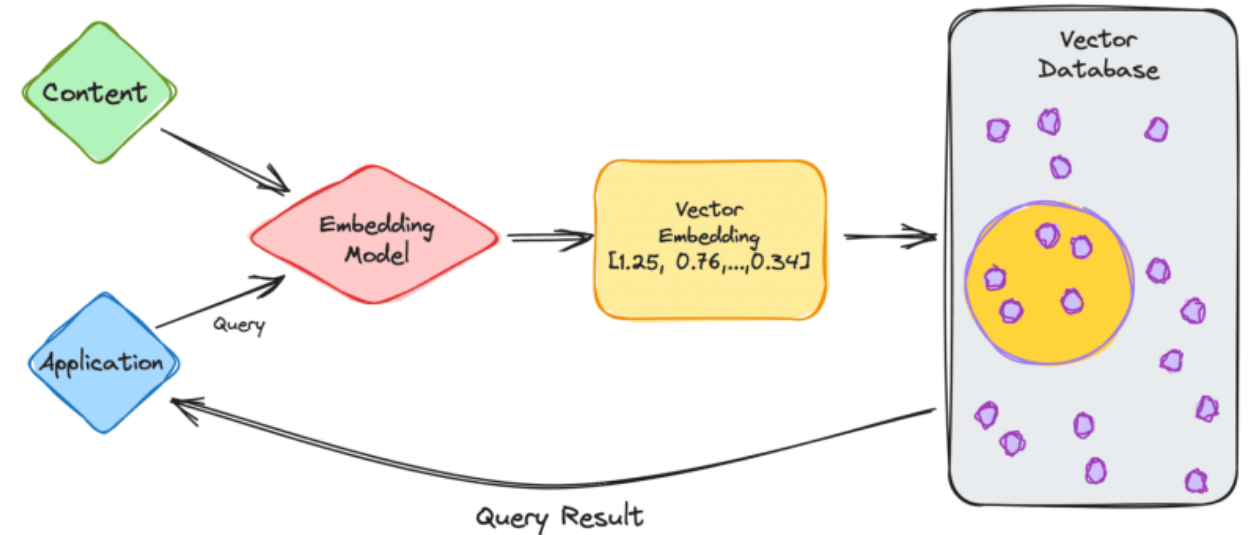
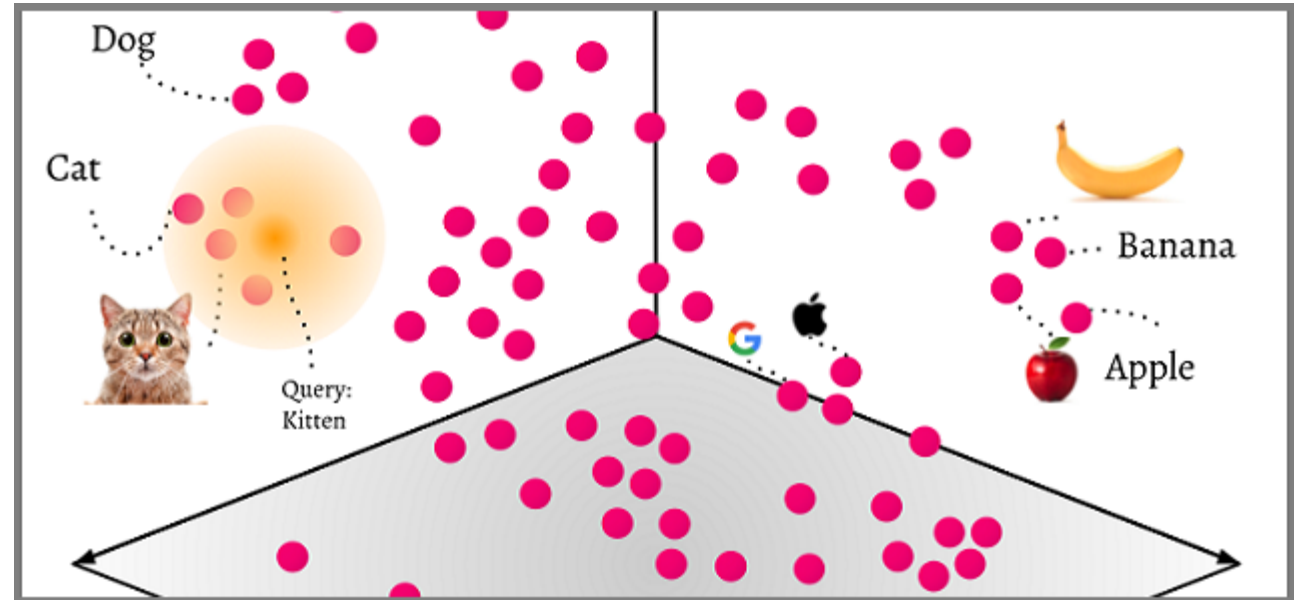
LangChain

- 도구들을 묶어서 쉽게 사용하게 해주는 프레임워크
- LangServe
 - Deploying
- LangSmith
 - LLMOps
- (+) Streamlit



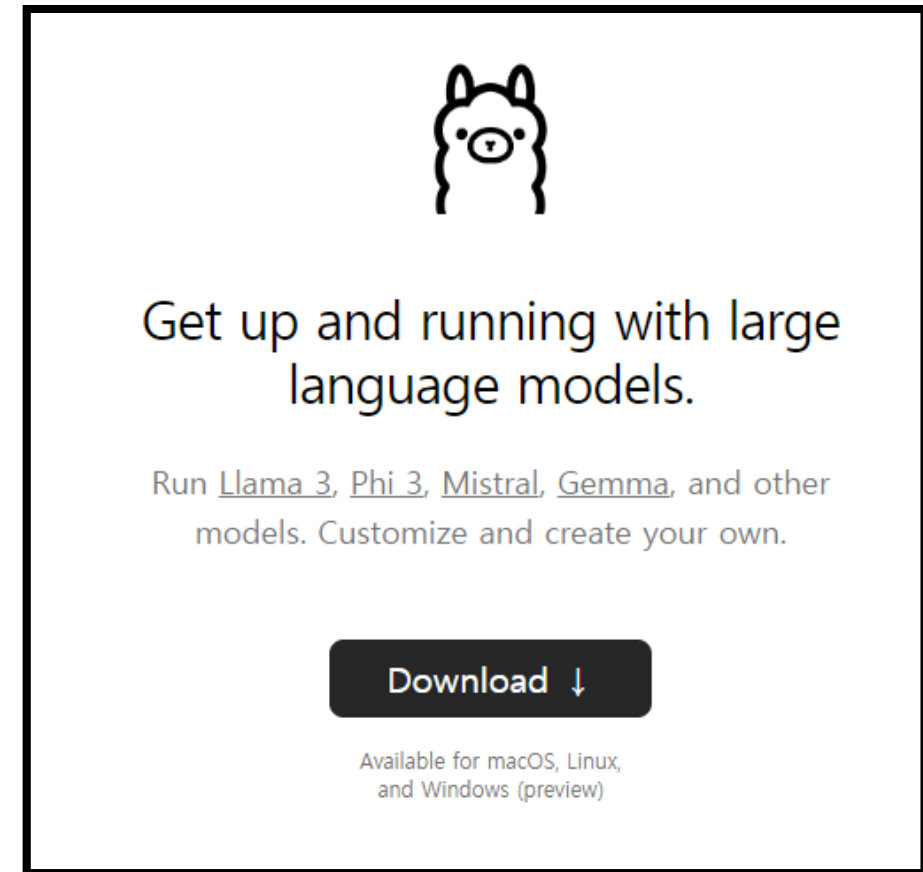
VectorDB

- Database for Vectors (Embeddings)
- Embedding 이 상용화된 2023년에 모두 투자 완
- Chroma, Weviate, Pinecone, ...



Ollama – Make the models run in local

- Phi3 - <https://ollama.com/library/phi3>
- Llava - <https://ollama.com/blog/vision-models>
- 실습 - Download 받아서 수행해보기
 - Phi3 3.8b 를 추천! 가장 작기 때문



Prompting Guide

- Google's Prompting 101 ([ref](#))
- OpenAI's Prompting Engineering ([ref](#))
- 고려할 점 - Hallucination

There are four main areas to consider when writing an effective prompt. You don't need to use all four, but using a few will help!

- Persona
- Task
- Context
- Format

Here is an example of a prompt using all four areas that could work well in Gmail and Google Docs:



You are a Google Cloud program manager. Draft an executive summary email to [persona] based on [details about relevant program docs]. Limit to bullet points.

Here are quick tips to get you started with Gemini for Workspace:

1. **Use natural language.** Write as if you're speaking to another person. Express complete thoughts in full sentences.
2. **Be specific and iterate.** Tell Gemini for Workspace what you need it to do (summarize, write, change the tone, create). Provide as much context as possible.
3. **Be concise and avoid complexity.** State your request in brief — but specific — language. Avoid jargon.
4. **Make it a conversation.** Fine-tune your prompts if the results don't meet your expectations or if you believe there's room for improvement. Use follow-up prompts and an iterative process of review and refinement to yield better results.

Google Prompting Guide

- Role Playing (= Persona)
- Task & Context 정보!
- Formatting
- And Examples for situation

There are four main areas to consider when writing an effective prompt. You don't need to use all four, but using a few will help!

- Persona
- Task
- Context
- Format

Here is an example of a prompt using all four areas that could work well in Gmail and Google Docs:



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OpenAI's Guide - Experts

- deeplearning.ai 1st course
- Provide Examples
 - Few-shot
- Provide Reference Text
 - RAG
- Split complex tasks into simpler subtasks & Give the model time to “think”
 - Chain of Thought

Few-shot Prompting

- Zero-shot

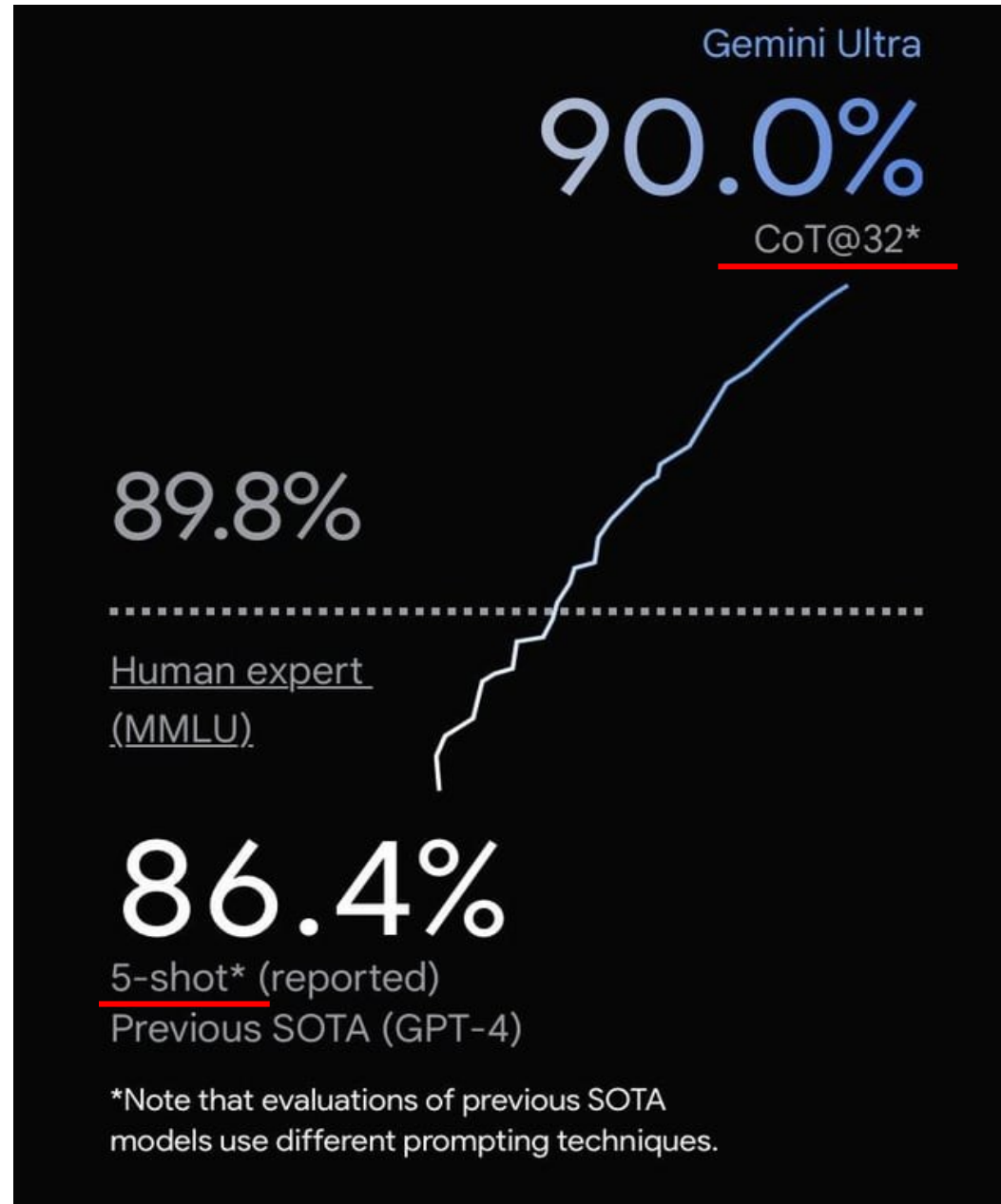
Classify the text into neutral, negative or positive.
Text: I think the vacation is okay. Sentiment:
Neutral

- Few-shot

This is awesome! // Positive This is bad! // Negative Wow that movie was rad! // Negative What a horrible show! //
Positive

Chain of Thought

- 논란! - Gemini (2023)



Chain of Thought (Few-shot)

Standard Prompting

Model Input

Q: Roger has 5 tennis balls. He buys 2 more cans of tennis balls. Each can has 3 tennis balls. How many tennis balls does he have now?

A: The answer is 11.

Q: The cafeteria had 23 apples. If they used 20 to make lunch and bought 6 more, how many apples do they have?

Model Output

A: The answer is 27. ❌

Chain-of-Thought Prompting

Model Input

Q: Roger has 5 tennis balls. He buys 2 more cans of tennis balls. Each can has 3 tennis balls. How many tennis balls does he have now?

A: Roger started with 5 balls. 2 cans of 3 tennis balls each is 6 tennis balls. $5 + 6 = 11$. The answer is 11.

Q: The cafeteria had 23 apples. If they used 20 to make lunch and bought 6 more, how many apples do they have?

Model Output

A: The cafeteria had 23 apples originally. They used 20 to make lunch. So they had $23 - 20 = 3$. They bought 6 more apples, so they have $3 + 6 = 9$. The answer is 9. ✅

Chain of Thought (Zero-shot)

(a) Few-shot

Q: Roger has 5 tennis balls. He buys 2 more cans of tennis balls. Each can has 3 tennis balls. How many tennis balls does he have now?

A: The answer is 11.

Q: A juggler can juggle 16 balls. Half of the balls are golf balls, and half of the golf balls are blue. How many blue golf balls are there?

A:

(Output) The answer is 8. **X**

(b) Few-shot-CoT

Q: Roger has 5 tennis balls. He buys 2 more cans of tennis balls. Each can has 3 tennis balls. How many tennis balls does he have now?

A: Roger started with 5 balls. 2 cans of 3 tennis balls each is 6 tennis balls. $5 + 6 = 11$. The answer is 11.

Q: A juggler can juggle 16 balls. Half of the balls are golf balls, and half of the golf balls are blue. How many blue golf balls are there?

A:

(Output) The juggler can juggle 16 balls. Half of the balls are golf balls. So there are $16 / 2 = 8$ golf balls. Half of the golf balls are blue. So there are $8 / 2 = 4$ blue golf balls. The answer is 4. **✓**

(c) Zero-shot

Q: A juggler can juggle 16 balls. Half of the balls are golf balls, and half of the golf balls are blue. How many blue golf balls are there?

A: The answer (arabic numerals) is

(Output) 8 **X**

(d) Zero-shot-CoT (Ours)

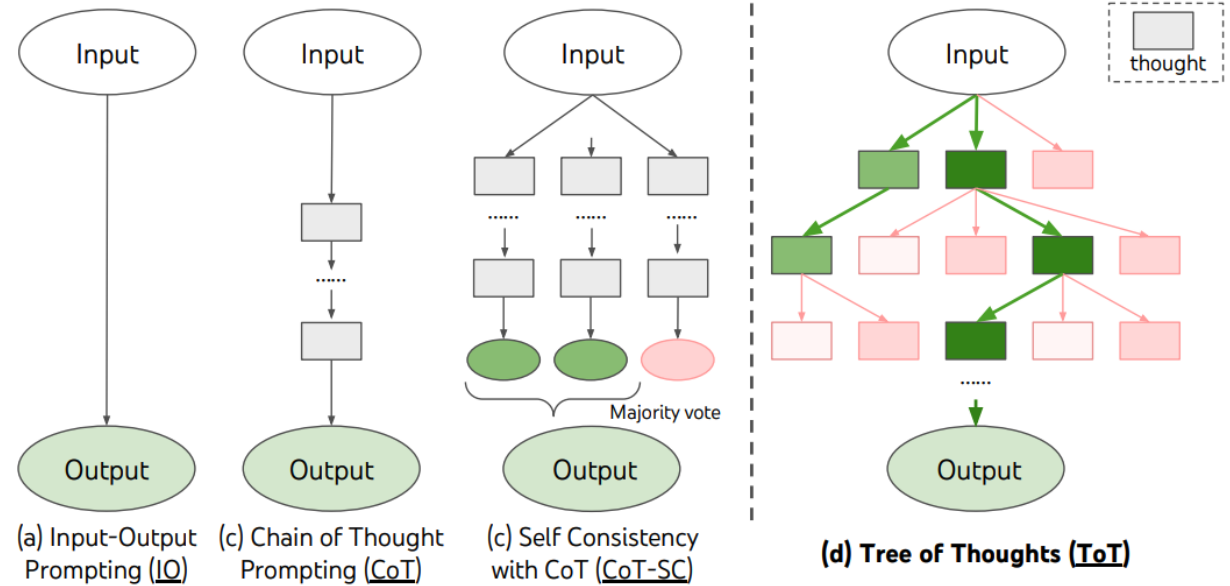
Q: A juggler can juggle 16 balls. Half of the balls are golf balls, and half of the golf balls are blue. How many blue golf balls are there?

A: **Let's think step by step.**

(Output) There are 16 balls in total. Half of the balls are golf balls. That means that there are 8 golf balls. Half of the golf balls are blue. That means that there are 4 blue golf balls. **✓**

Furthermore Prompting

- Tree of Thought
- Prompt Chaining
- Automatic Reasoning
- Agents
- Self-Consistency
- Automatic Prompt Engineer



GPTs / Fine Tuning

- GPT Store opened – <https://openai.com/blog/introducing-the-gpt-store>
- Capabilities
 - Prompting
 - Knowledge (RAG, Retrieval Augmented Generation)
 - Web Browsing
 - DALL·E Image Generation
 - Code Interpreter
 - Actions
- Fine Tuning through APIs
 - Formatting the Natural Language to Query
 - (Example) food delivery chat bot
- Memory ! (coming soon) - <https://openai.com/blog/memory-and-new-controls-for-chatgpt>

Vision Transformer

- Image Patches → embeddings → transformer
- Could (Should) learn a looooooooooooooot of data
 - Inductive bias
- Every Data → Encode → vector
- Vector → Decode → Every Data
- Image, Language, Speech, Time-signal, etc.

Cropped Image



Image Patches

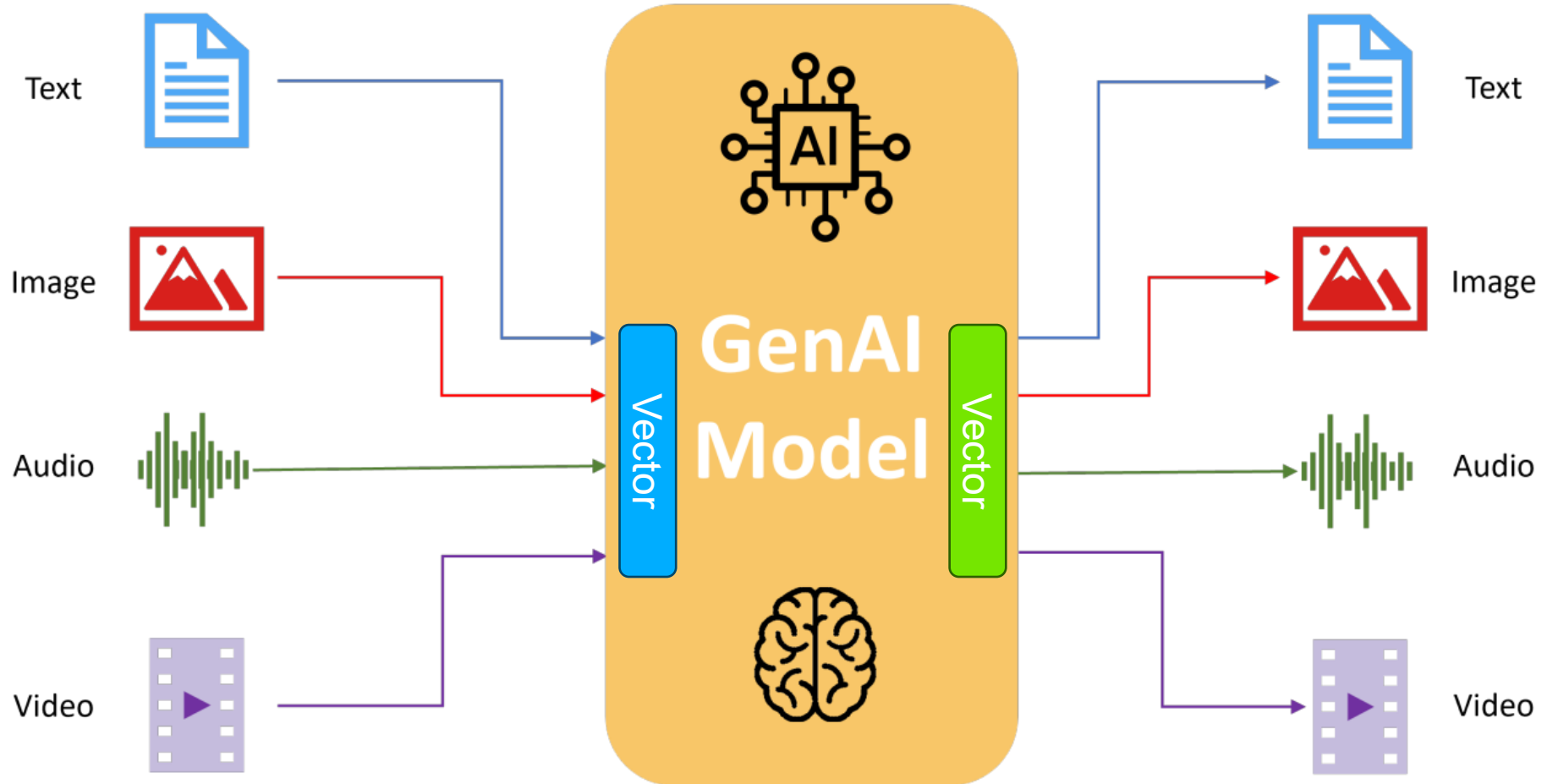


Flattened Image Patches



Multimodal

- <https://deepmind.google/technologies/gemini/#gemini-1.0>



GPT4V (2023)

ChatGPT 4

You



이 이미지에 대해 설명을 해주고, 재밌게 꾸며서 다른 이미지로 만들어줘

ChatGPT

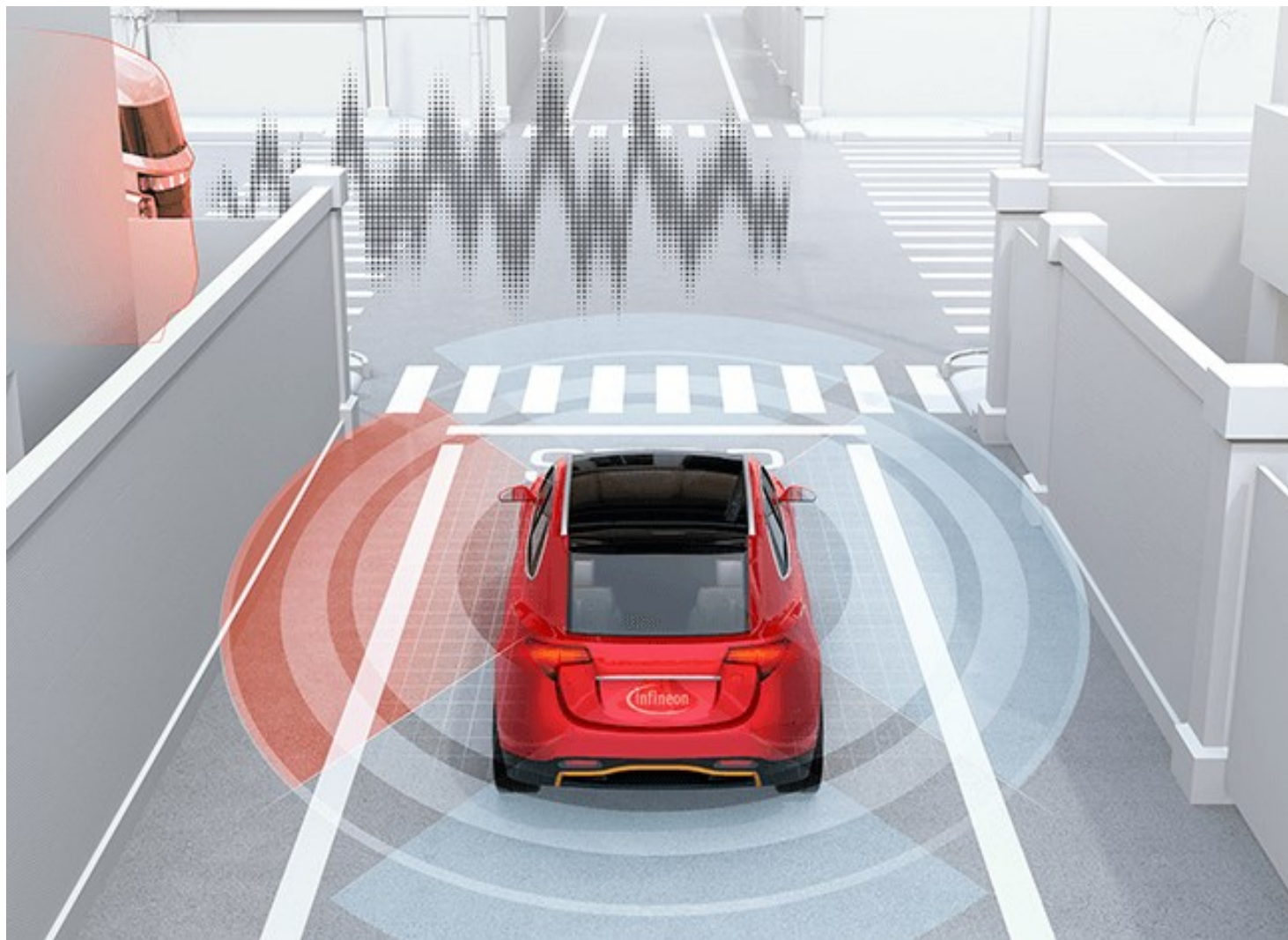
이 사진은 한 남자가 강연을 하고 있는 모습을 담고 있습니다. 그는 무대 위에 서서 마이크를 들고, 오른손으로 무언가를 설명하고 있는 제스처를 취하고 있네요. 그가 입고 있는 티셔츠엔 다른 사람의 얼굴이 프린트되어 있고, 목에는 이벤트 이름과 년도가 적힌 ID 카드를 걸고 있습니다. 배경은 밝은 색상으로, 이벤트 또는 컨퍼런스의 일환으로 보입니다.

이 이미지를 재밌게 변형해보자면, 이 남자를 마법사로 꾸며서 마법의 학교에서 강의하는 장면으로 만들 수 있겠네요. 그의 티셔츠 대신에 긴 로브를 입히고, 마이크 대신에 마법 지팡이를 들게 하며, 배경으로는 고딕 양식의 거대한 서재나 강의실을 넣어주면 분위기가 완전히 달라질 것 같아요. 그리고 마법사의 손 제스처는 마법 주문을 외우고 있는 것처럼 보일 겁니다. 이 장면에 어울리는 애니멀 컴패니언도 추가해보죠, 어쩌면 올빼미나 작은 드래곤 같은 것을 어깨에 올려놓을 수 있을 거예요. 이렇게 상상력을 발휘해본다면, 전혀 다른 분위기의 이미지가 탄생할 거예요. 지금 바로 이 아이디어로 이미지를 만들어 볼게요.



여기 마법의 학교에서 강의하는 마법사의 모습을 담은 이미지가 있습니다. 이 마법사는 오래된 서적으로 가득한 고딕 양식의 서재에서 자신감 있는 자세로 주문을 외우고 있어요. 어깨 위에는 올빼미가,

Autonomous Vehicle with Sounds



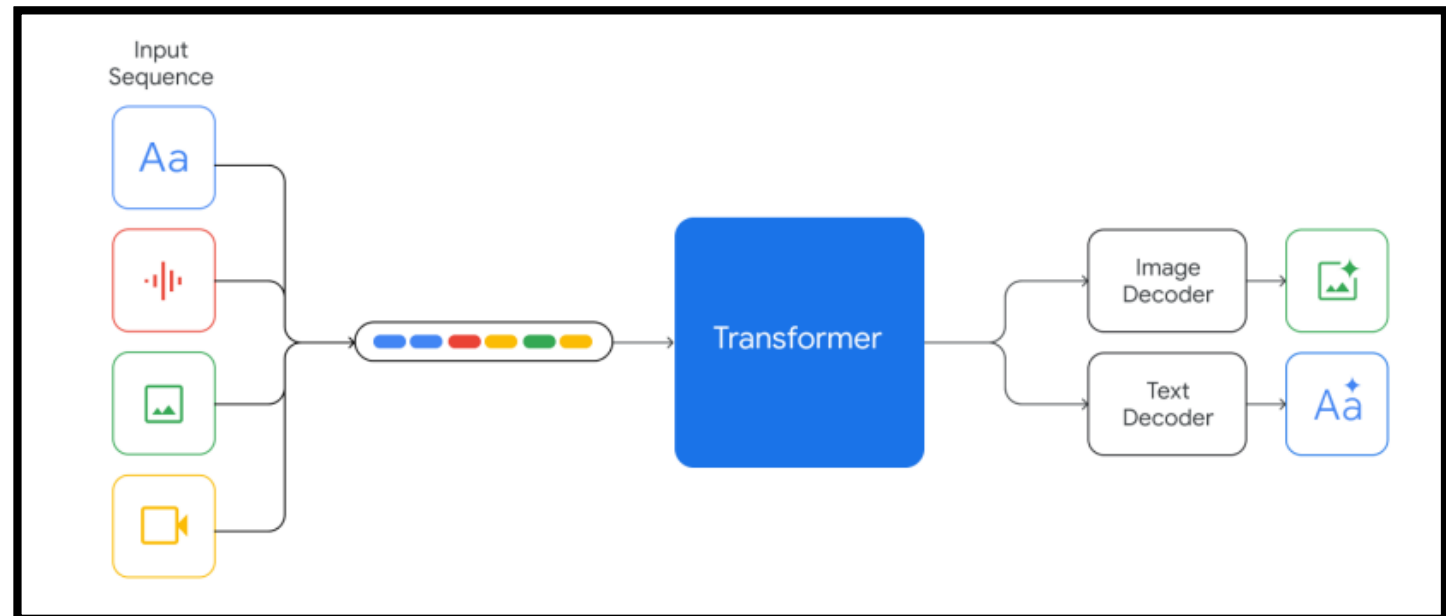
GPT4o (2024)

- [OpenAI Live Demo \(2024.05\)](#)
- ChatGPT 24.09 ~



Google Gemini

- Multimodal model from the scratch
 - <https://deepmind.google/technologies/gemini>
 - <https://arxiv.org/abs/2312.11805>
- Bard → Gemini
 - <https://blog.google/intl/ko-kr/products/explore-get-answers/bard-gemini-advanced-app-kr/>
- NANO! For On-Device AI



Multimodal Models Example

- Google Projects Astra
 - <https://deepmind.google/technologies/gemini/project-astra/>
- Meta Embodied EQA
 - <https://ai.meta.com/blog/openeqa-embodied-question-answering-robotics-ar-glasses/>
 - <https://ai.meta.com/research/movie-gen/>
- Infographic PDF Analyzer
- LangSmith Trace
 - <https://smith.langchain.com/public/62bf6669-ec96-4e0b-9482-1a89daf536a9/r>

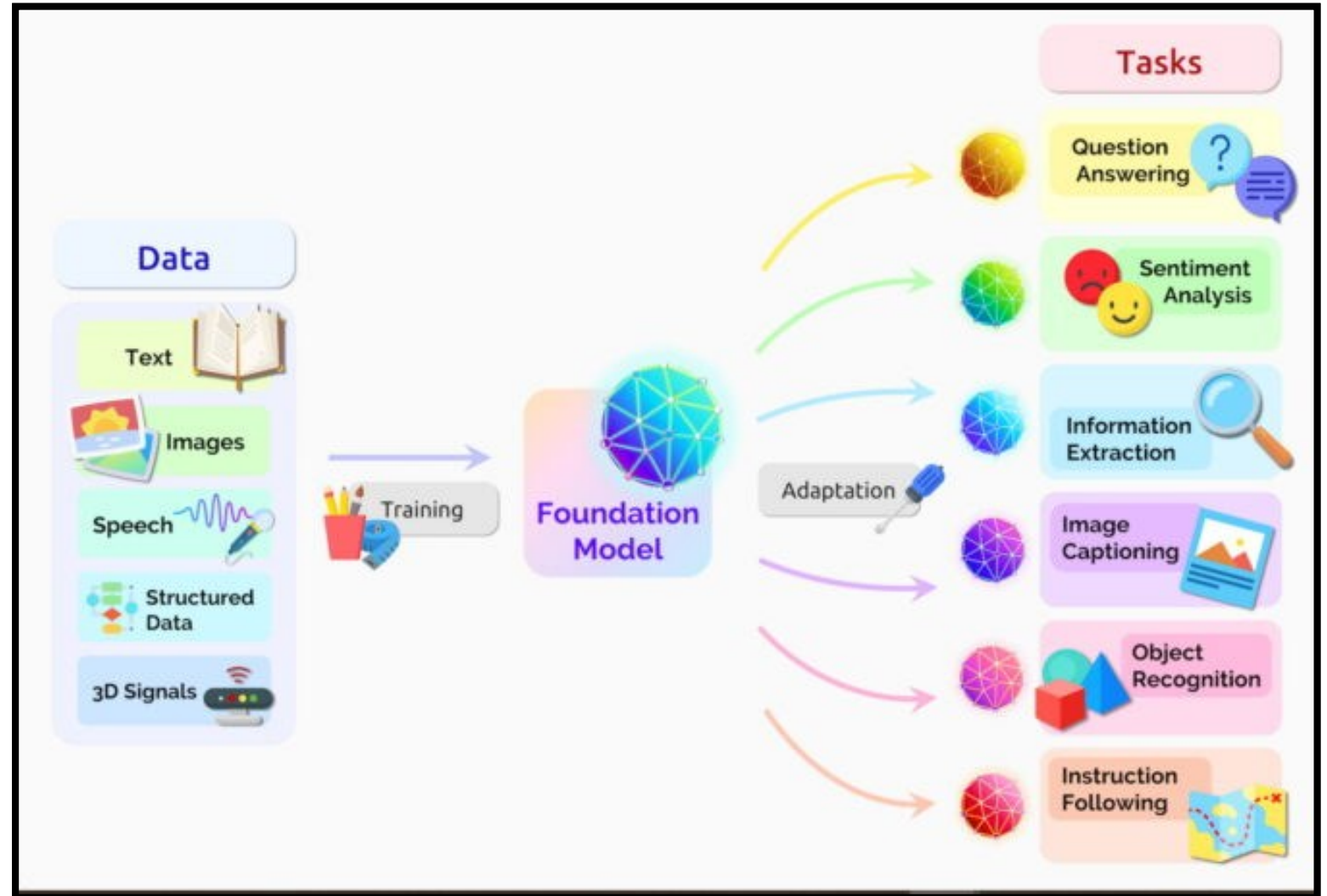
Copilot+PC & phi-3 (24.05)

- Copilot + PC : Next Windows?
 - <https://blogs.microsoft.com/blog/2024/05/20/introducing-copilot-pcs/>
- Phi-3 : On Device AI – Snapdragon NPU
 - <https://github.com/microsoft/Phi-3CookBook>



Foundation Model

- Model can do EVERYTHING!
- LLM / LMM 은 지능이 있는가?



OpenAI DevDay 2024.10

- <https://openai.com/devday/content/>
- Realtime API
 - Speak (<https://www.speak.com/ko>)
 - ChatGPT 고급 음성 모드
 - 데모 시연 !
- Vision Fine Tuning
- Prompt Caching
- Model Distillation

Video, 2023, and now?



Short Film by SORA

- <https://openai.com/blog/sora-first-impressions>

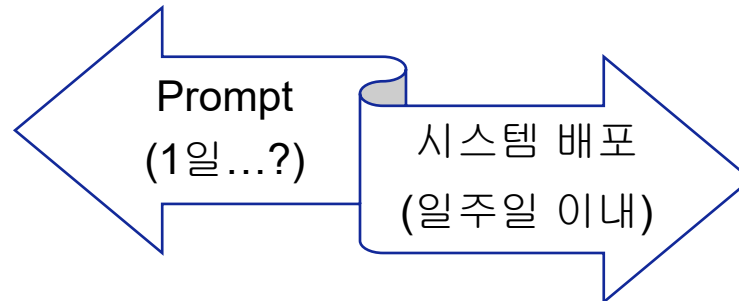


Paradigm 변화

- Model 학습 기반 AI



- Prompting 기반 AI



딥러닝에 관한 고찰

- 결국 딥러닝이 잘하는 것은 무엇일까?

- Data 의 Feature 를 찾는 일!
 - 이전에는 인간만이 가능하고 알고리즘으로는 불가능 했었다
 - Data 의 종류는 가리지 않음, Image, Audio, Language, Time-series 등등

- 딥러닝의 탄생 이후의 breakthrough?

- 더 많은 data를 학습할 능력을 얻었을 때마다 breakthrough 가 있었음
- Alexnet 의 등장, Resnet 의 성공 이유, Transformer 의 성공 이유 모두 같음
- 구조는 간단해야 하고, 데이터 파이프라인을 잘 만들어야 한다
 - 테슬라에서 Lidar 가 필요 없다고 주장하는 이유

추천 자료 링크

▪ 1. 학습

- 공식 홈페이지 – Tensorflow, PyTorch, Kaggle, HuggingFace (쉬운 것 부터 어려운 순)
- 빠르게 대략적인 이해를 하고 싶다면! – 3Blue1Brown Youtube Channel
- 정말 자세히 동작을 알고 싶다면! - <https://karpathy.ai/zero-to-hero.html>
- 양질의 강의 모음 (광고가 섞여 있음) – <https://www.deeplearning.ai/>

▪ 2. 모니터링

- HackerNews – <https://news.ycombinator.com/>
- GeekNews (한국어 버전) – <https://news.hada.io/>

▪ 3. 문제 해결

- Starts From HuggingFace Models

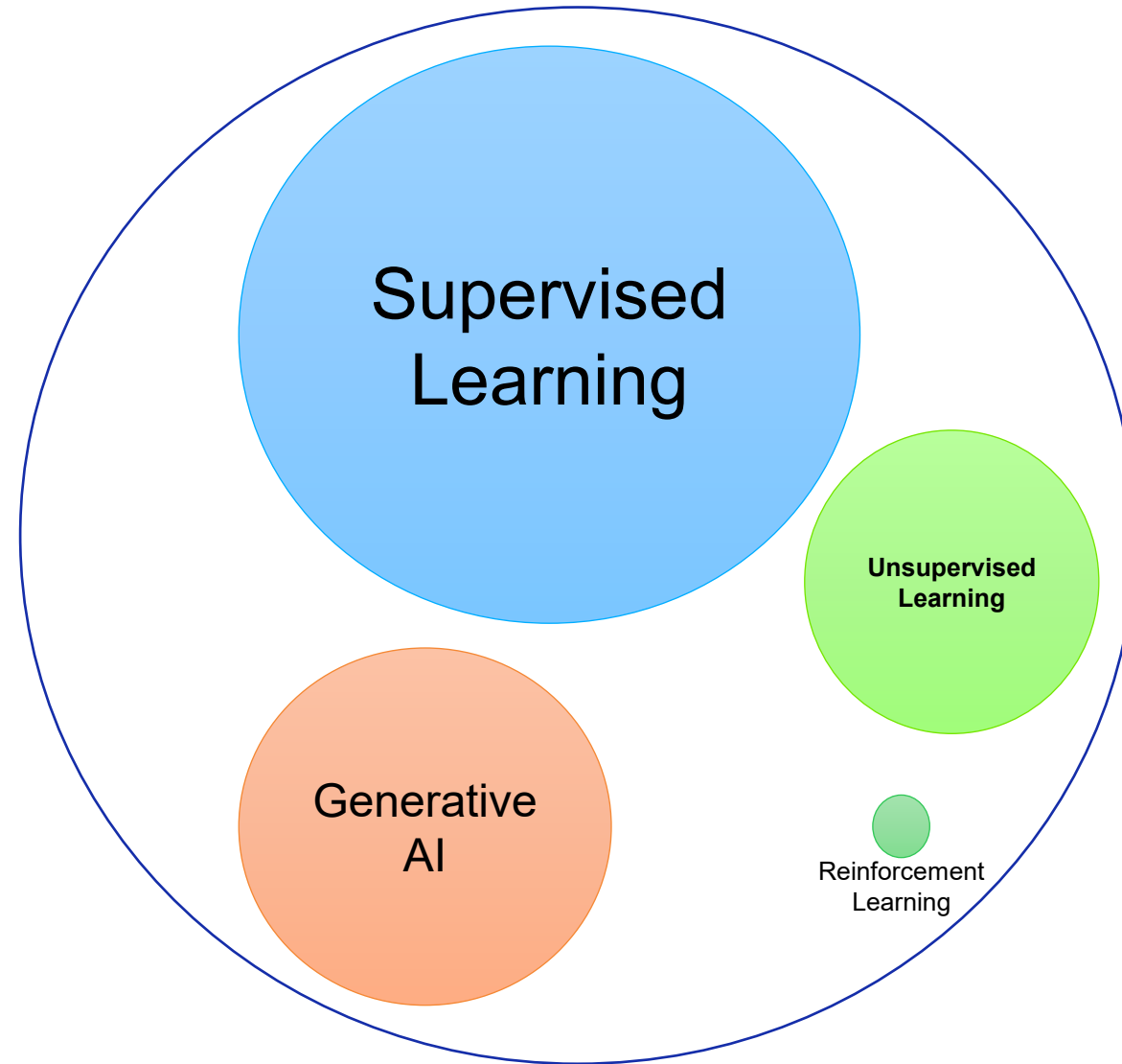
▪ 4. 기타

- Awesome Links - <https://github.com/aishwaryanr/awesome-generative-ai-guide>

문제 해결 알고리즘

- 1. 문제를 정의한다 : $X \rightarrow Y$
- 2. X 의 데이터 타입이 100개 미만의 숫자인가?
 - Yes $\rightarrow$ Machine Learning 을 쓰면 된다, chatgpt조차도 알고리즘을 만들 수 있는 수준!
하지만, 정말 잘하려면 데이터 엔지니어링이 추가적으로 필요함
- 3. X 의 데이터 타입이 다양한가? (이미지, 음성, 시계열, 텍스트 등등이 섞여 있는가)
 - X 를 분리해서 각각에 맞는 Vector 로 치환하고, 2번으로 문제를 치환해서 풀어야 함
 - 1년 후에는 멀티모달 모델의 발전으로 위 방법이 필요 없어질 수 있음
- 4. 데이터가 충분한가?
 - 다른 사례들을 통해 문제의 난이도를 가늠해서 데이터가 충분한지 판단해 봐야 함
 - 데이터를 추가 확보하는 것이 어렵다면... 문제 정의를 비트는 아이디어가 필요함, 도메인 지식에 기반하는 것이 가장 쉬운 길

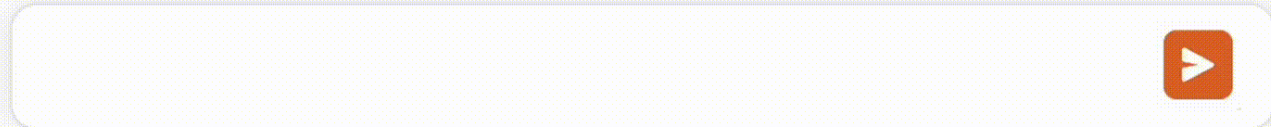
2023 – The Era of Generative AI



2023 – The Era of Generative AI

- ChatGPT
- Stable Diffusion, Dall-E
- Midjourney
 - [미술 대회 우승](#)

ChatGPT●



Andrej Karpathy

- Toronto Univ → Stanford, Geoffrey Hinton ?
- The Reference Human – ImageNet 에서의 휴먼 에러!
- ConvNetJS
- Stanford CS 231n 강의 개발자 (현 최고 인기 강의)
- OpenAI 창립멤버
- Tesla Autopilot 수장
- OpenAI 복귀
 - “Hallucination” 정의
- 다시 퇴사, 차기 행보는?
 - <https://karpathy.ai/zero-to-hero.html>



Andrej Karpathy

I like to train deep neural nets on large datasets 🧠🔥



2023 -  Back at [OpenAI](#).

2017 -  I was the Sr. Director of AI at Tesla, where I led the computer vision team of [Tesla Autopilot](#). This includes in-house data labeling, neural network training, the science of making it work, and deployment in production running on our custom inference chip. Today, the Autopilot increases the safety and convenience of driving, but the team's goal is to develop and deploy [Full Self-Driving](#) to our rapidly growing fleet of millions of cars. Our Aug 2021 [Tesla AI Day](#) provides the most detailed and up-to-date overview of this effort.

2015 -  I was a research scientist and a founding member at [OpenAI](#).

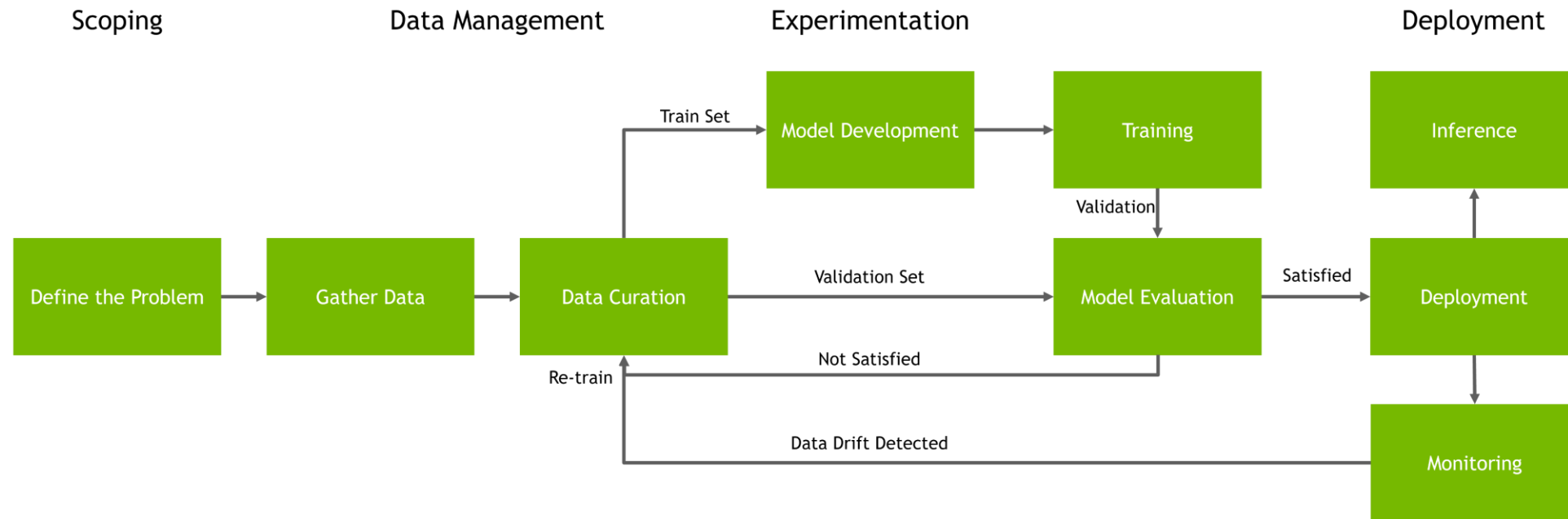
2011 -  My PhD was focused on convolutional/recurrent neural networks and their applications in computer vision, natural language processing and their intersection. My adviser was [Fei-Fei Li](#) at the Stanford Vision Lab and I also had the pleasure to work with [Daphne Koller](#), [Andrew Ng](#), [Sebastian Thrun](#) and [Vladlen Koltun](#) along the way during the first year rotation program.

I designed and was the primary instructor for the first deep learning class Stanford - [CS 231n](#).

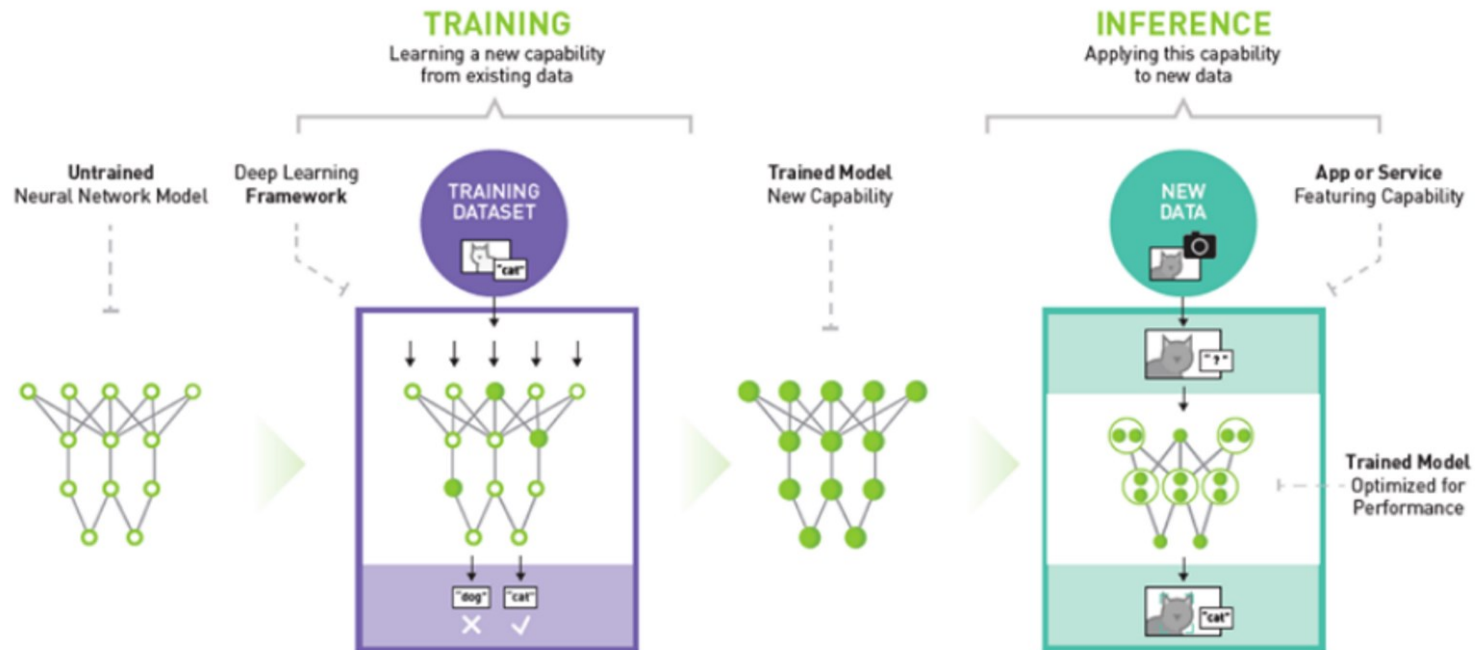
OPENAI SORA

- Transformer Architecture & Training like Diffusion
- Technical Details
 - <https://openai.com/research/video-generation-models-as-world-simulators>

Full Workflow

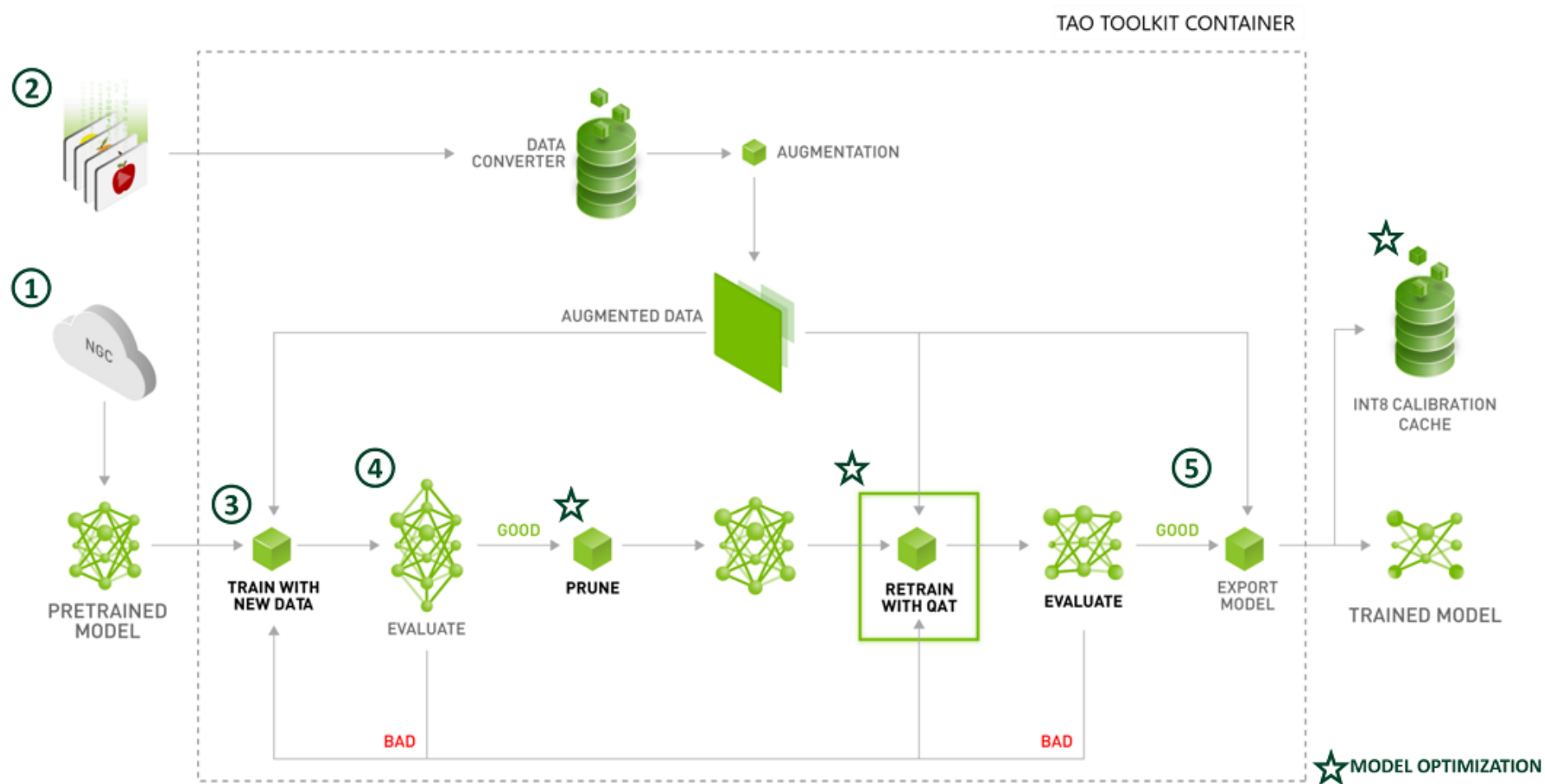


Training Workflow



Deployment

- Pruning
- Quantization



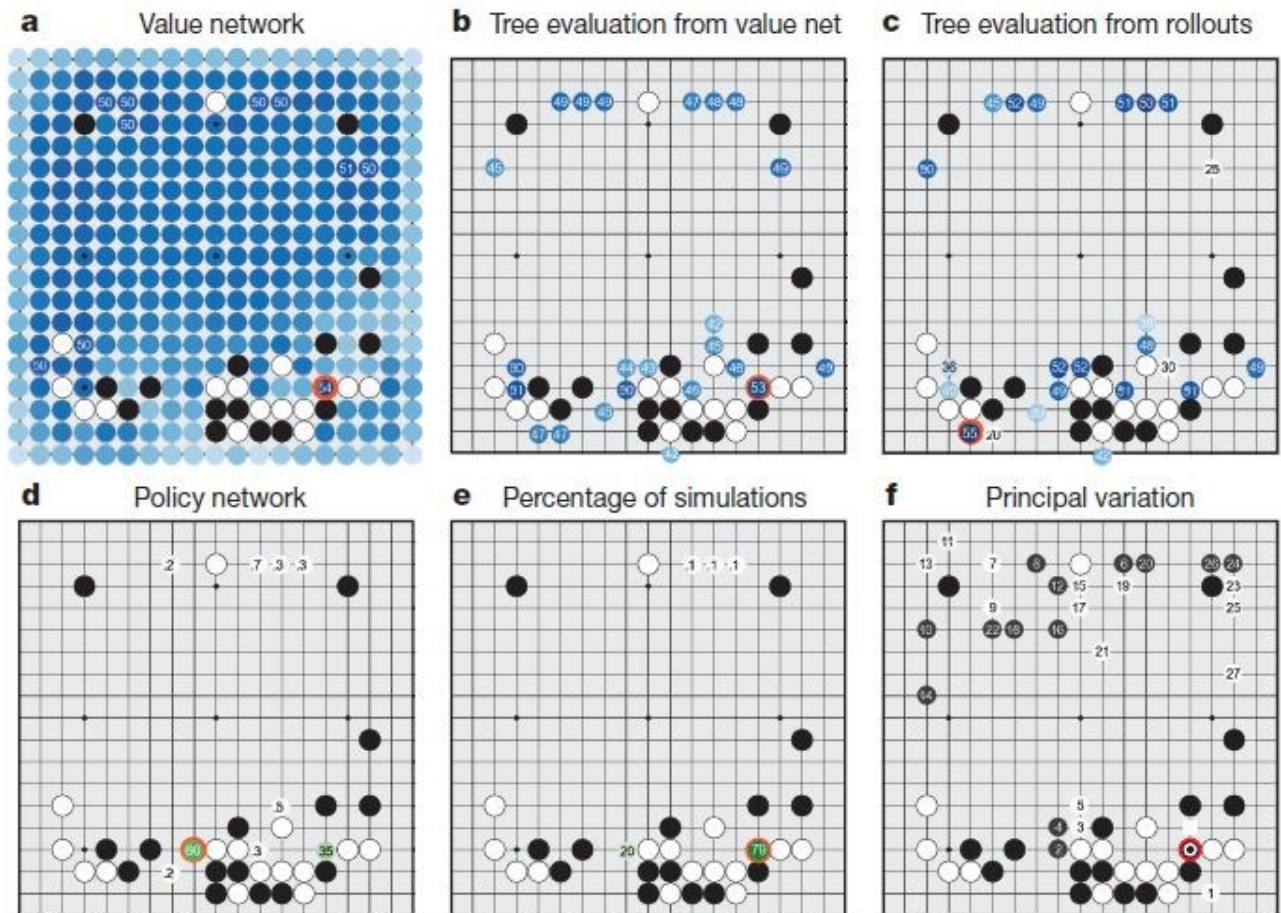
AlphaGo – How it works

- **Value Network**

- CNN 을 이용한 바둑판 현황 분석, 가치가 높은 위치 산정

- **Tree Evaluation from value net, rollouts**

- Value network를 통하여 몇 수를 뒤 보고 승률을 다시 판단하여 어디가 좋을지 결정
 - rollouts(계산 대충해서 빠르게 뒤 보는 것)으로 끝까지 뒤 보고 승률을 다시 판단하여 어디가 좋을지 결정



AlphaGo – How it works

- **Supervised Learning**

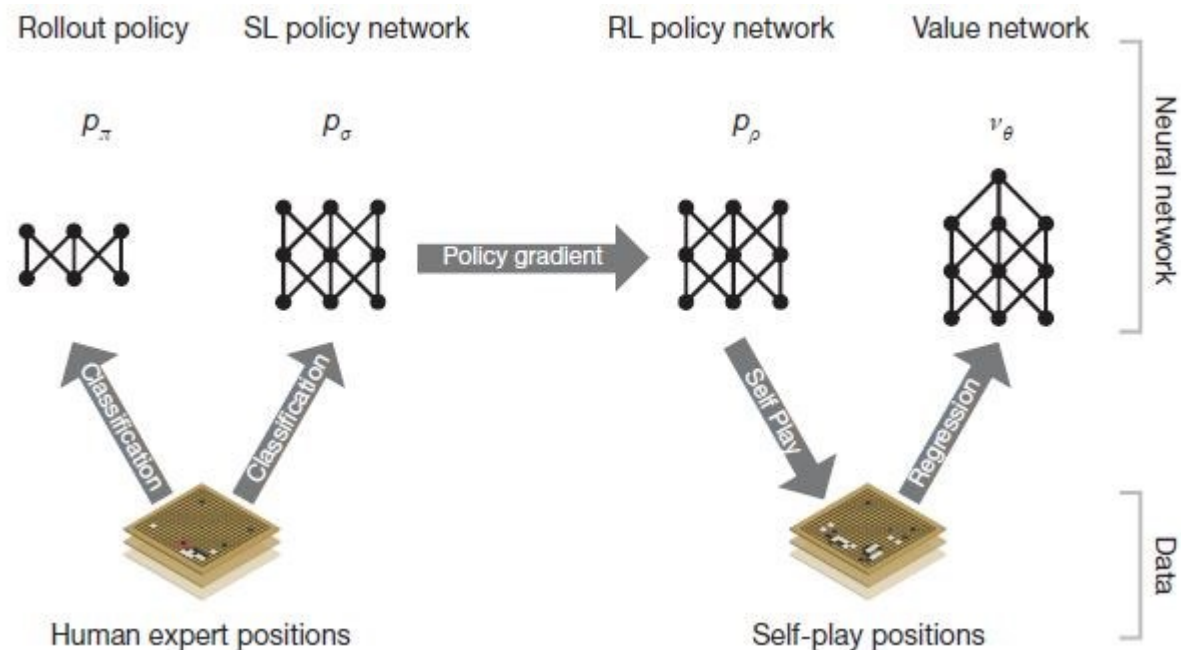
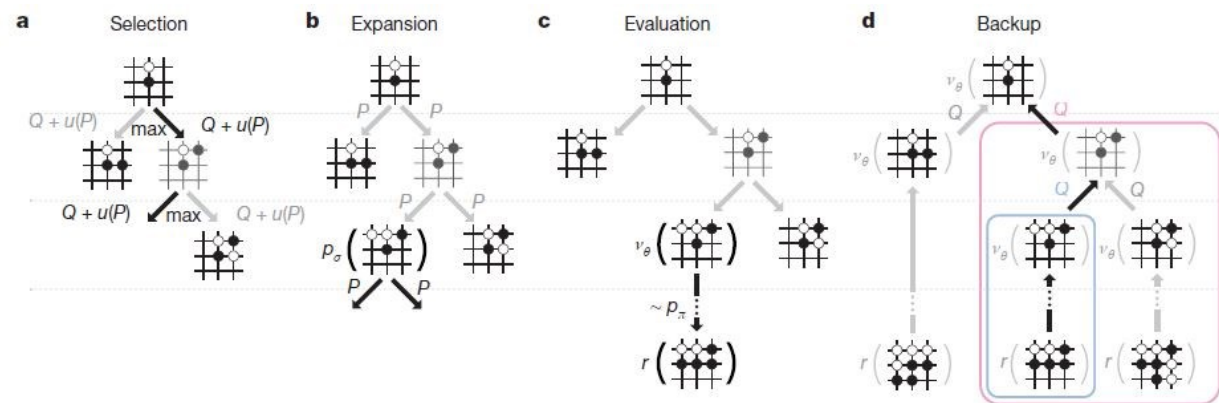
- 선수들의 기보를 보고 학습

- **Reinforcement Learning**

- 자체적으로 바둑을 뒀 보면서 학습

- **Value Network**

- 현재 상황 기준 승률 예측



Contents (detail)

Introduction

딥러닝의 탄생 이야기
Learning Path
딥러닝 최신 트렌드

Neural network

동작 이해
학습, 평가
(실습) Keras 뉴럴넷 만들기
(실습) 시각화 도구로 관찰하며 뉴럴넷 학습시키기

Outro

Multi-modal Data
Paradigm의 변화, 학습에서 prompting으로

Appendix

- Training workflow
- Deployment workflow - Quantization, QAT, Pruning
- AlphaGO - Reinforcement Learning
- Andrej Karpathy - Data Pipeline 의 중요성

Computer Vision

Convolution
MNIST data visualization
CNN - Lenet
(실습) 개/고양이 이미지 분류하기
딥러닝의 시작 - AlexNet & 원리
발전 - VGGNet, GoogLeNet, ResNET & ...
실전 - Transfer Learning (Fine Tuning)
(실습) Transfer Learning
Object Detection
Segmentation
Generative (deprecated) - style transfer, VAE, GAN
Generative - Diffusion
(실습) stable diffusion
Vision Transformer

NLP

Word2Vec
RNN, LSTM, GRU
Seq2Seq
Attention
Transformer
LLM(GPT vs BERT)
AGI를 향한 도전 - Embeddings & VectorDB
(실습) - GPT 응용 - Embeddings

Reference

- [1] ImageNet (<https://www.image-net.org/about.php>)
- [2] DallE3 (<https://openai.com/dall-e-3>)
- [3] 3Blue1Brown (<https://www.youtube.com/@3blue1brown>)
- [4] Assembly.ai (<https://www.assemblyai.com/blog/pytorch-vs-tensorflow-in-2023/>)
- [5] 한땀한땀 딥러닝 컴퓨터 비전 백과사전 (<https://wikidocs.net/book/6651>)
- [6] Yann Lecun (<http://yann.lecun.com/exdb/lenet/>)
- [7] Alex et al. (https://proceedings.neurips.cc/paper_files/paper/2012/file/c399862d3b9d6b76c8436e924a68c45b-Paper.pdf)
- [8] Wei et al. (<https://icmlviz.github.io/icmlviz2016/assets/papers/4.pdf>)
- [9] Karen et al. (<https://arxiv.org/pdf/1409.1556.pdf>)
- [10] Christian et al. (<https://arxiv.org/pdf/1409.4842.pdf>)
- [11] Kaiming et al. (<https://arxiv.org/pdf/1512.03385v1.pdf>)
- [12] Joseph et al. (<https://arxiv.org/abs/1506.02640>)
- [13] Ronneberger et al. (<https://arxiv.org/pdf/1505.04597.pdf>)
- [14] Mets et al. (<https://arxiv.org/pdf/1611.02163.pdf>)
- [15] NVIDIA (<https://developer.nvidia.com/blog/improving-diffusion-models-as-an-alternative-to-gans-part-2/>)
- [16] Mikolov et al. (<https://arxiv.org/pdf/1301.3781.pdf>)
- [17] Vaswani et al (<https://arxiv.org/abs/1706.03762>)
- [18] Zhao et al (<https://arxiv.org/pdf/2303.18223.pdf>)

Additional Resources

State of AI 2023 ([ref](#))

Introducing GPTs ([ref](#))

Open Compute Narrow Precision ([ref](#))

LLM Applications today, 23.10.30 ([ref](#))

AI야 우리 딸 그림 좀! ([ref](#))

CIVIT AI

stable video diffusion ([ref](#))

Krita ai plugin ([ref](#))

Animated ai ([ref](#))

H100 tour (https://www.youtube.com/watch?app=desktop&v=a_tXcmEeGxo)

Stable diffusion 3d ([ref](#))

Microsoft VASA ([VASA-1 - Microsoft Research](#))

GENAI Awesome Links ([ref](#))

ChatGPT 4o (<https://openai.com/index/hello-gpt-4o/>)

GPU cloud lambdaLabs (<https://lambdalabs.com/>)